09/06/2025

The Cranial Base

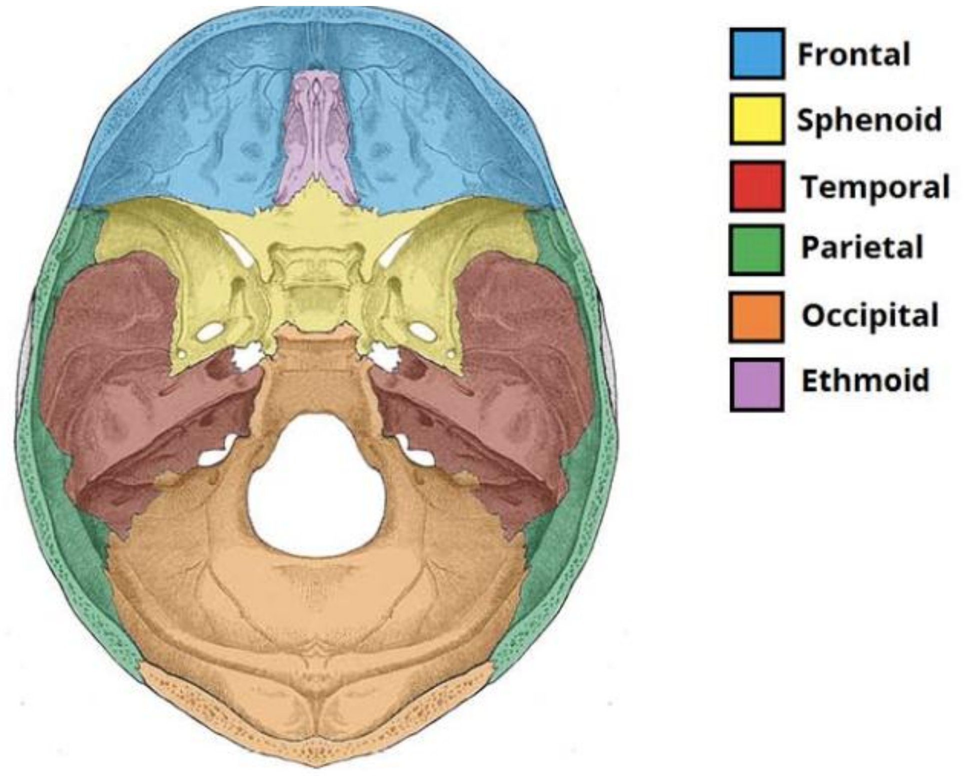
LEE ANN CUNY, DO

Objectives – Base (Sphenoid and Temporal)

- Identify the individual bones of the cranial base, their developmental parts and origin (cartilage or membrane) and locations in the adult osteology
- Describe the physiologic motion of the individual bones with the primary respiratory mechanism (PRM) axis of motion
- Identify patient presentations that would indicate a need to evaluate and possibly treat the bones of the cranial base, including possible cranial nerve impingements
- List the foramen present in the bones of the cranial base and the relevant structures that traverse.

Cranial Base

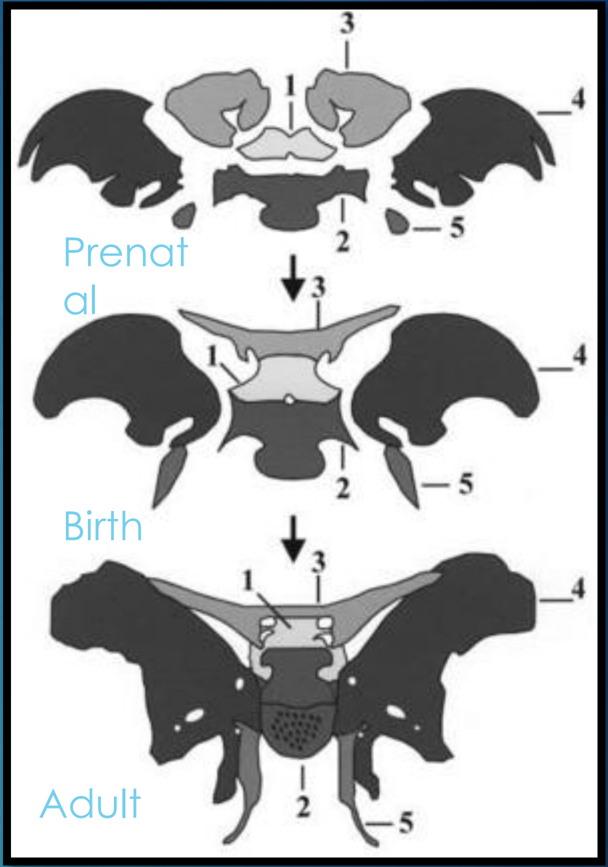
- Bone formed in cartilage secondary to compression
  - Ethmoid
  - Basi-Sphenoid, medial parts of greater and lesser wings
  - Basi-occiput, condylar parts, supraocciput
  - Temporal x 2
    - Petrous portion
    - Mastoid portion



Teachmeanatomy.info

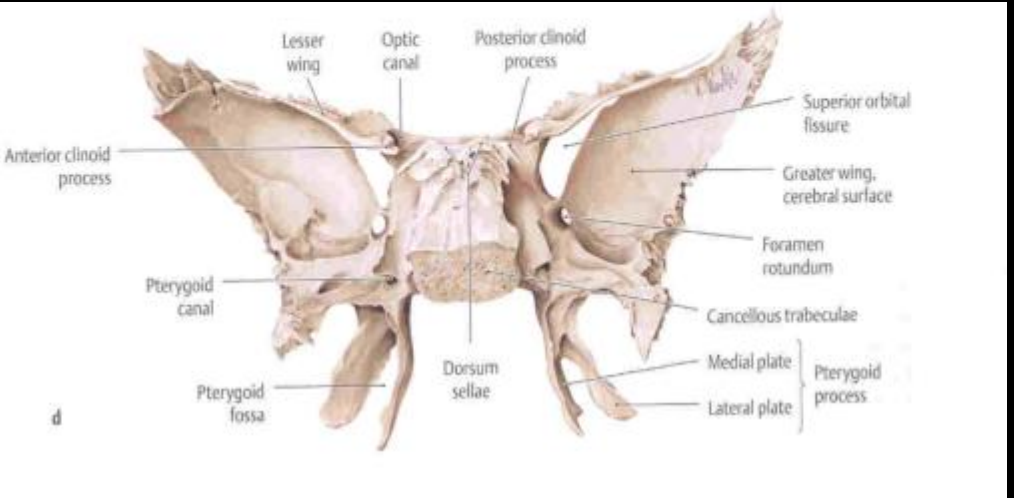
Sphenoid

- **Cartilaginous Ossification**
  - Sphenoid Body- Develops in 2 parts
    - Presphenoid
      - Tuberculum Sellae with lesser wings
      - 2 ossification centers of the body 1 for each wing
    - Postsphenoid
      - Sella Tursica, Dorsum Sellae
      - 2 ossification centers for the body portion, 1 for each lingula
  - The Lesser Wings
  - Proximal aspects of the greater wings and pterygoid plates
- **Membranous Ossification**
  - Distal aspects of the greater wings and pterygoid plates

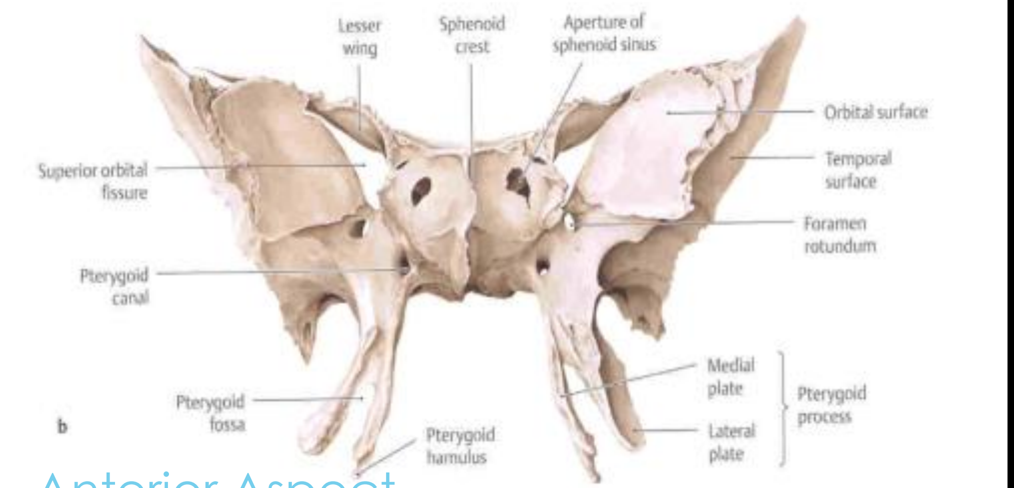


Embryologic Origin/Fetal Vs Adult Sphenoid

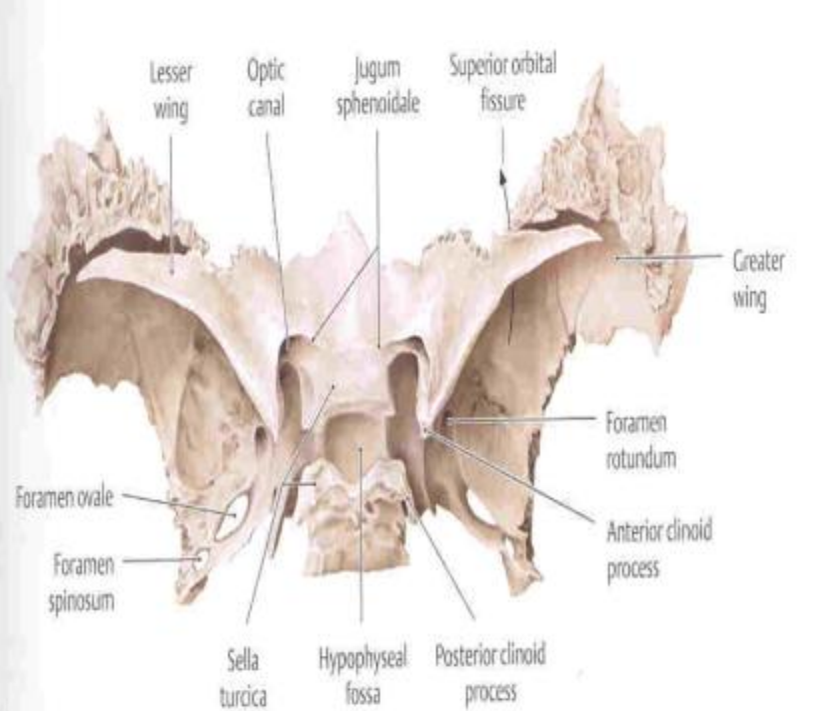
Parts of the Sphenoid



Posterior Aspect

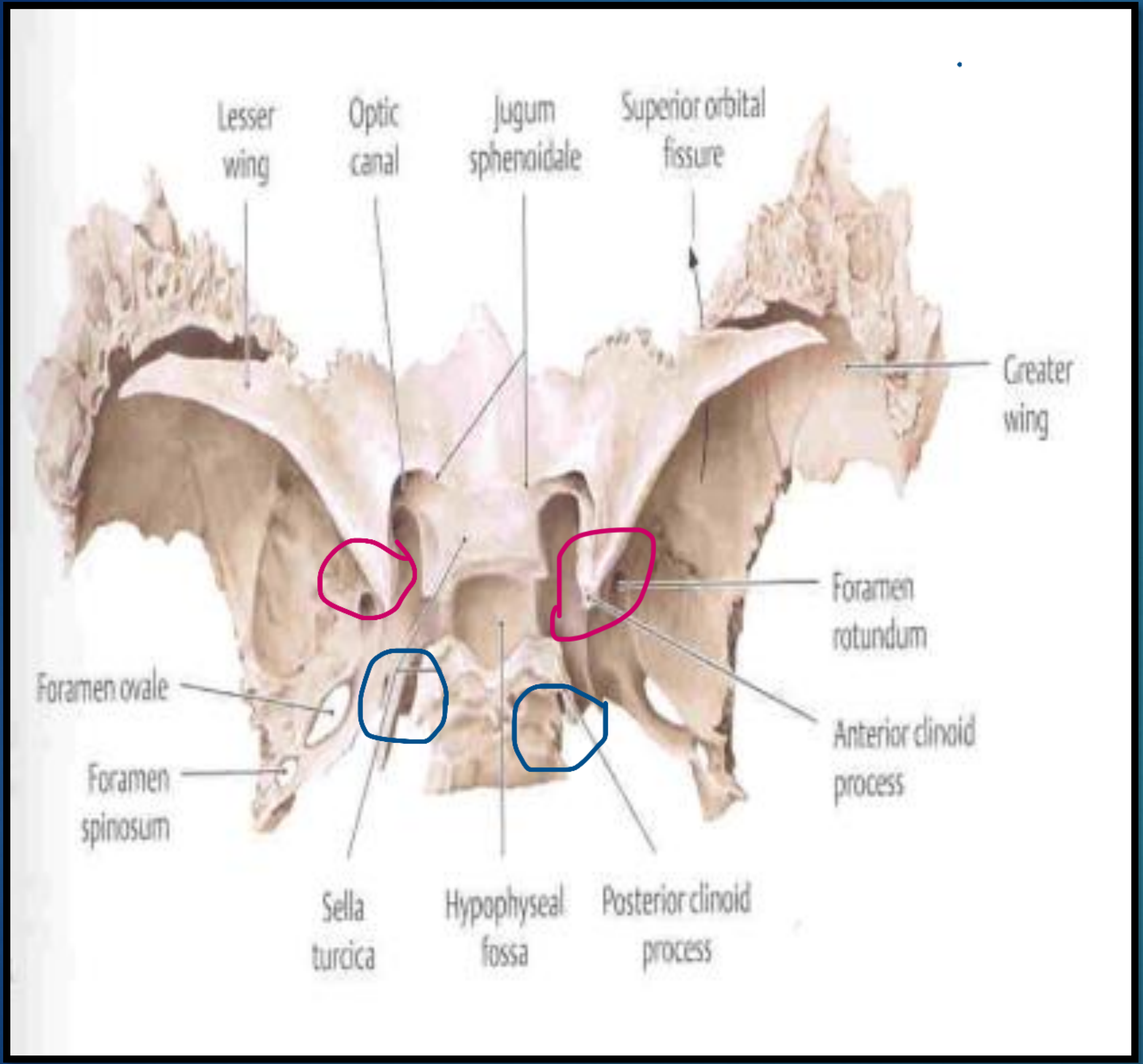


Anterior Aspect



Superior Aspect

greater wings, lesser wings, optic canal, superior orbital fissure, ant post clinoid processes, hypophyseal fossa, lateral medial plat of pterygoid fossa



- **Tentorium Cerebelli**
  - Free border
    - Attaches to the anterior clinoid process
  - Fixed border
    - Attaches to the posterior clinoid process

Attachment of the Falx Cerebelli

Identification of the Sphenoid Parts & Articulations

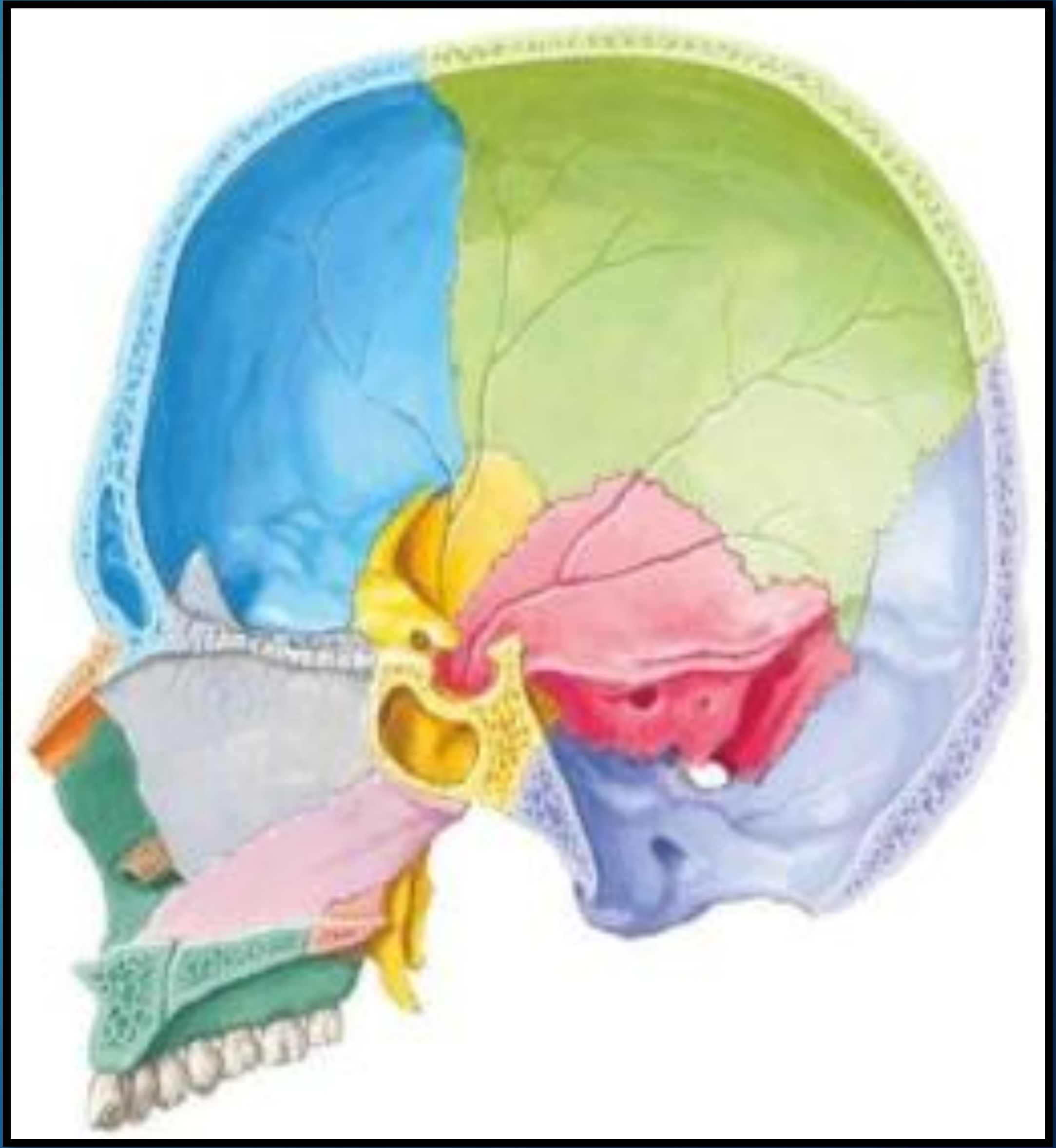
Parts of the Sphenoid

- Body-Sella Turcica
- Two lesser wings
- Two greater wings
- Two pterygoid processes

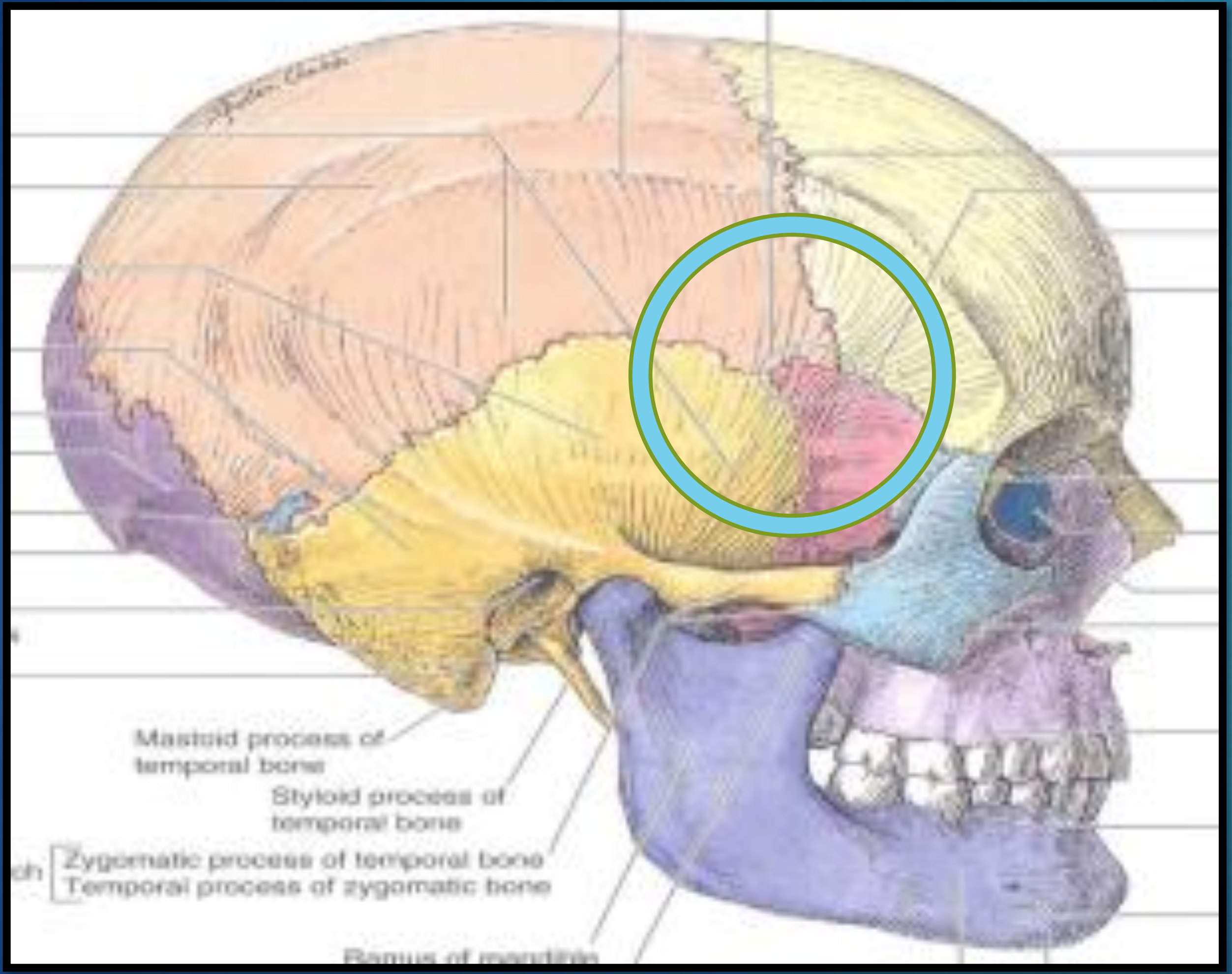
Articulates with:

1. Occipital
2. Temporal
3. Parietal
4. Frontal
5. Zygomae
6. Palatine
7. Vomer
8. Ethmoid

be aware of the articulations



Identification of Sphenoid Parts & Articulations

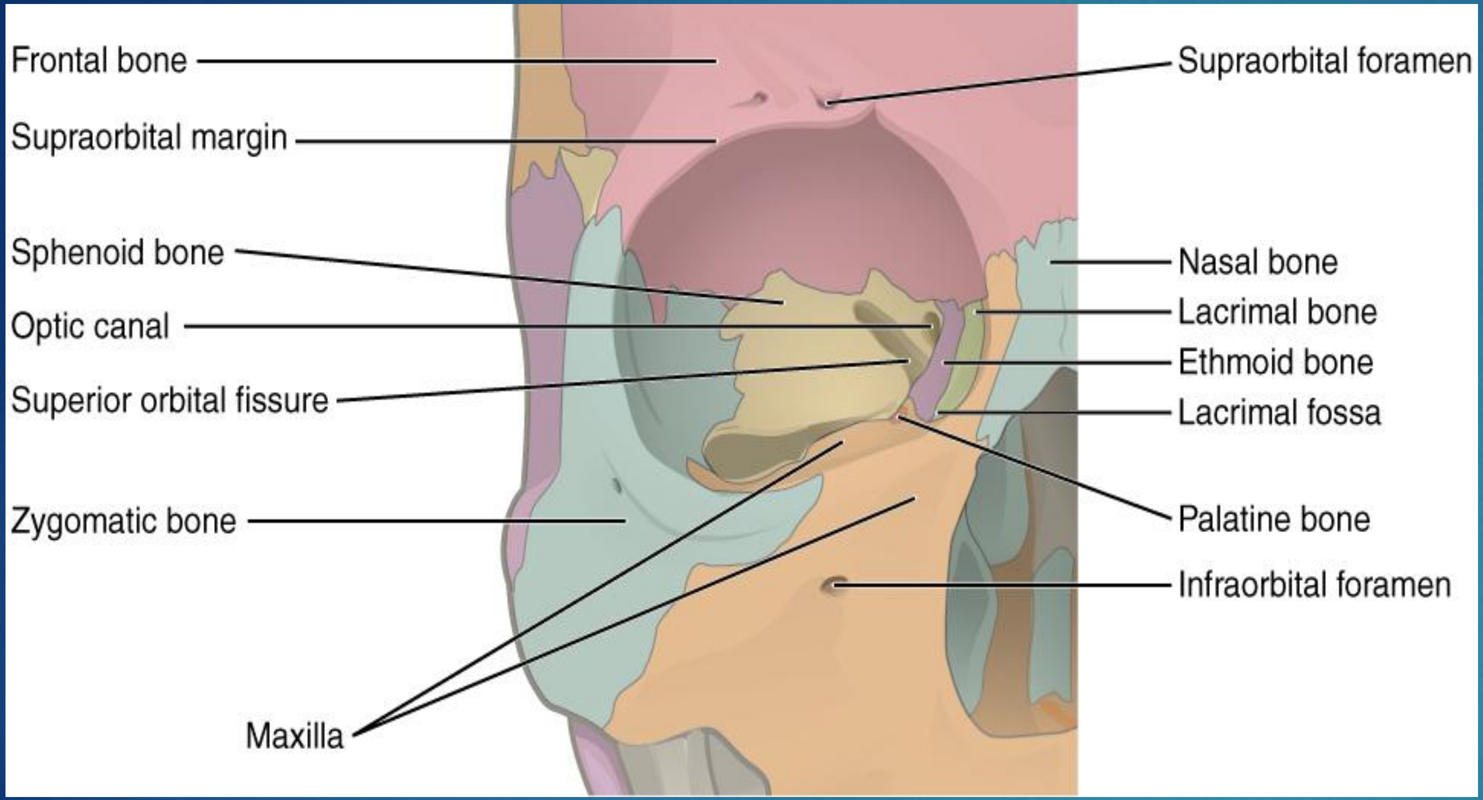


Articulations at Pterion

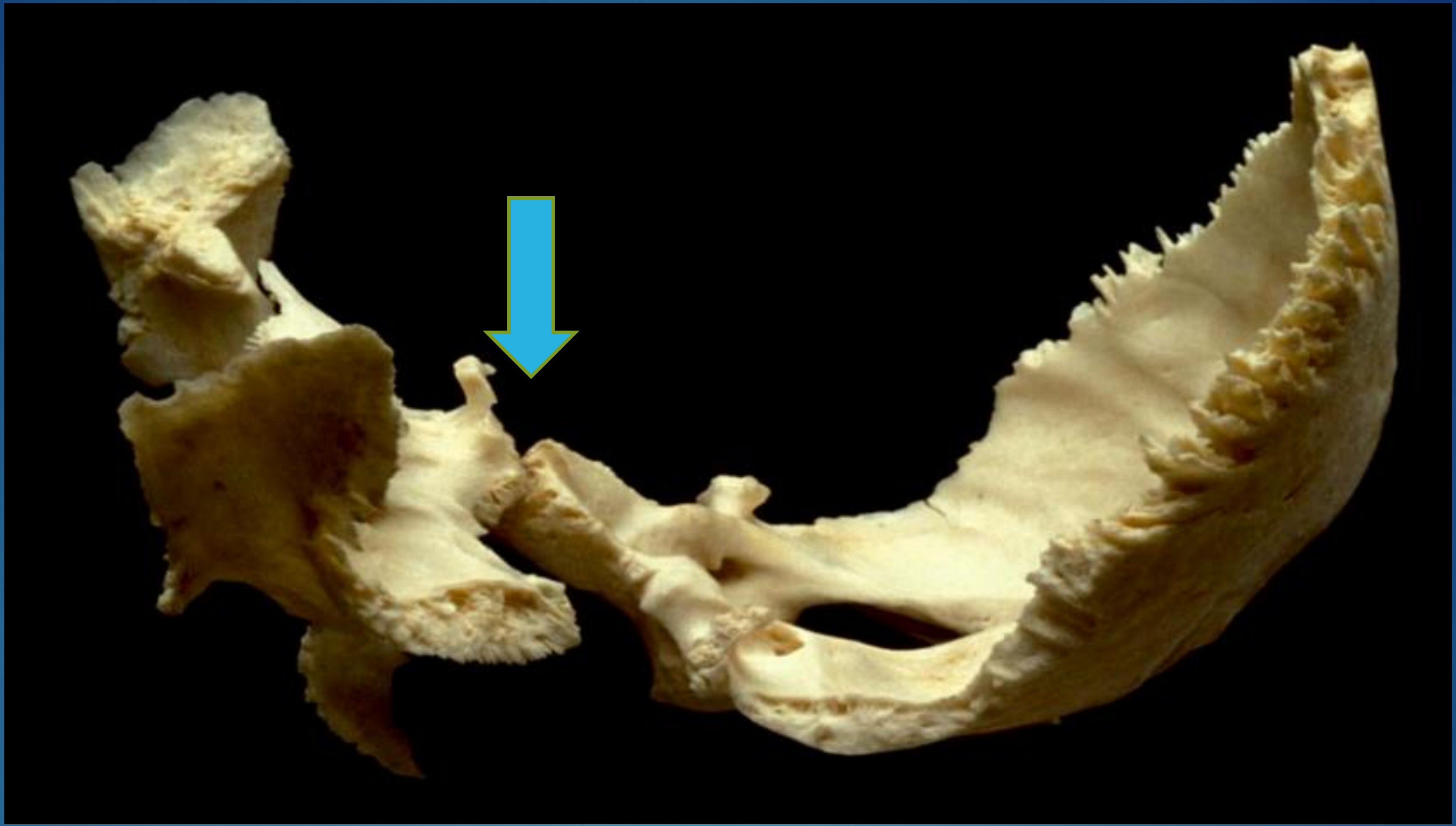
- Temporal
- Parietal
- Frontal
- Sphenoid

Superficial → Deep
Temporal, Sphenoid, Parietal, Frontal

Identification of Sphenoid Parts & Articulations



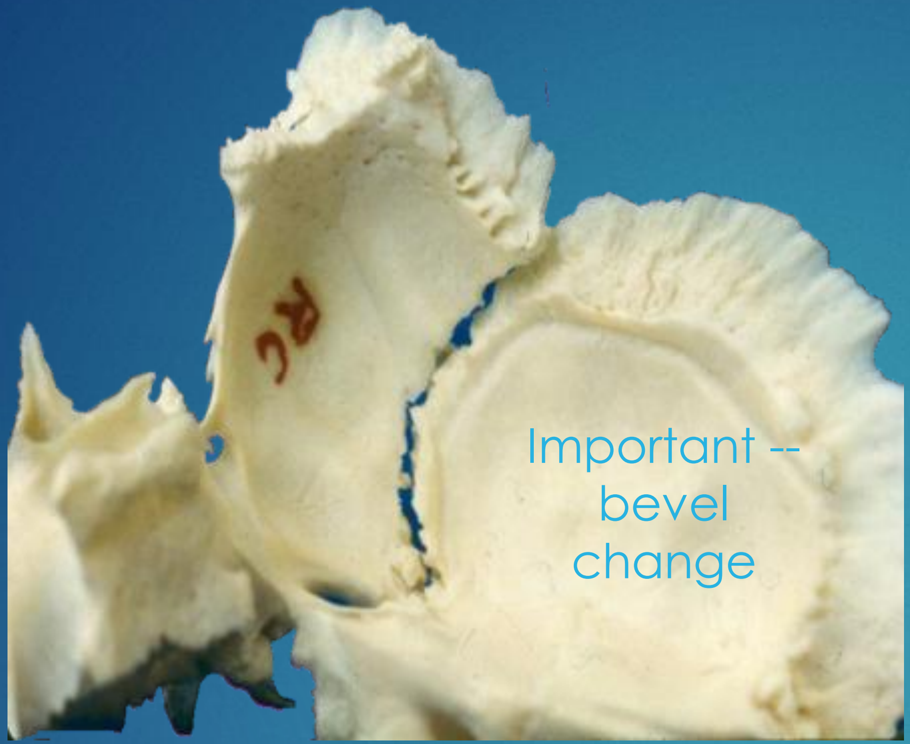
Frontal (red)
Maxilla (orange)
Sphenoid (yellow)
Zygoma (blue-green)
Lacrimal (lime green)
Palatine (barely visible red-orange)
Ethmoid (purple)

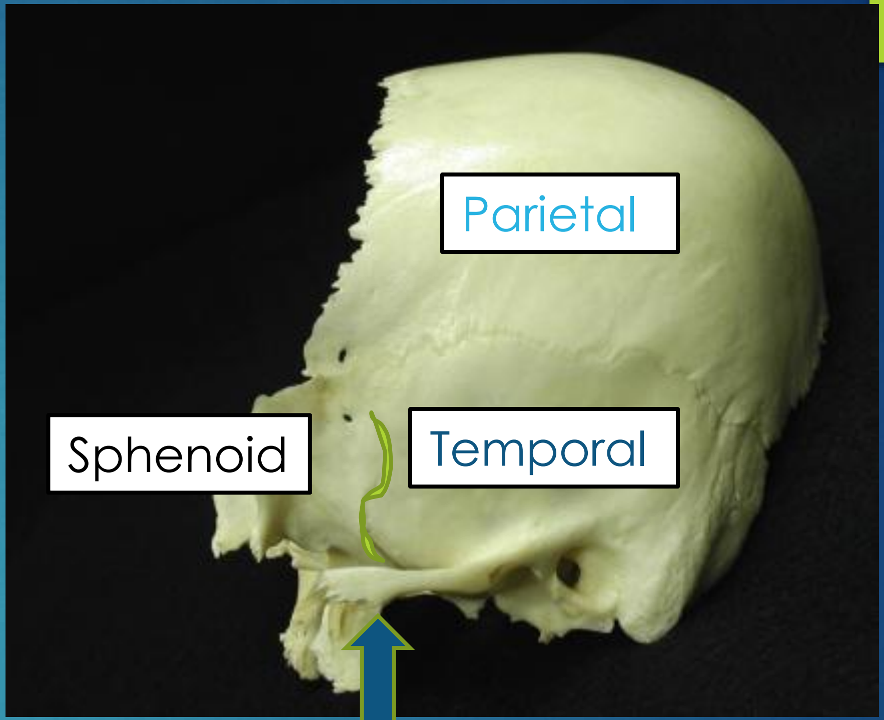


Sphenobasilar Synchondrosis (SBS)

Between body of sphenoid and basilar portion of occiput

normal motion and f/e in vault

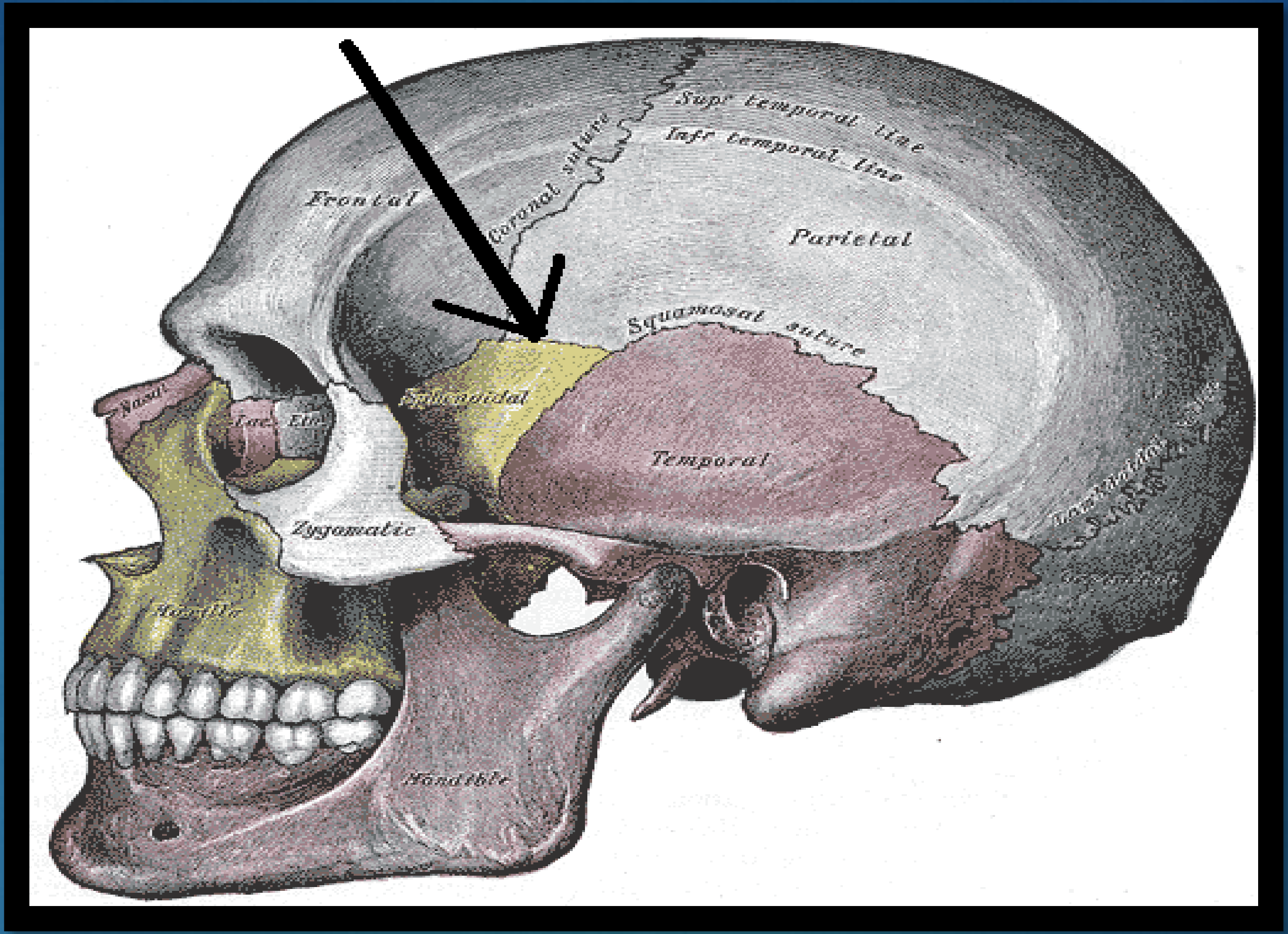




Sphenosquamous Articulation

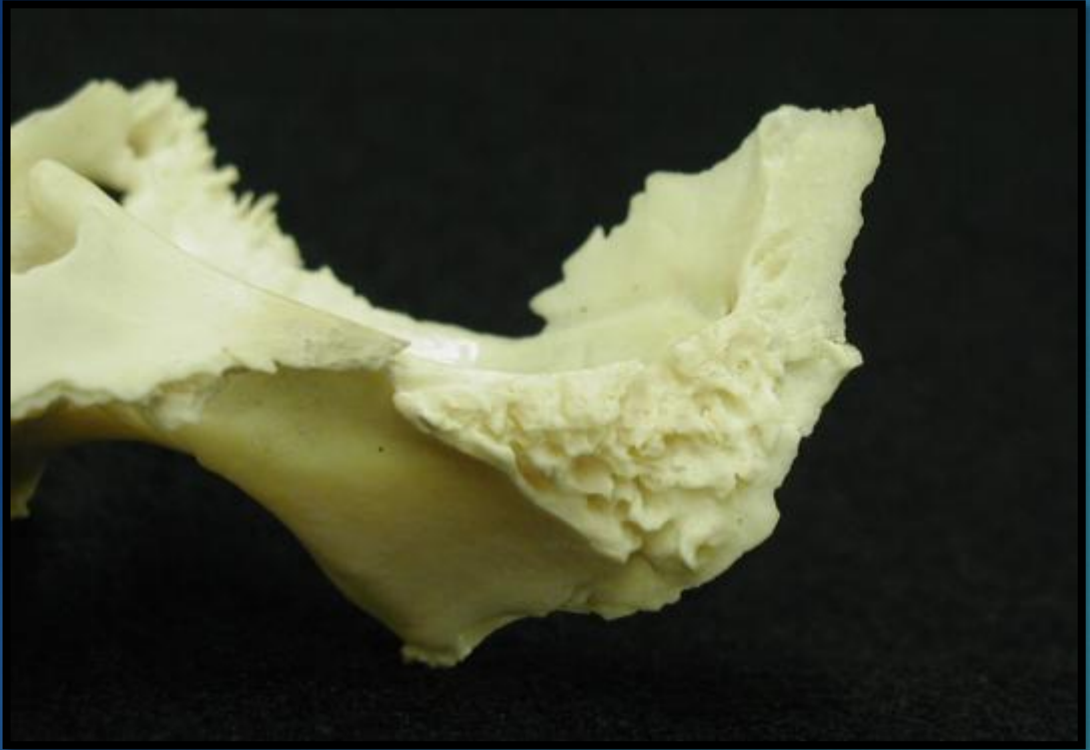
ss pivot Between sphenoid and at squamous temporal

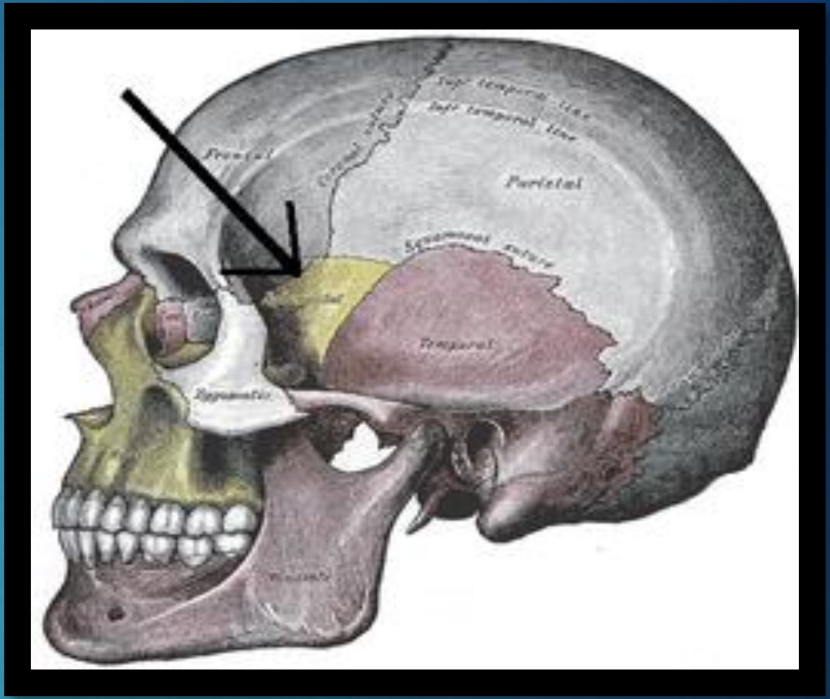
** bevel change at the level of the zygomatic process



Sphenoparietal Suture

Articulation with the parietal bone

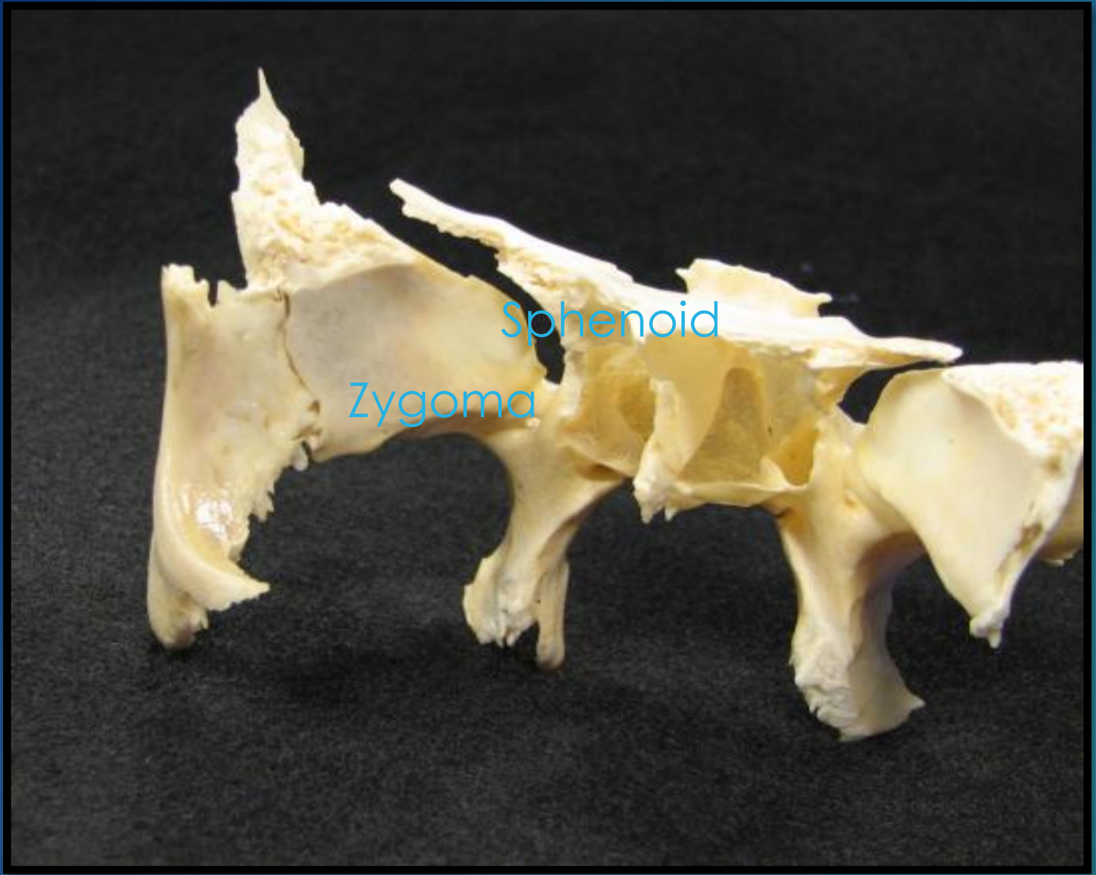


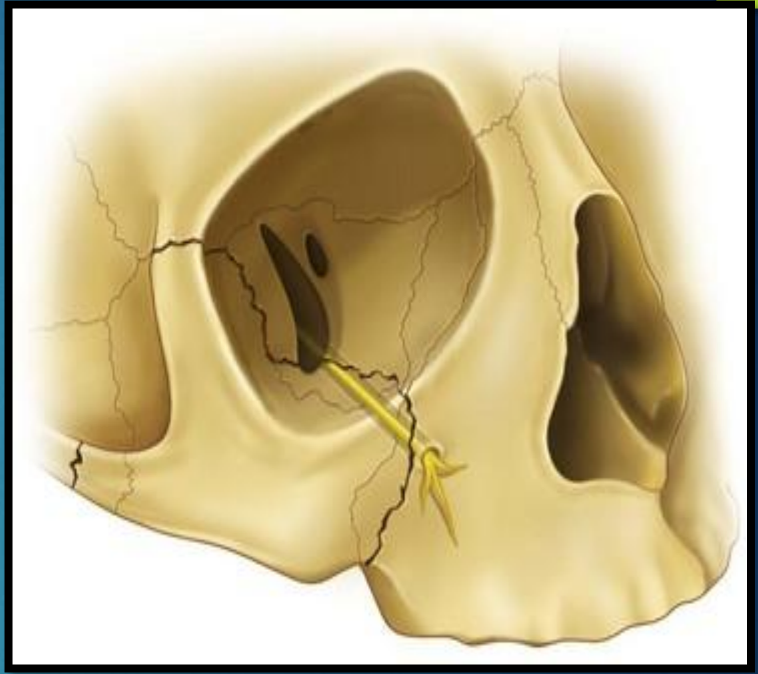


Frontosphenoidal Articulation

triangular shaped

Articulation with the frontal bone

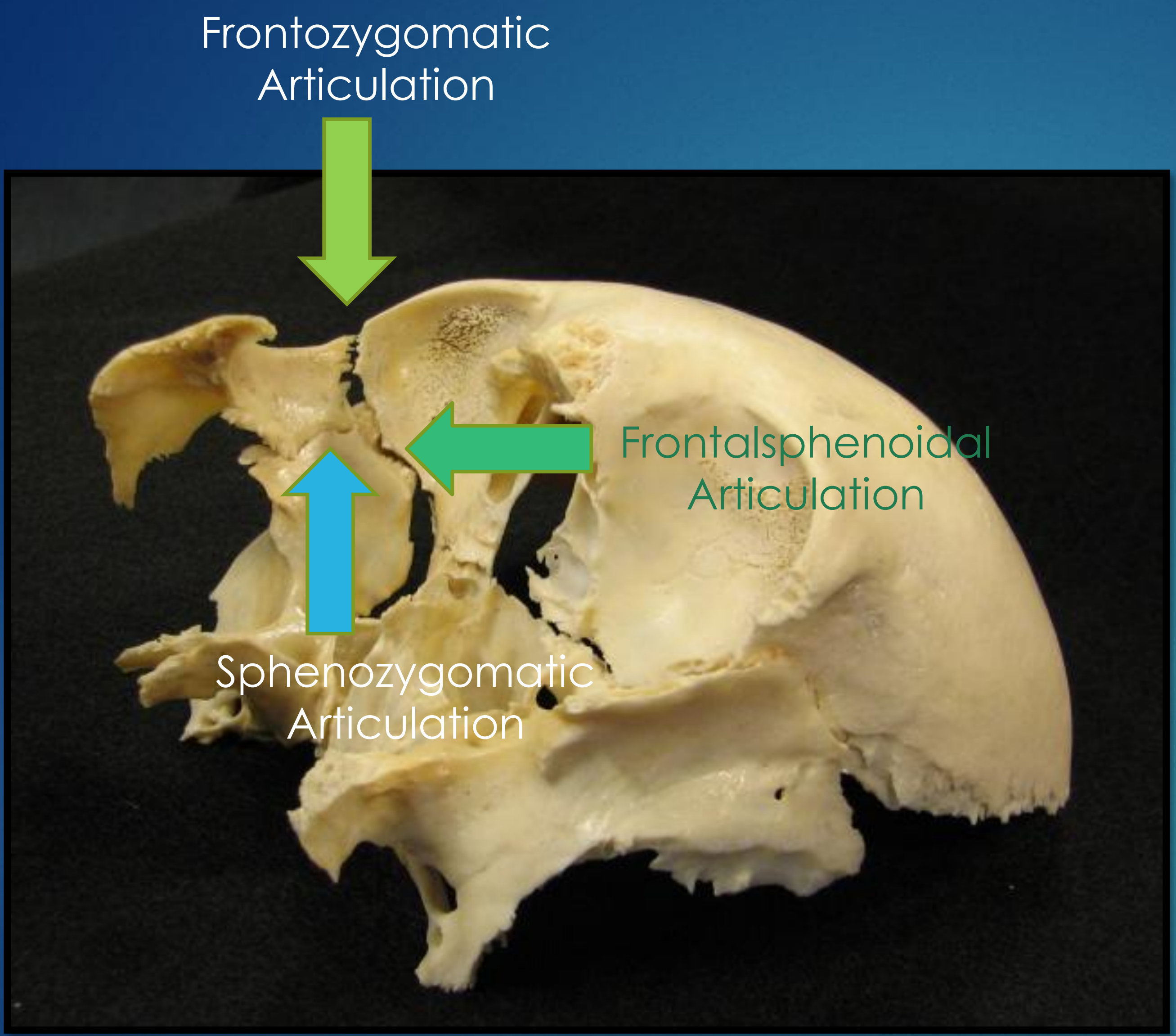


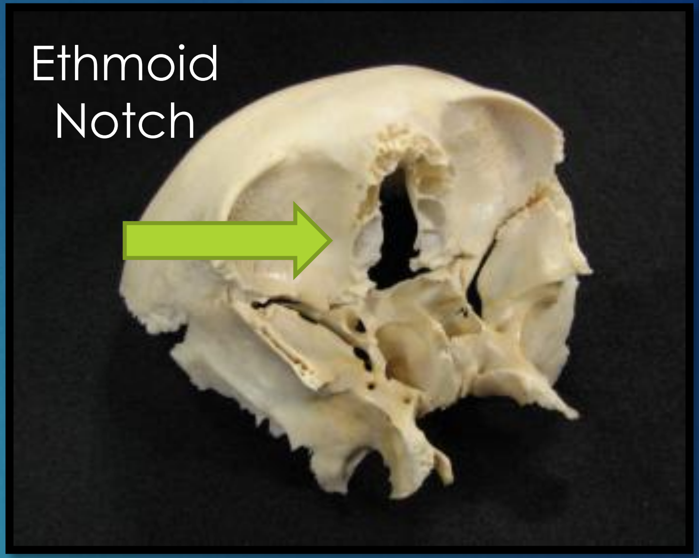


https://pocketdentistry.com/16-fractures-of-the-zygomatic-complex-and-arch/

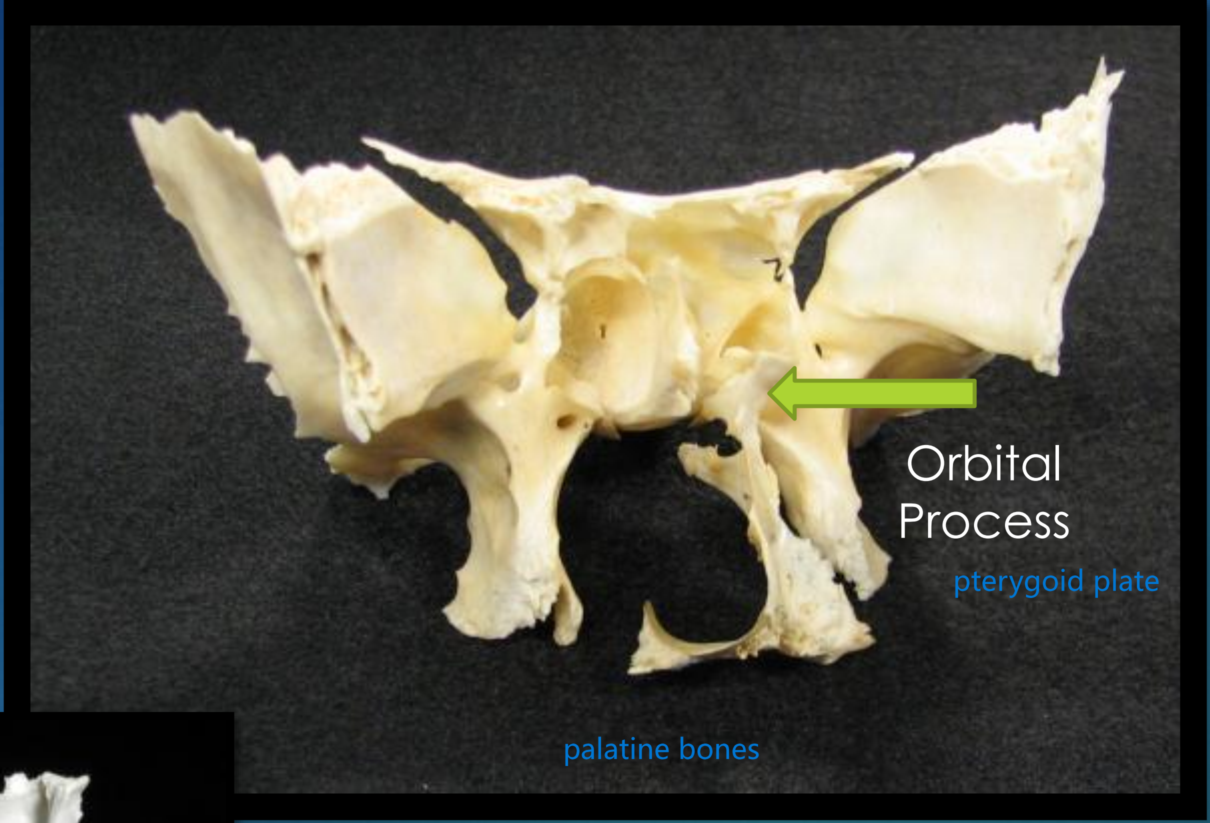
Sphenozygomatic Articulation

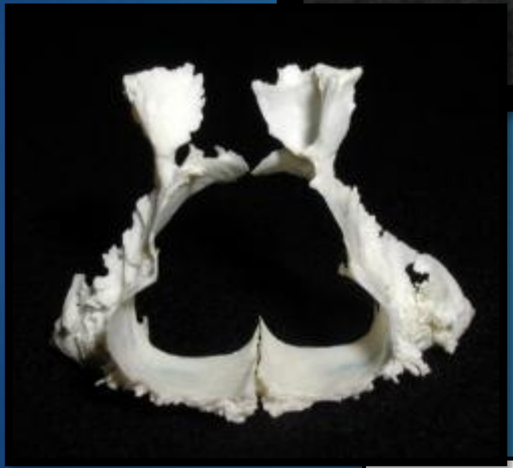
Articulation with the Zygoma

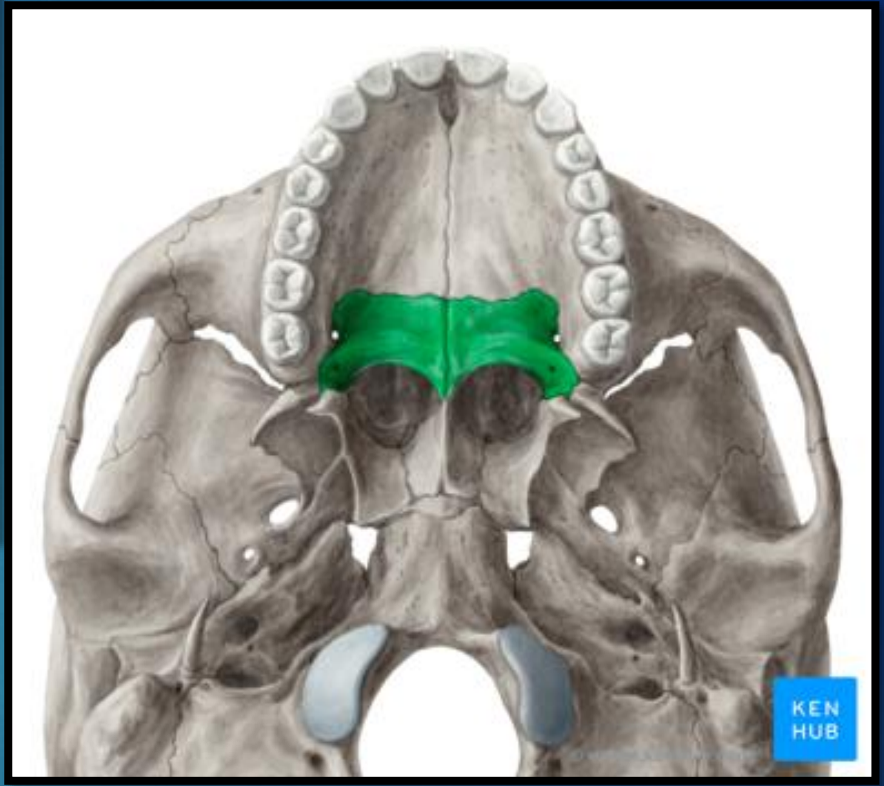




Putting the Articulations Together

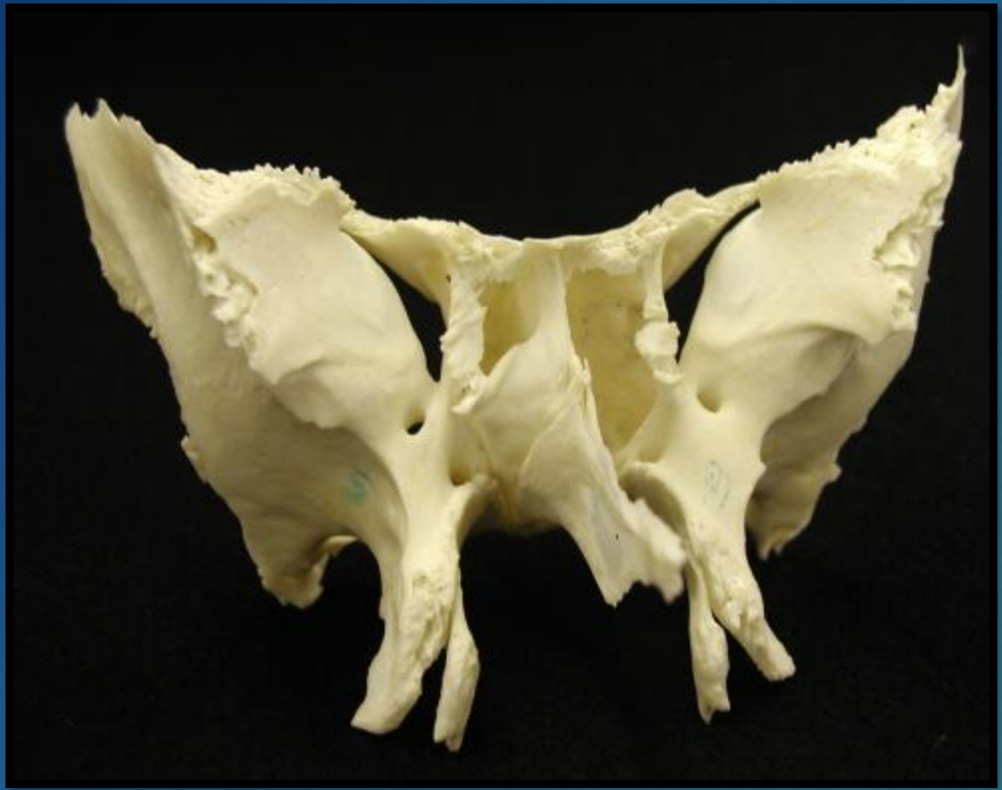


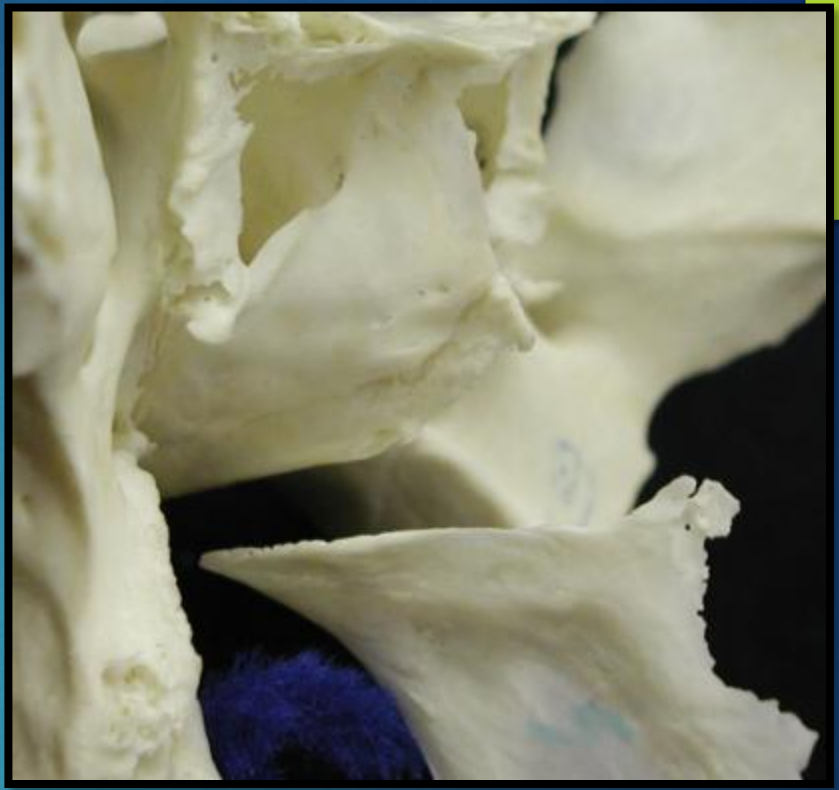




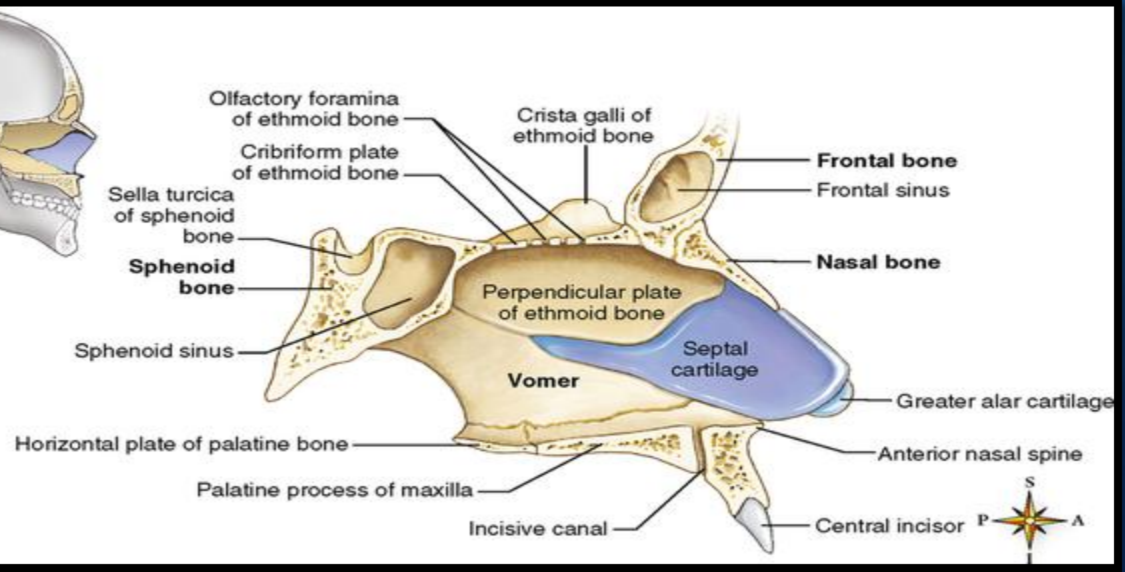
Sphenoid and Palatines

Articulation with the palatine bones



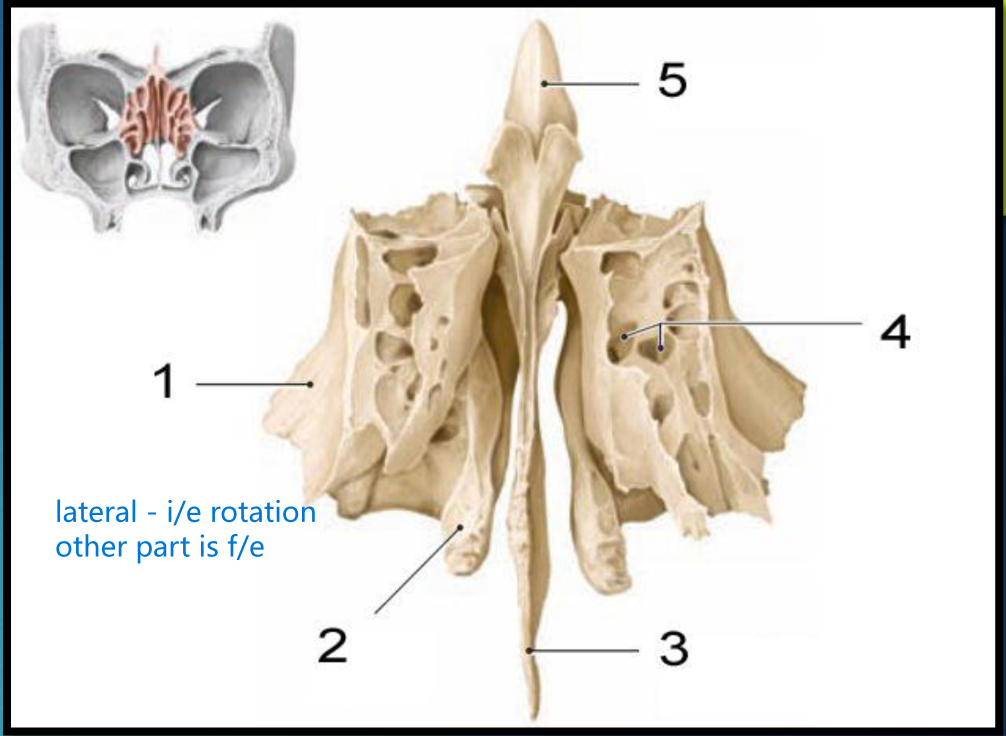


Sphenoid-Vomer Articulation



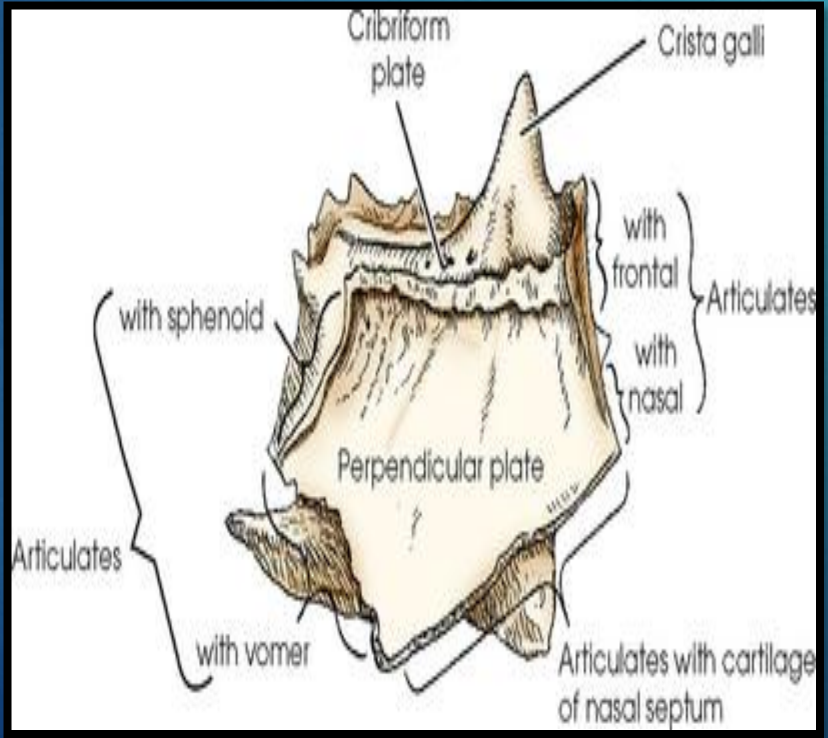
sphenoid drive motion of vomer

1. Orbital plate
2. Middle Concha
3. Perpendicular Plate
4. Ethmoid Air Cells
5. Crista Galli

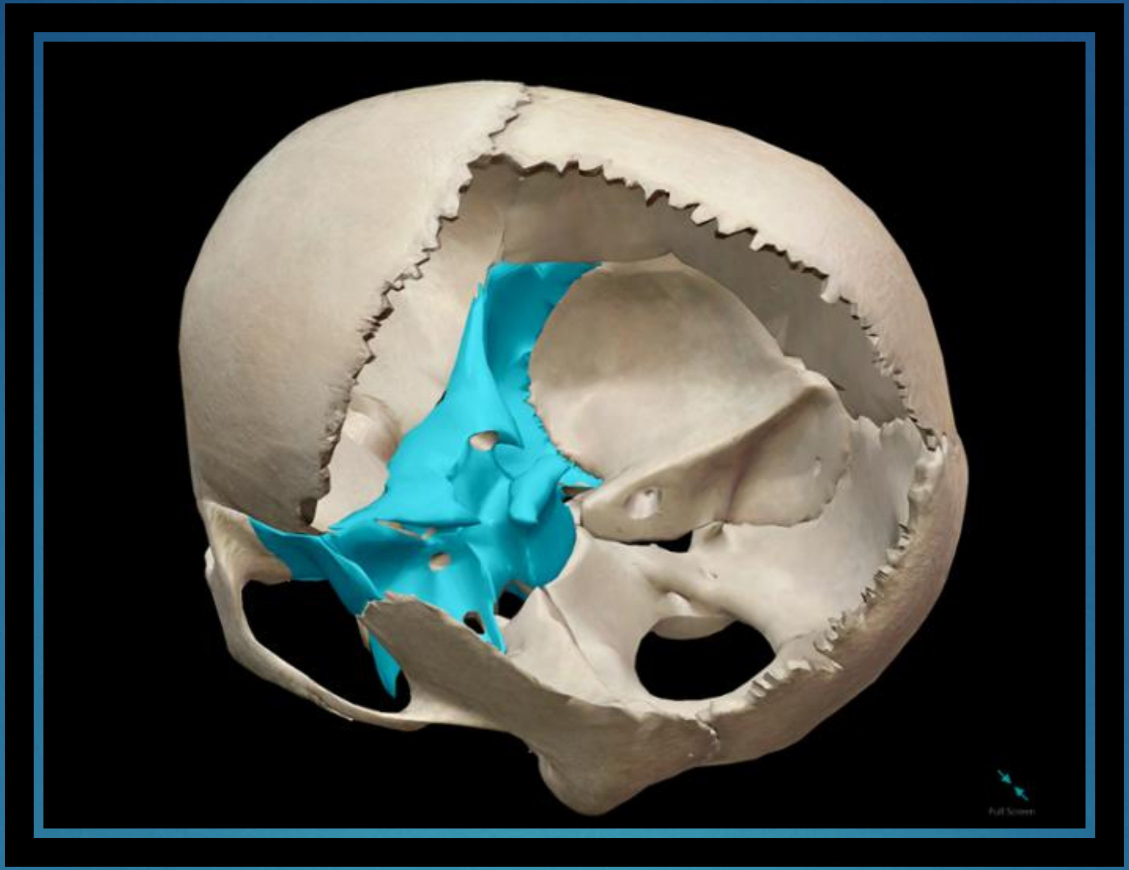


Posterior View Ethmoid

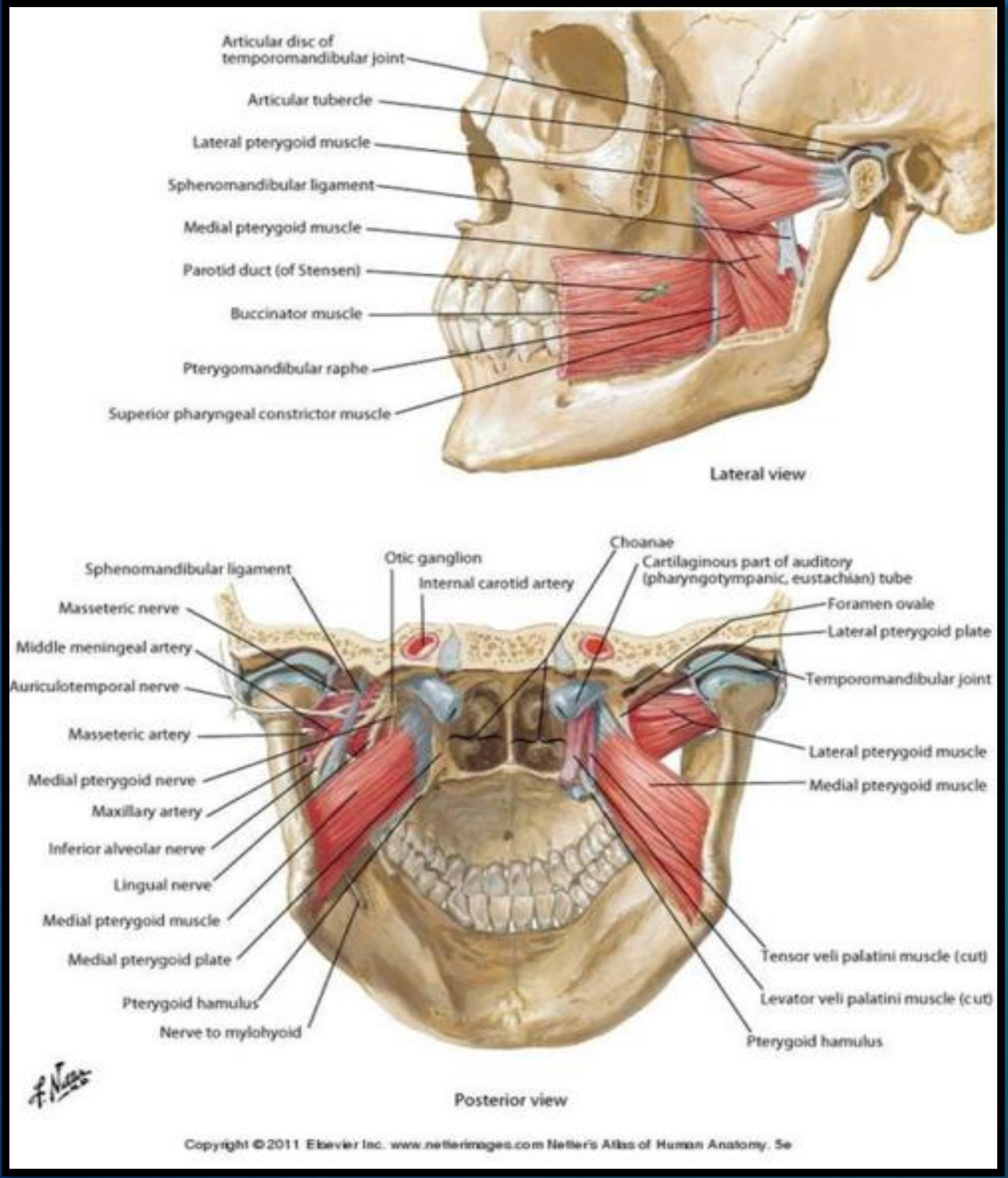
Lateral View Ethmoid



Spheno-Ethmoid Articulation



Important Muscular & Ligamentous Attachments to the Sphenoid



- The Pterygoid Processes
  - Separated by a fissure which articulates with the palatine bone
  - Diverge posteriorly creating a fossa
  - Attachment point for:

1. Lateral Pterygoid
   - Origin- greater wing of sphenoid and lateral pterygoid plate
   - Insertion- TMJ and mandible
   - V3 innervation
   - Jaw protrusion and closure
2. Medial Pterygoid
   - Origin: medial surface of the lateral pterygoid plate and pterygoid fossa
   - Insertion: mandibular angle
   - V3 innervation
   - Jaw closure
3. Tensor Veli Palatini
   - Origin- medial pterygoid plate
   - Insertion- palatine aponeurosis
   - V3 innervation

Muscular Attachments at the Sphenoid

- Middle ear through the petrous portion of the temporal bone

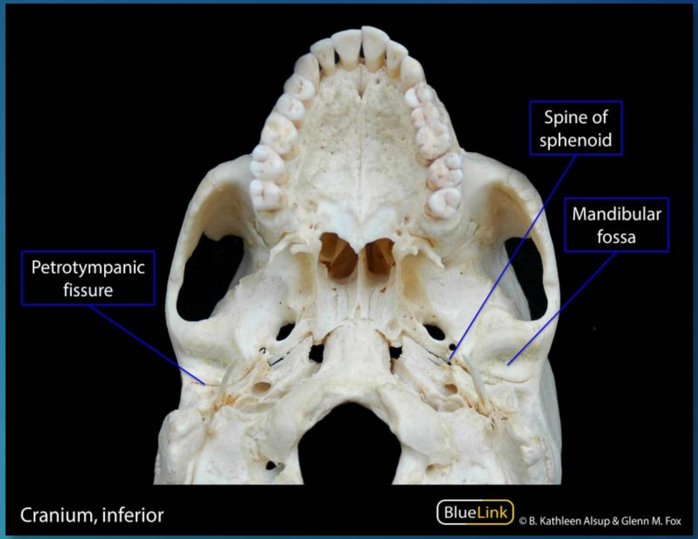
- Across the articulation between the sphenoid and the petrous tip
  - The groove lies medial to the spine of the sphenoid

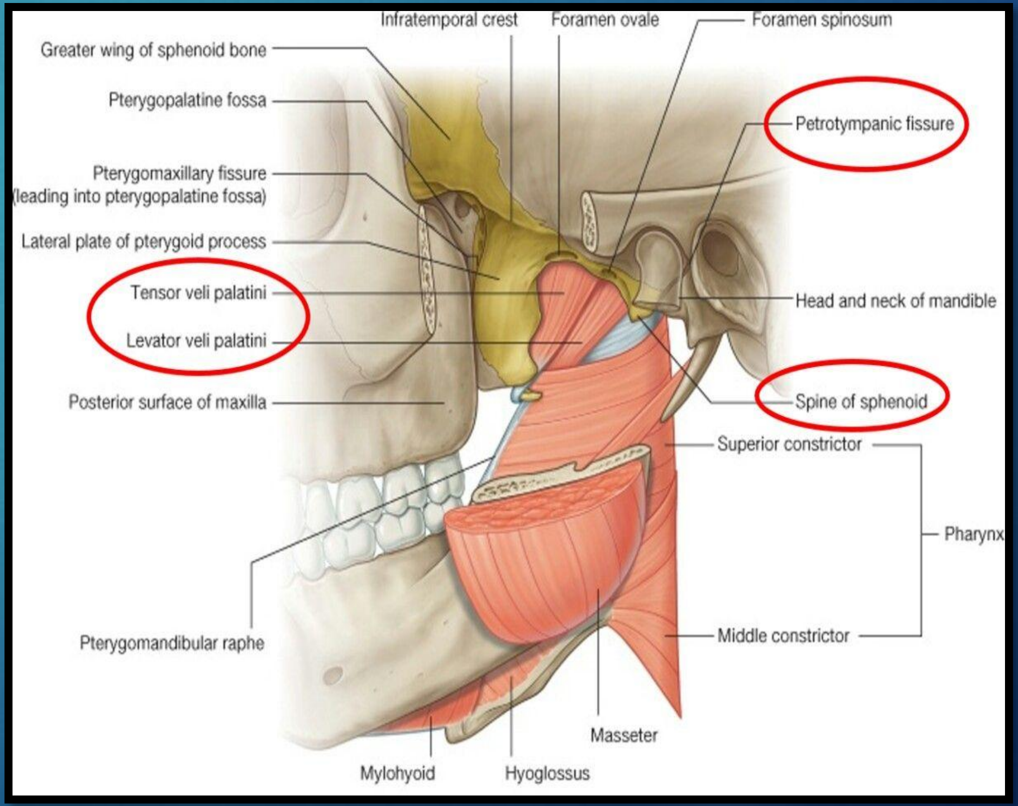
- Through pharyngeal musculature
  - Closely approximates tensor veli palatini and lavator veli palatini and salpingopharyngeus

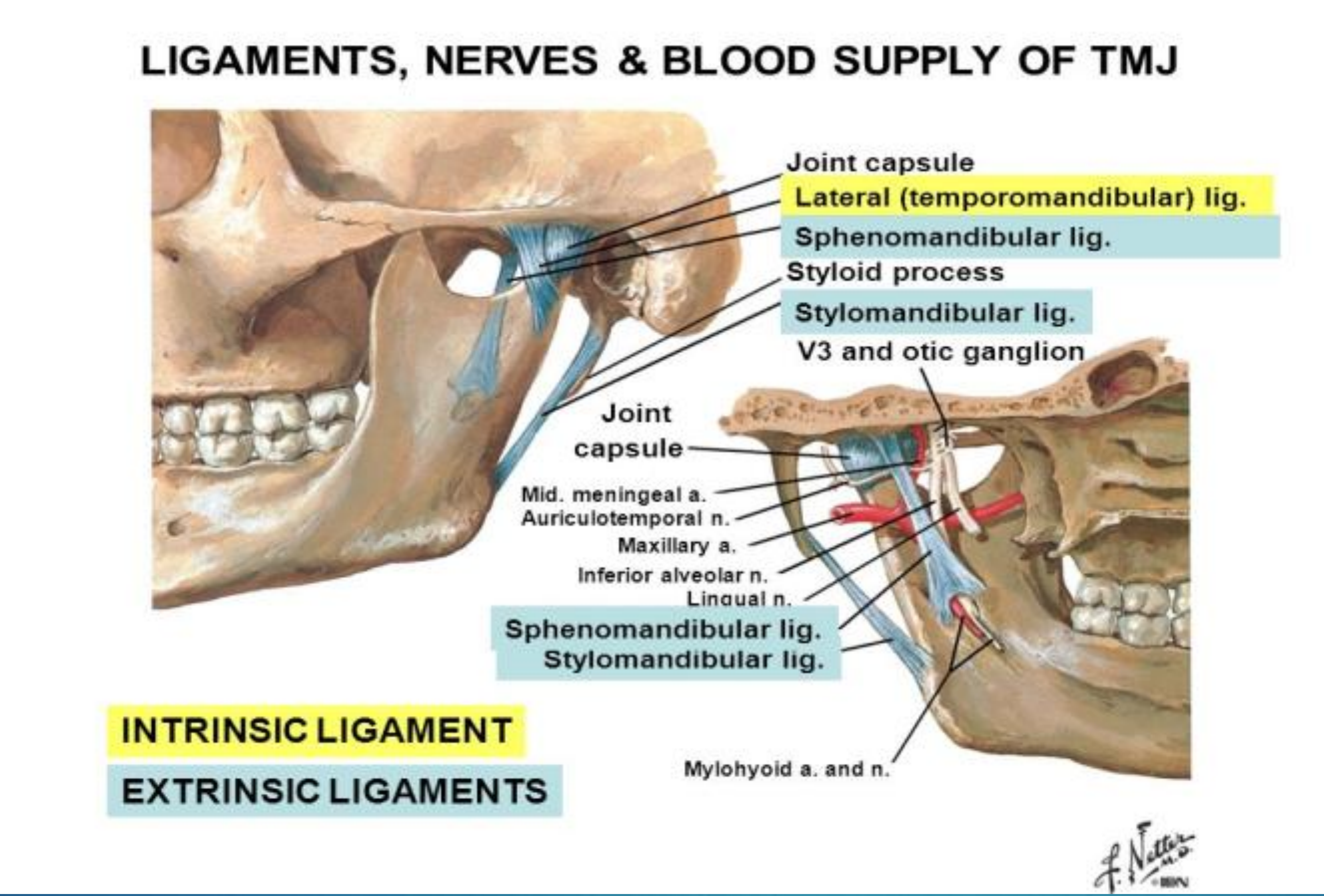
- Into the nasopharynx

Pharyngotympanic (eustachian) tube

can be impinged at any place during the passage

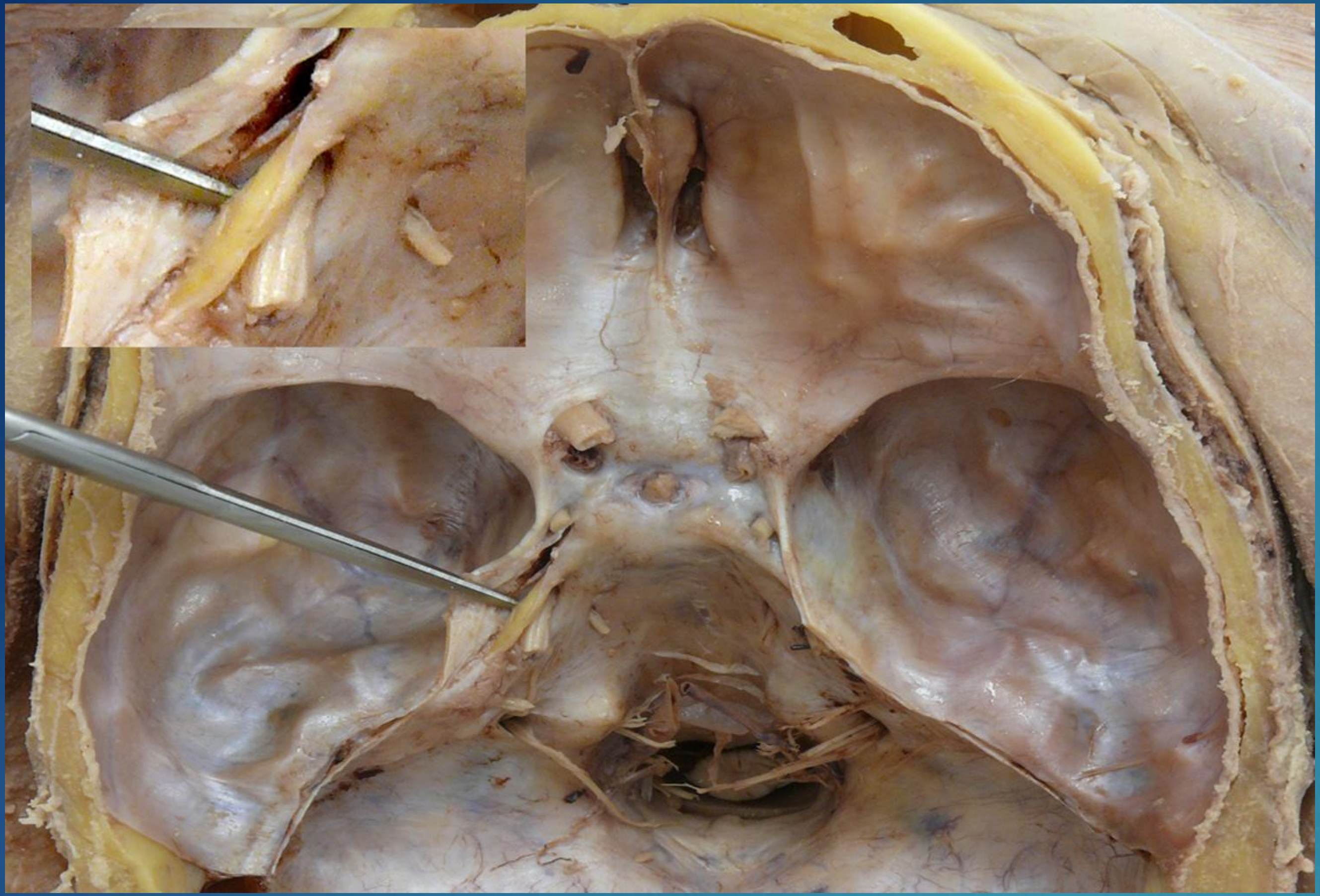






- Connects sphenoid to the lingula of the mandibular foramen
- Slackens when TMJ is closed, Taut when TMJ is open

Sphenomandibular Ligament



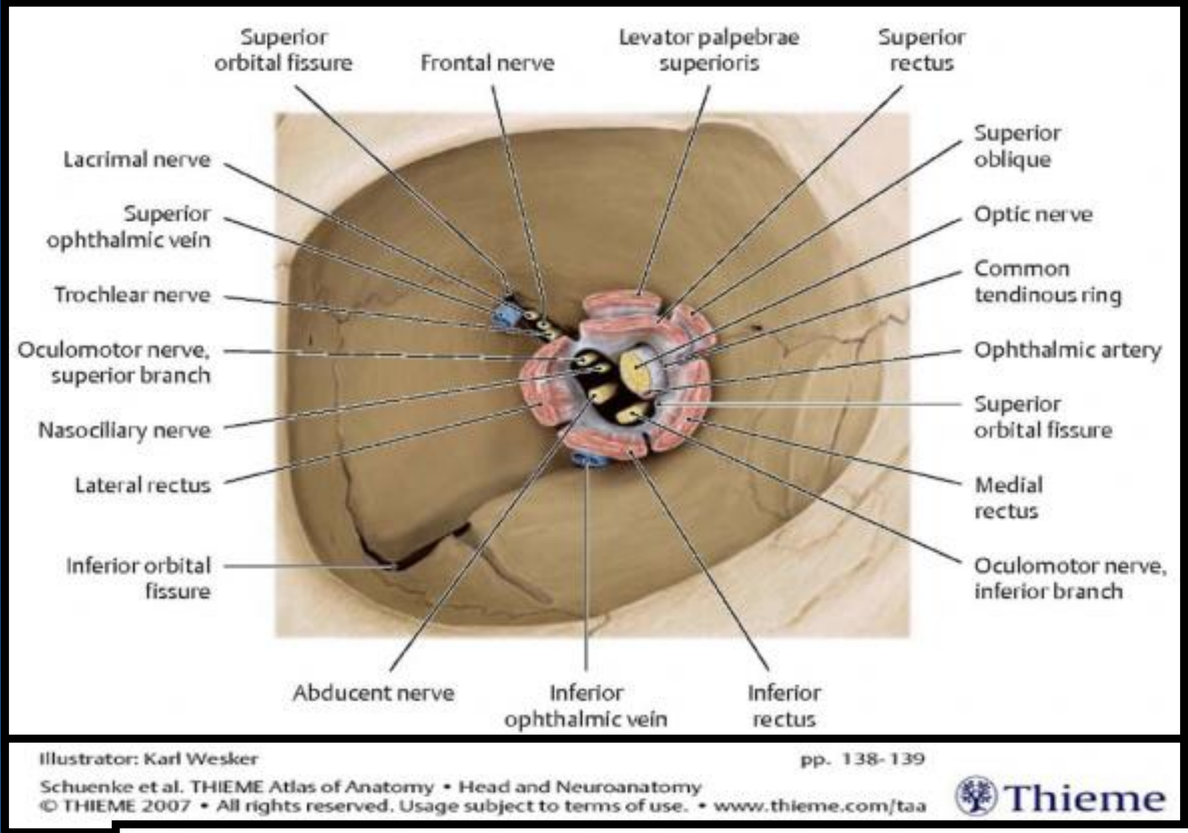
- From the petrous ridge of the temporal bone to the posterior clinoid process
- The Abducens Nerve Travels underneath it
- Tension here could effect contents of the cavernous sinus

Petrosphenoidal Ligament

(Gruber's Ligament)

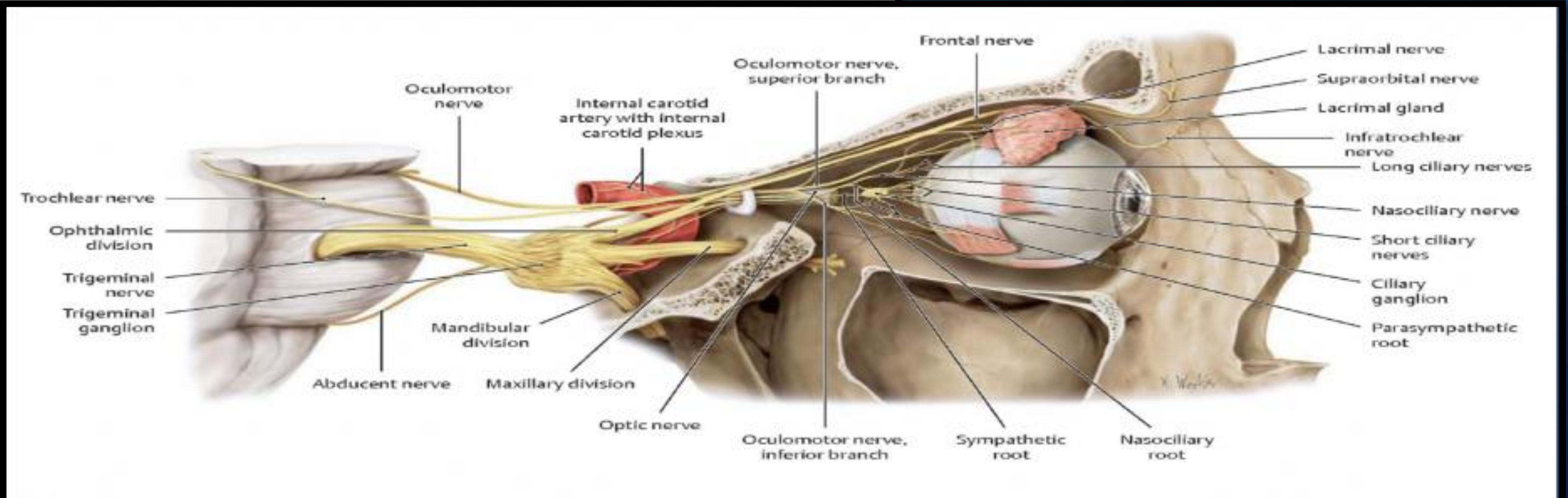
under probe

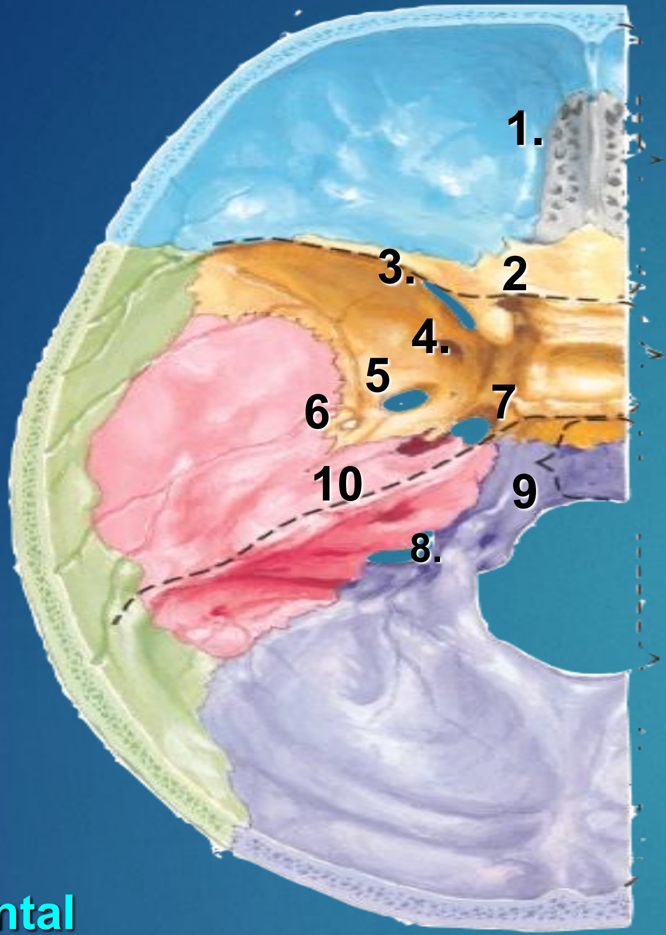
- O TOM CAT
- O: oculomotor
- T: trochlear
- O: ophthalmic branch of V
- M: maxillary branch of V
- C: Internal Carotid Artery
- A: Abducent nerve
- T: Trochlear Nerve



- Common Tendinous Ring (annulus of Zinn)
- Point of attachment of the four rectus muscles
- Opthalmic artery
- Sheath of CN II
- CN III, VI & V1 (nasociliary nerve)
  - Moore pg 902
- Crosses the optic canal and the superior orbital fissure and then attaches to the greater wing of the sphenoid
  - "It will be influenced by the mechanics of the lesser wing, greater wing and body of the sphenoid"

alteration in sphenoid can affect eye





1. cribriform plate
2. optic canal
3. superior orbital fissure
4. foramen rotundum
5. foramen ovale
6. foramen spinosum
7. foramen lacerum
8. jugular foramen
9. condylar canal
10. internal auditory meatus

Frontal

Ethmoid

Sphenoid

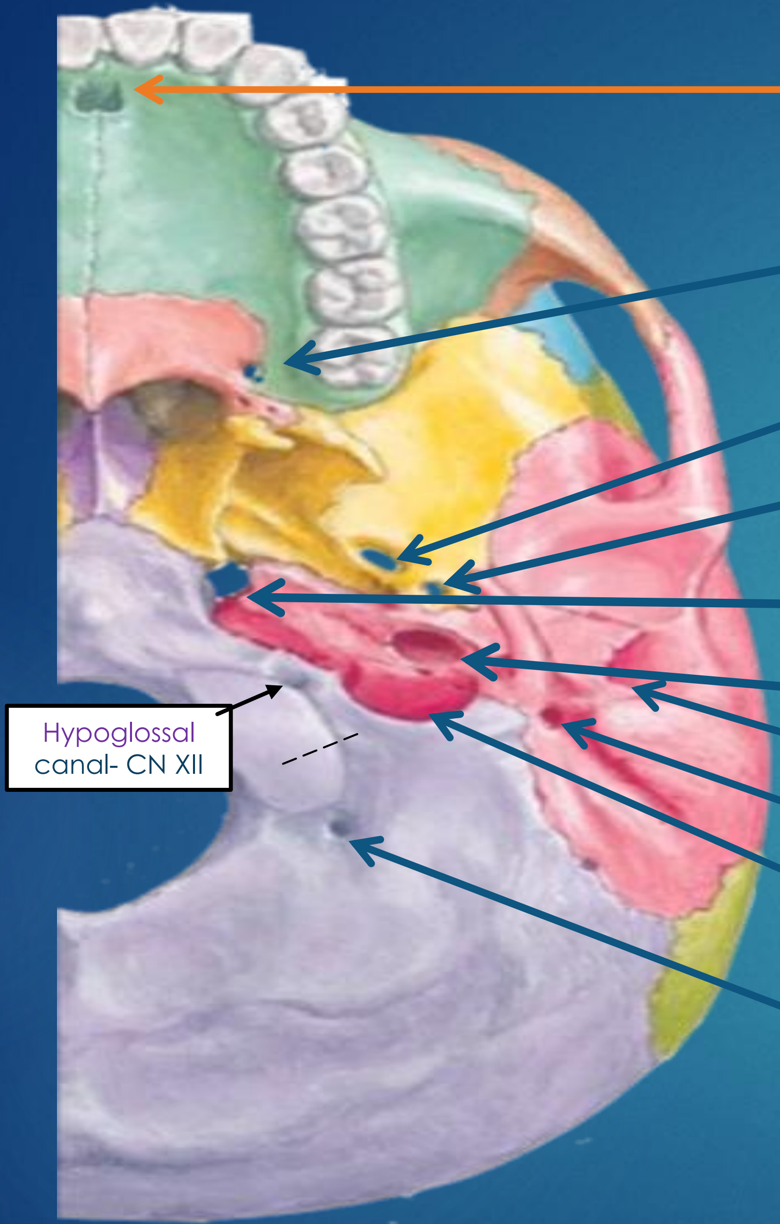
Temporal

Parietal

Occipital

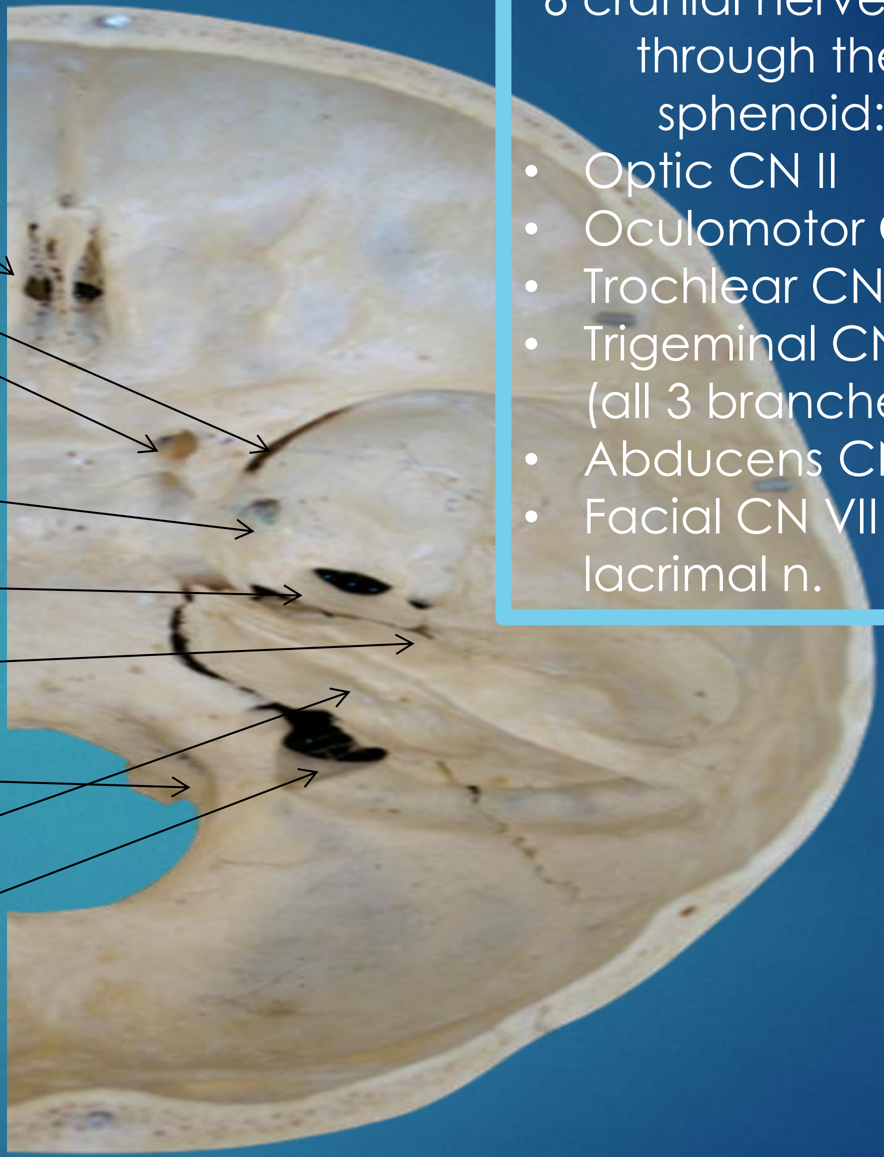
Foramina

CN IX, X, XI through the jugular foram



- Incisive Canal
  - Emissary vein
- Greater Palatine Foramen
  - CN V Maxillary- greater palatine n
- Foramen Ovale
  - CN V mandibular
- Foramen Spinosum
  - Middle Meningeal A.
- Foramen Lacerum
  - Symp fibers for deep petrosal nerve
- Carotid Canal
  - Internal Carotid Artery
- External Auditory Meatus
- Stylomastoid Foramen
  - CN VII motor branch
- Jugular Foramen
  - CN IX, X, XI
- Condylar Foramen
  - Emissary vein

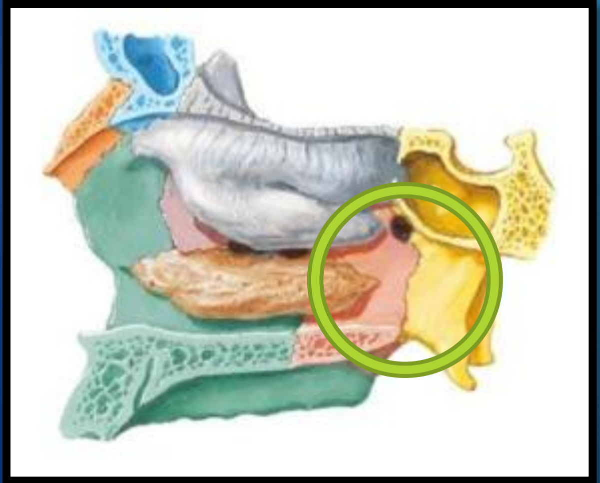
- Ethmoid Foramina
  - CNI- Olfactory
- Optic Canal
  - CNII- Optic
- Superior Orbital Fissure
  - CN III Opthalmic N.
  - CN IV Trochlear N.
  - CN VI Abducens N.
  - CN V: opthalmic N.
- Foramen Rotundum
  - CN V: maxillary division N.
- Foramen Ovale
  - CN V: mandibular division N.
- Foramen Spinosum
  - Middle meningeal artery
- Hypoglossal canal
  - CN XII: Hypoglossal N.
- Internal Auditory Meatus
  - VII: Facial N.
  - CN VIII: Vestibulocochlear
- Jugular Foramen
  - CN IX: Glossopharngeal N.
  - CN X: Vagus N.
  - CN XI: Accessory N.

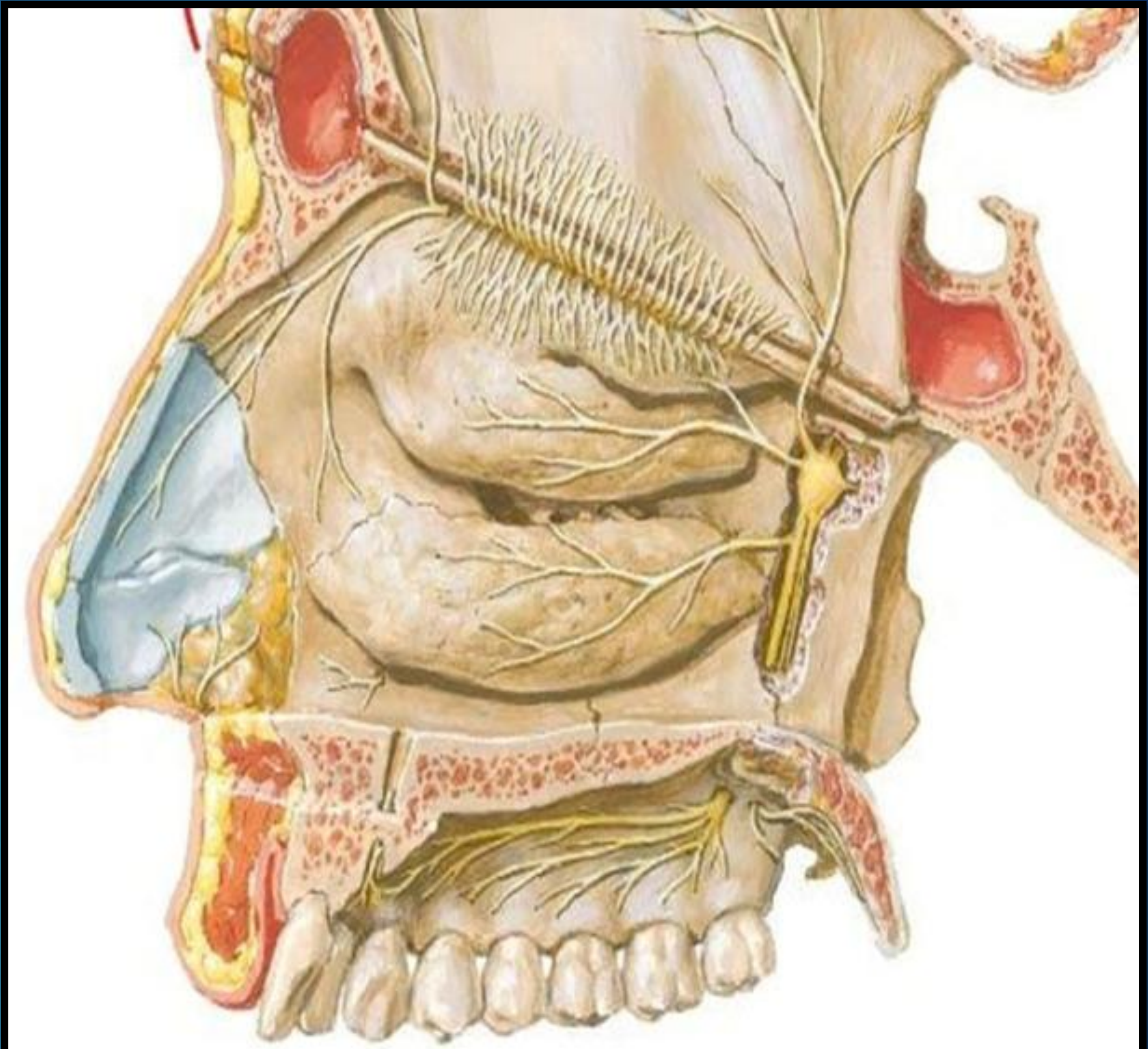


6 cranial nerves exit through the sphenoid:

- Optic CN II
- Oculomotor CN III
- Trochlear CN IV
- Trigeminal CN V (all 3 branches)
- Abducens CN VI
- Facial CN VII via lacrimal n.

Cranial Nerves that Traverse the Foramina, Internal Cranial View





P-ANS ganglia

The Pterygopalatine Fossa

posterior to teeth and superior

Clinical Indications for Evaluating/Treating Sphenoid

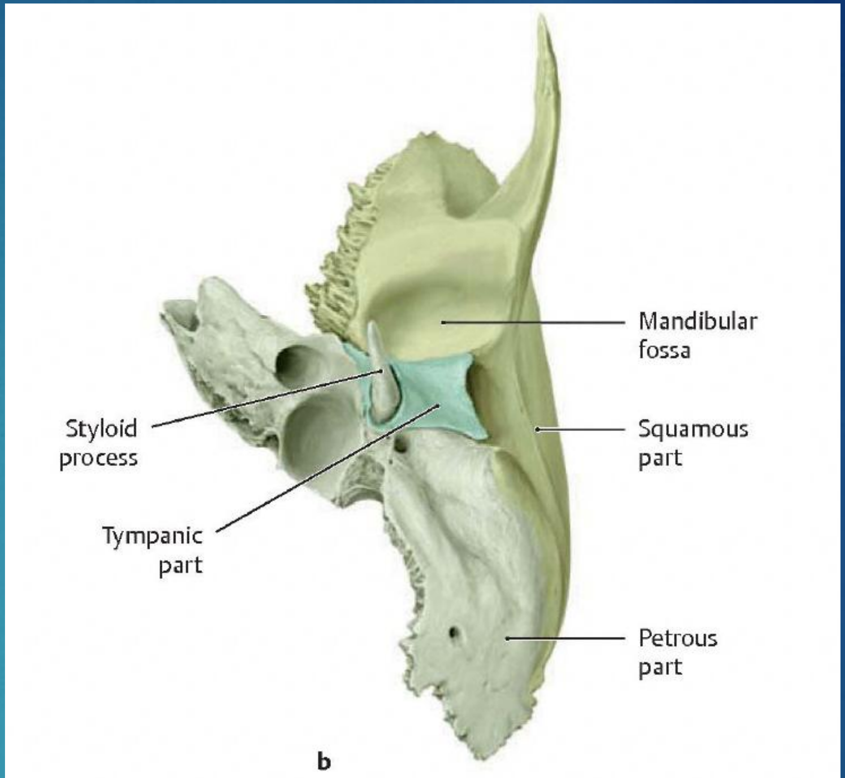
- Vision problems
- Strabismus
- Balance issues/vertigo
- Hearing problems
- URI/sinus congestion
- Sinus headache
- Bell's palsy

Temporal

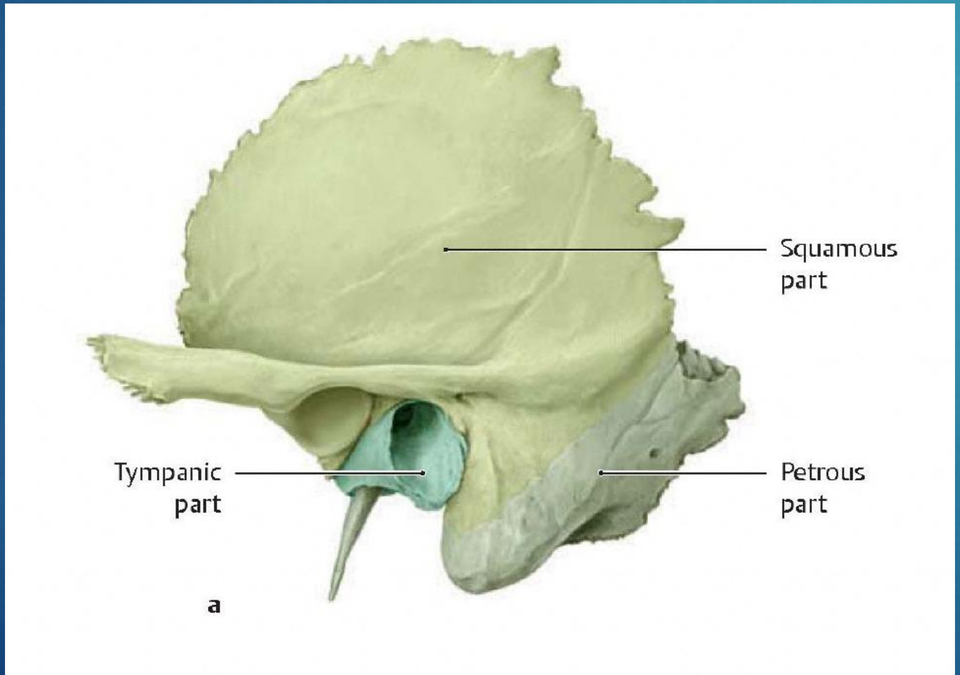
Origins

- Membranous origin
  - Squamous
  - Tympanic

Inferior View



Lateral View

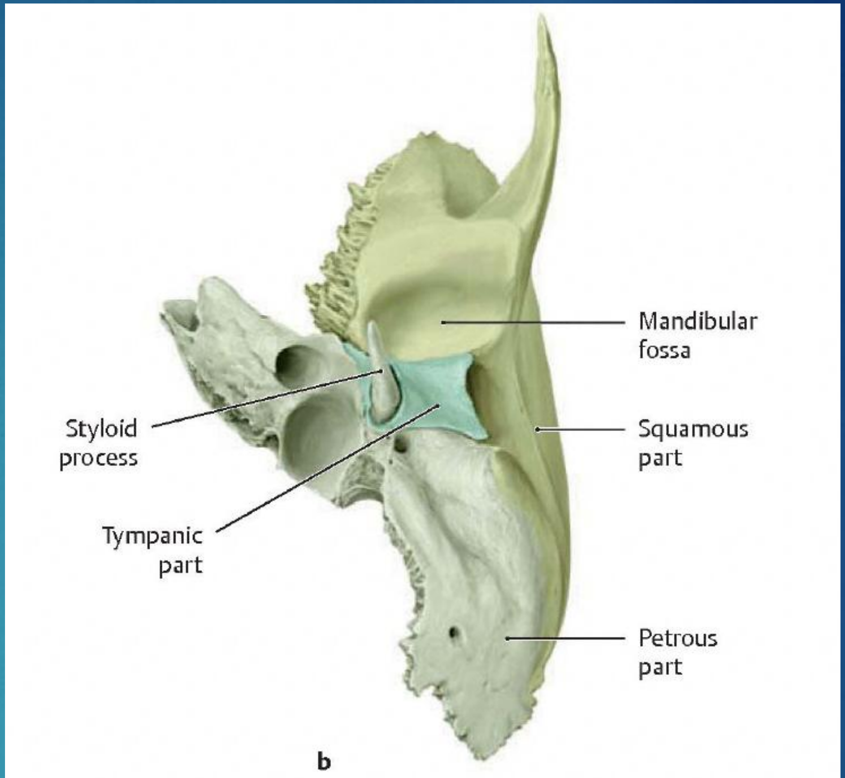


http://biologydictionary.net/temporal bone/ (accessed May 29, 2025).

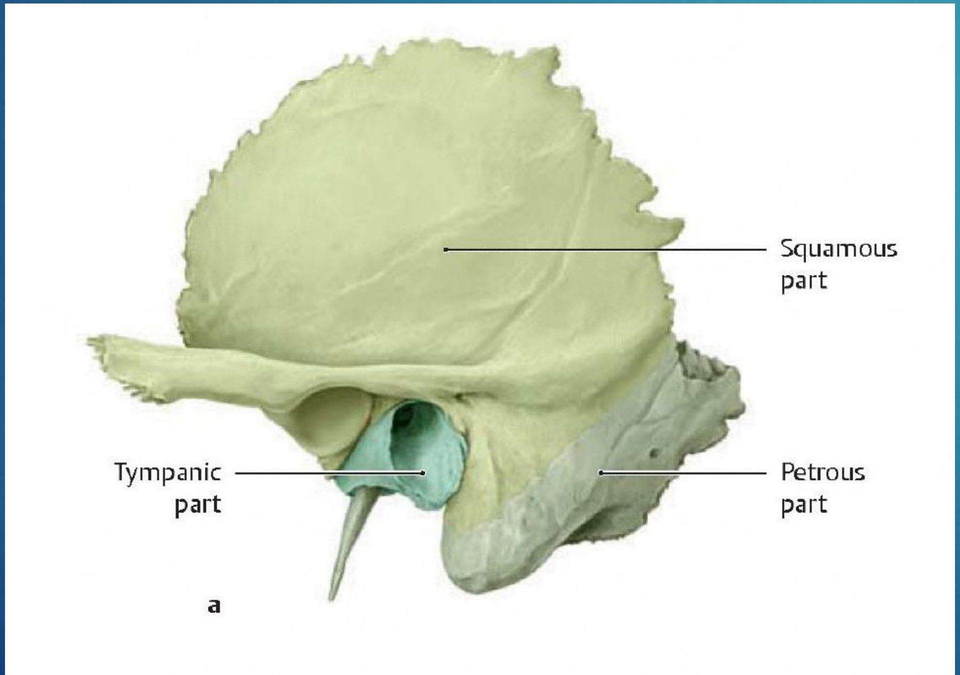
Origins

- Cartilaginous origin
  - Petrous
  - Mastoid
  - Styloid

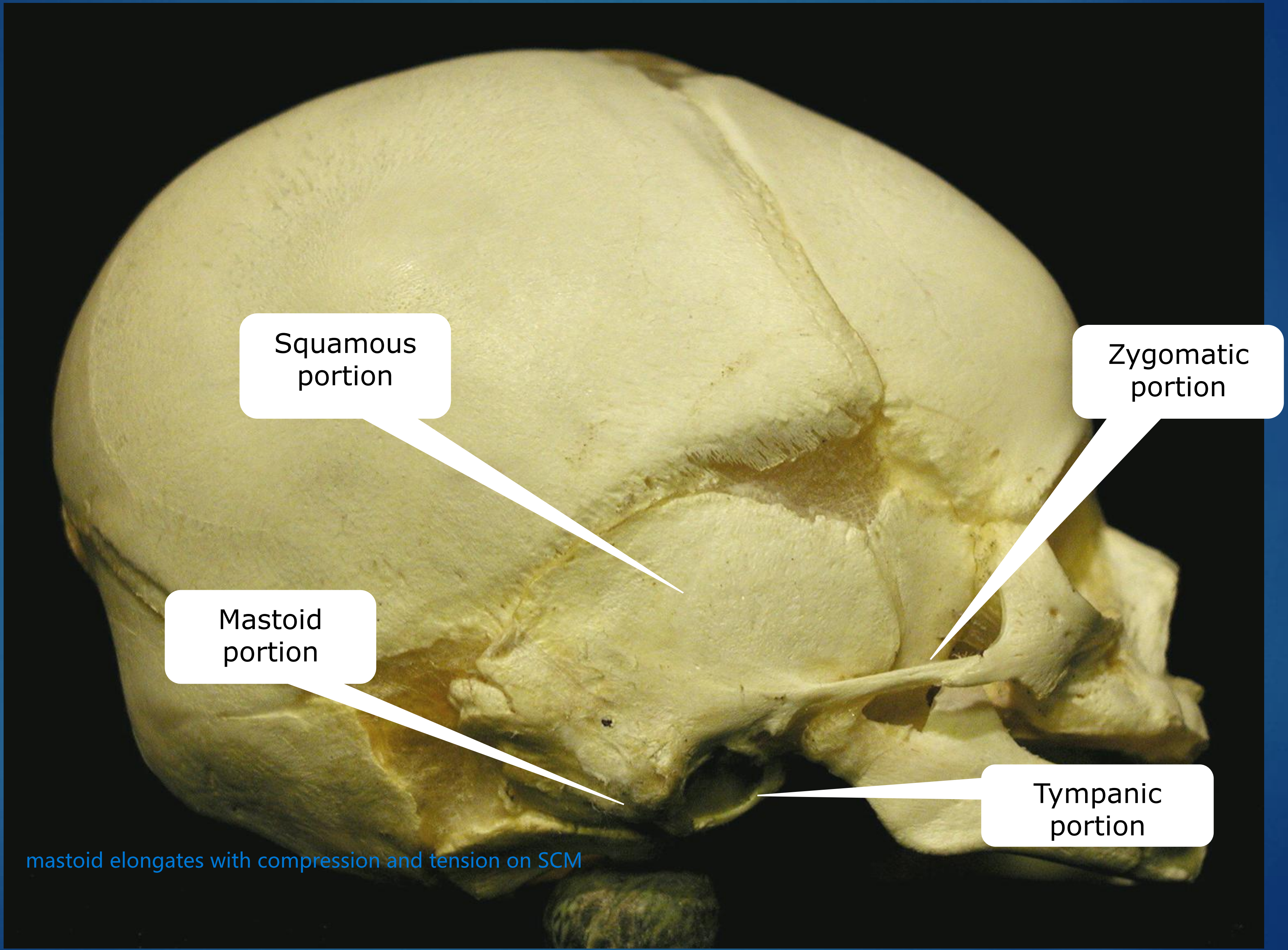
Inferior View



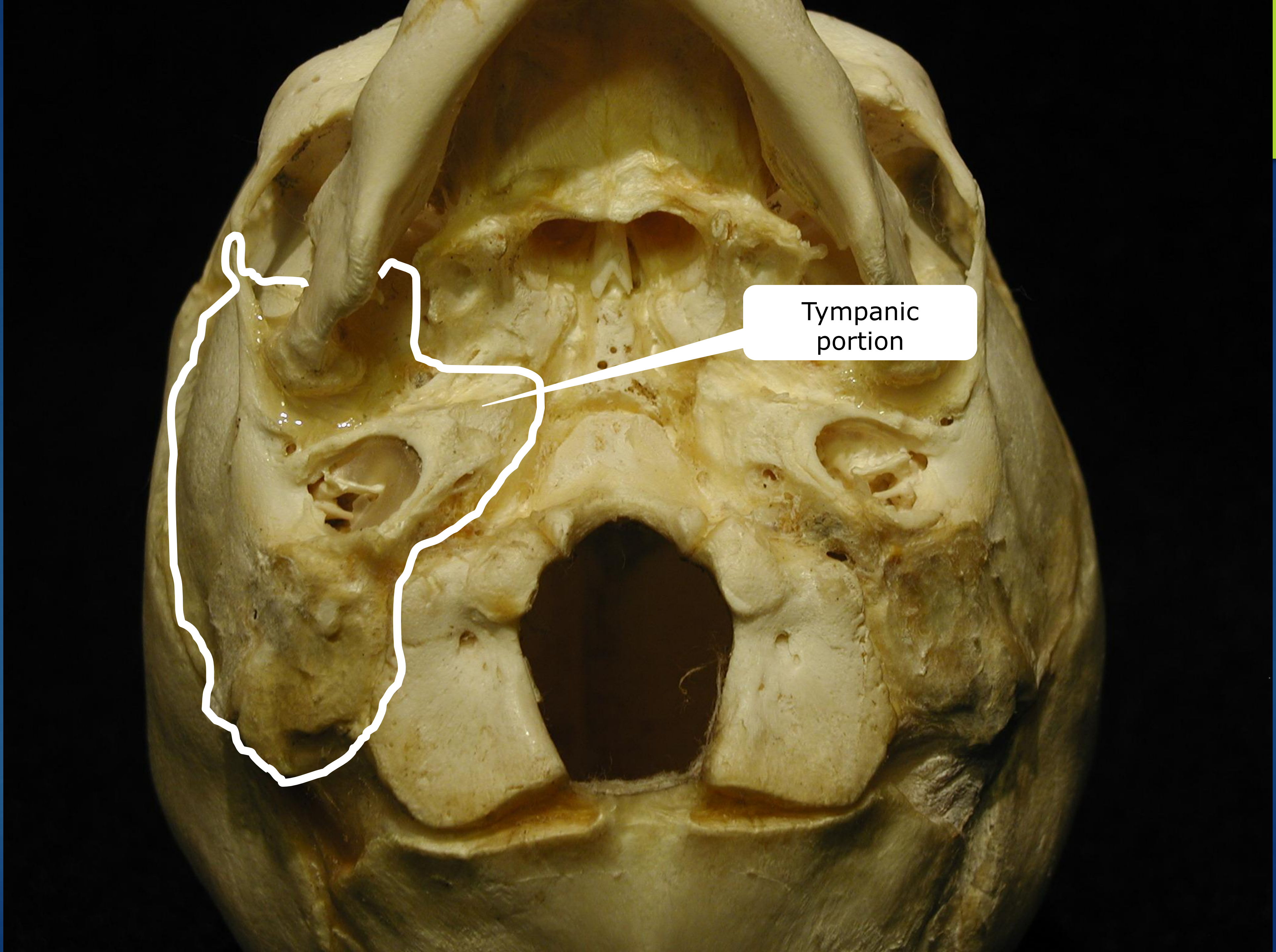
Lateral View



http://biologydictionary.net/temporal bone/ (accessed May 29, 2025).

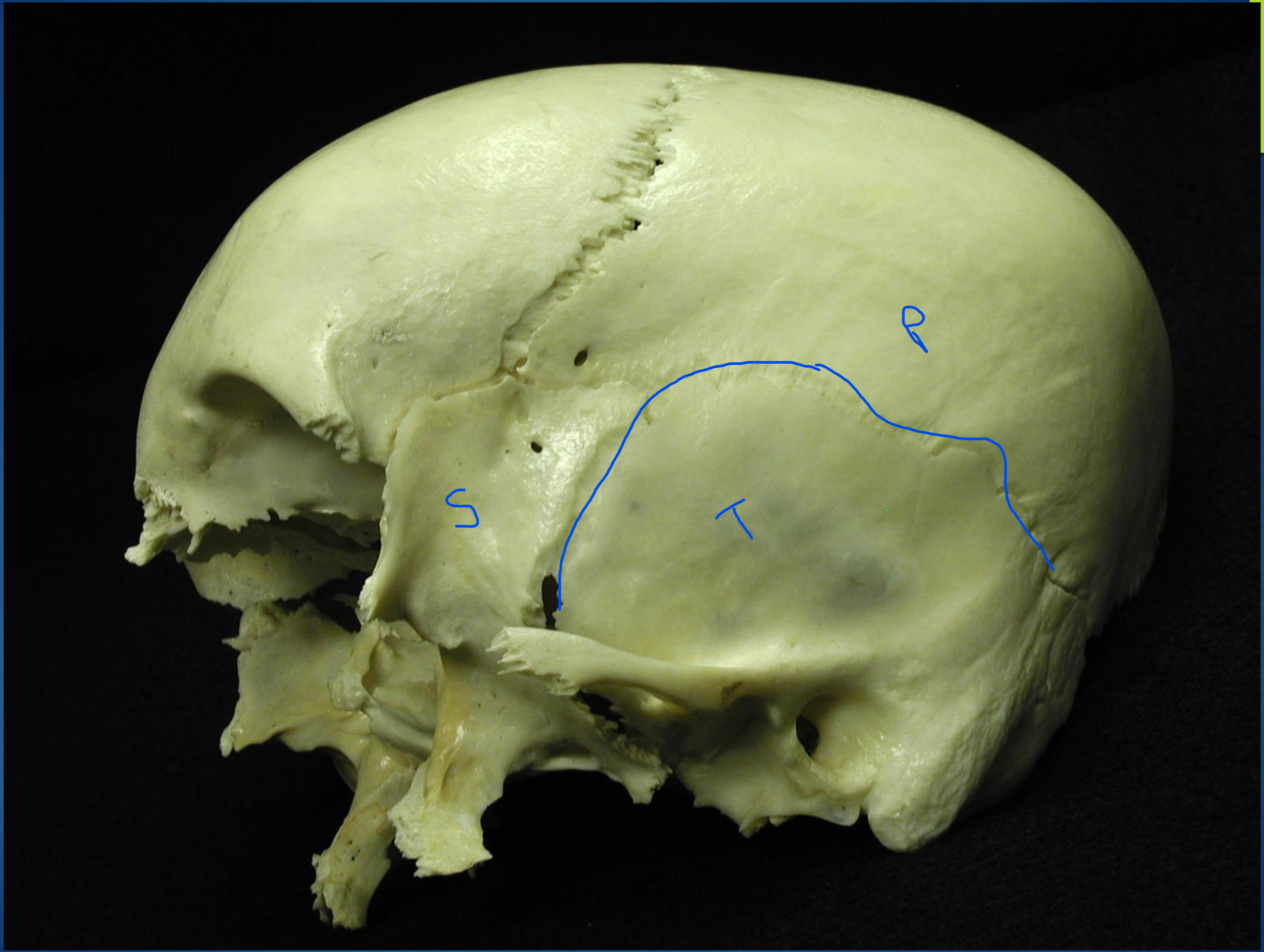


mastoid elongates with compression and tension on SCM


Tympanic
portion

Temporal Bone Articulations

- Occiput
- Parietal
- Sphenoid
- Zygoma
- Mandible



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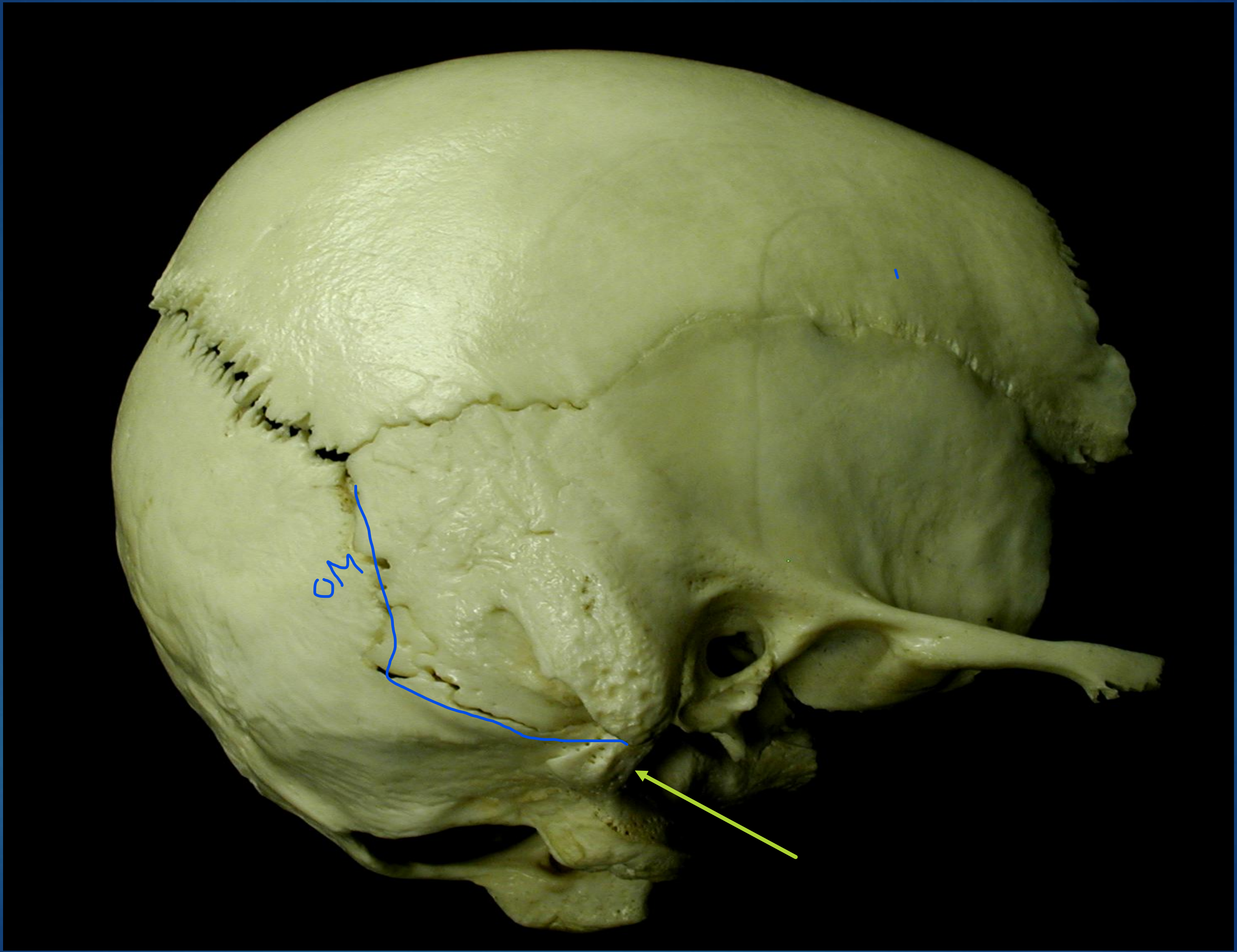
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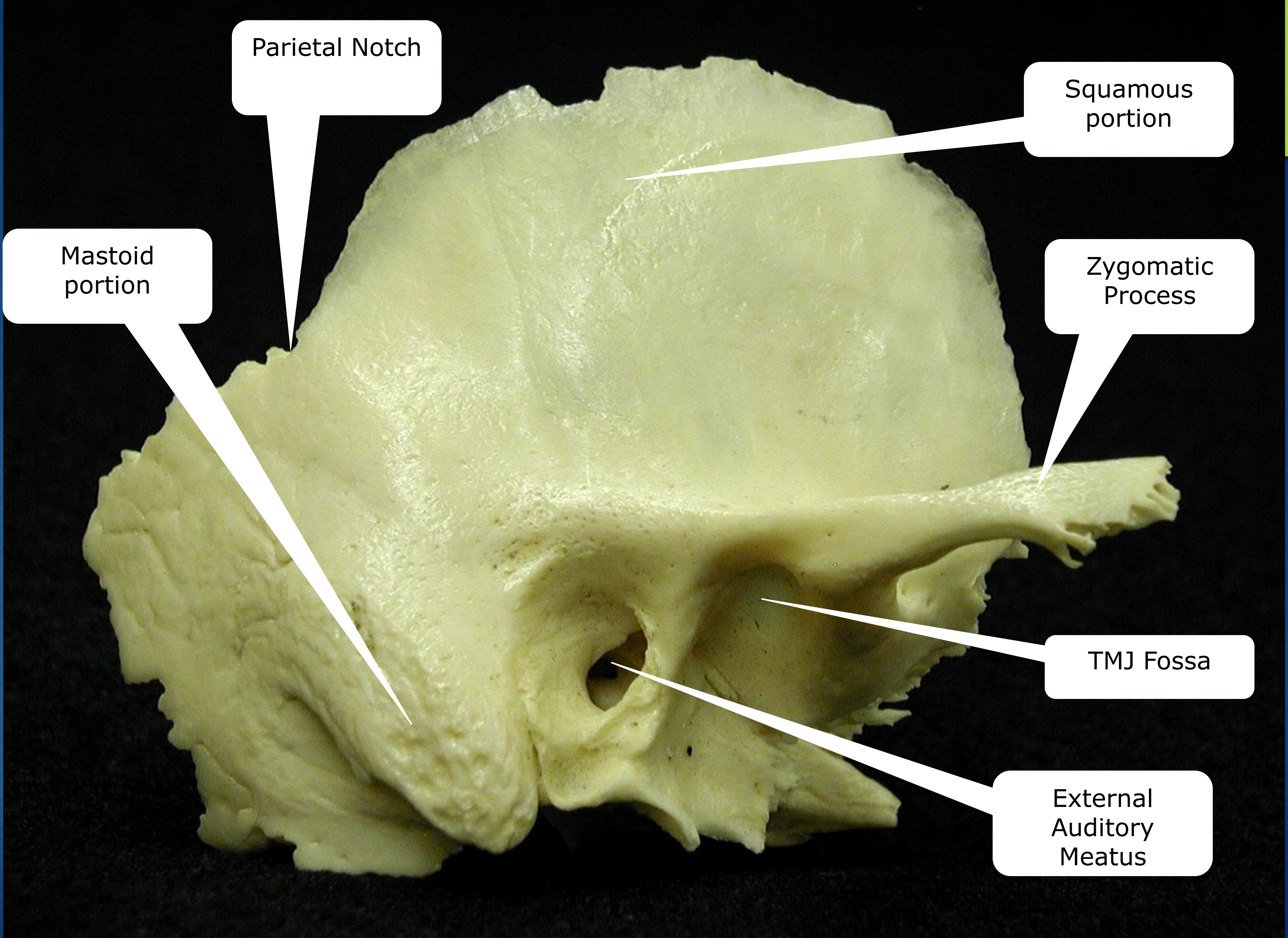
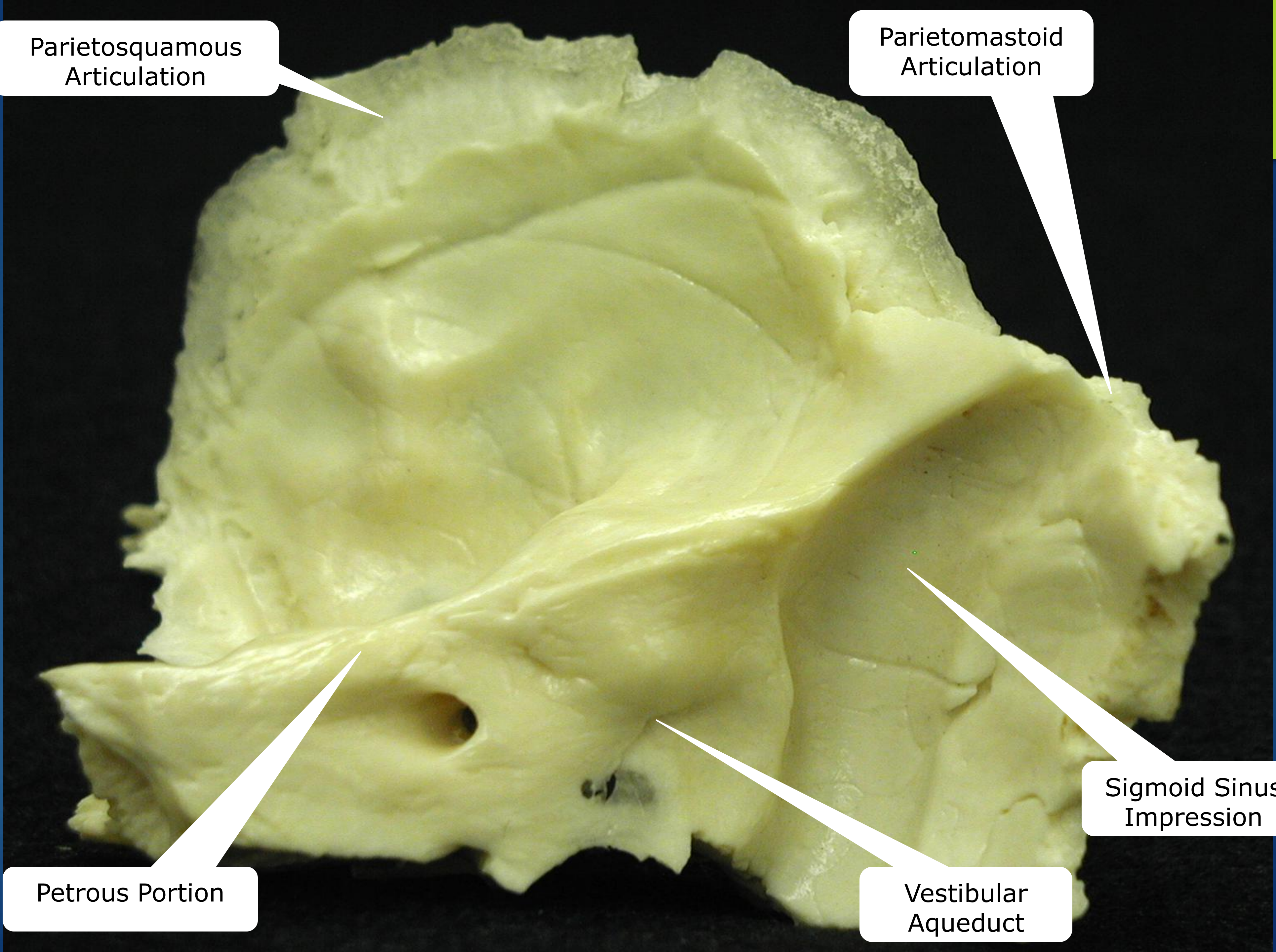
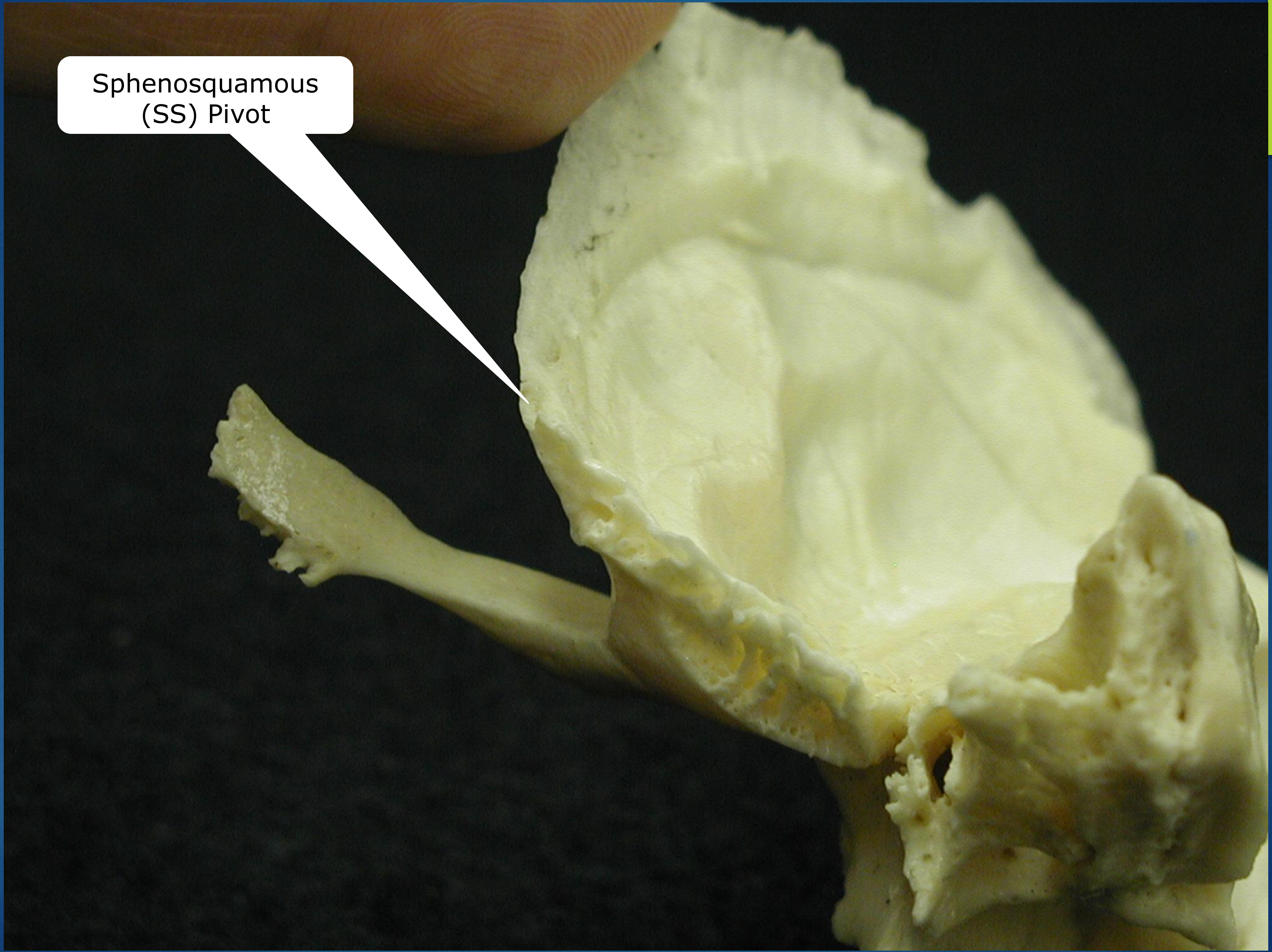
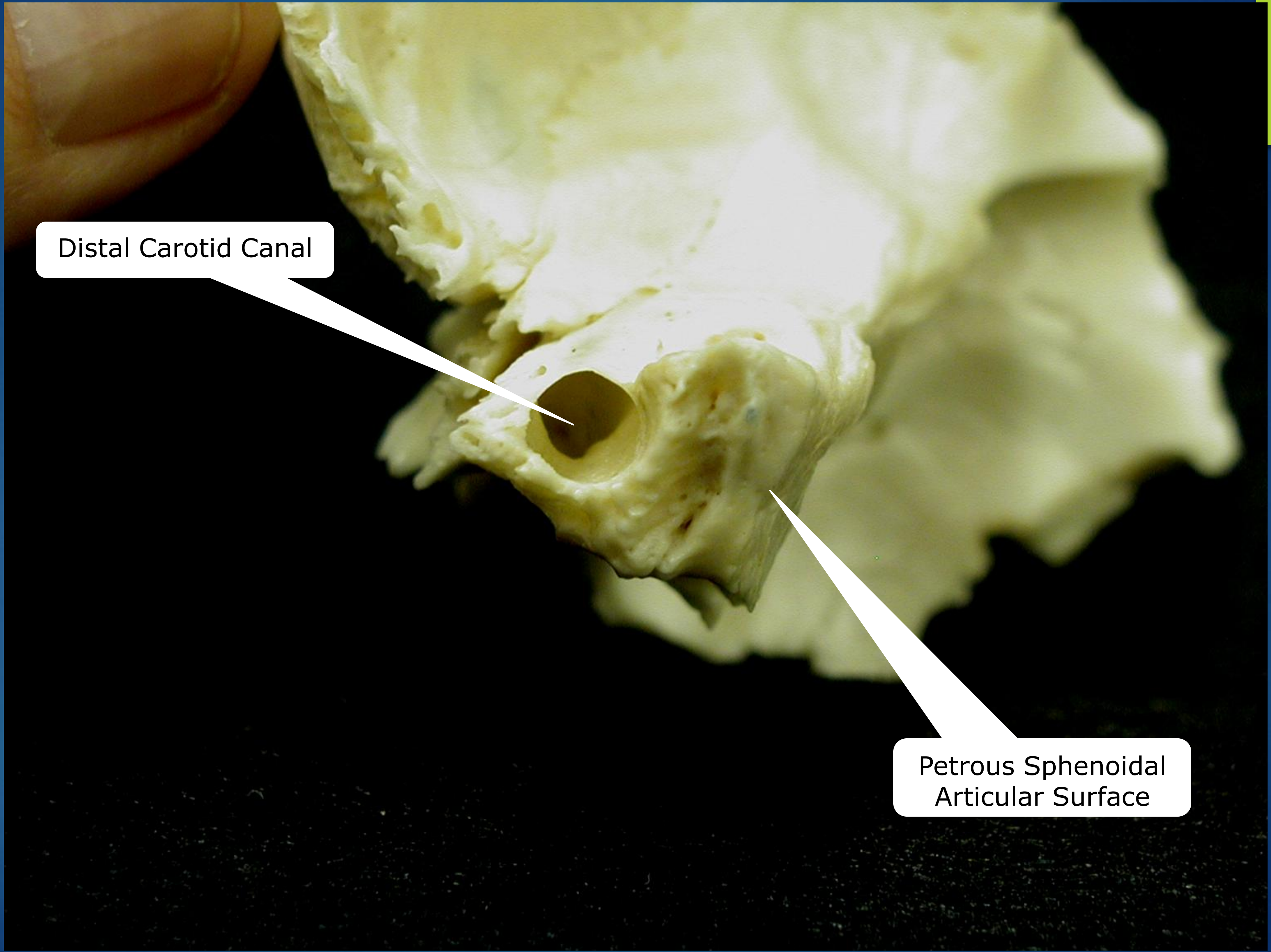
Adult Temporal Bone – Posterior, Medial & Anterior Features

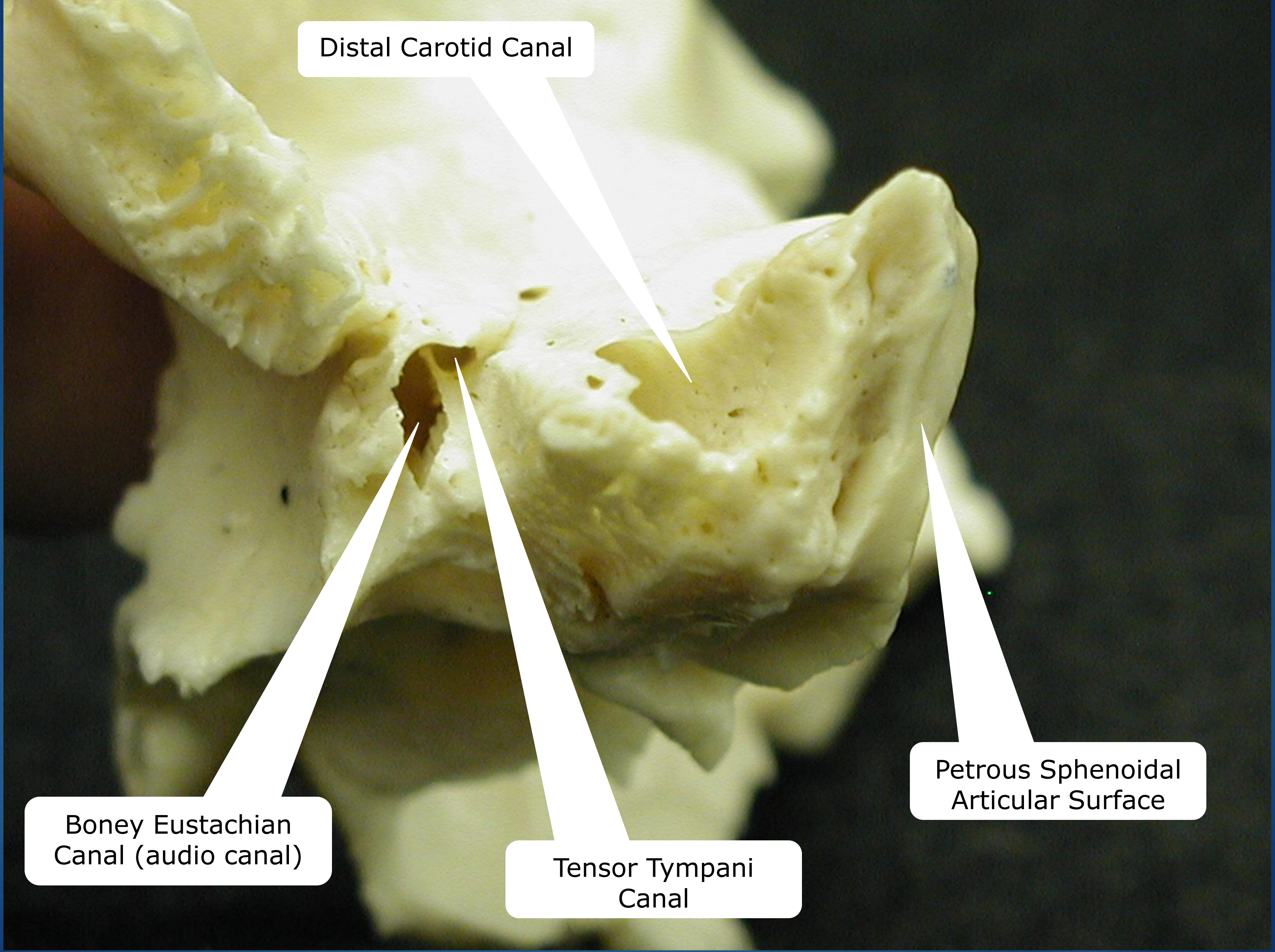
- Petrous Ridge
- Parieto-Squamous & Parieto-mastoid Articulations
- Sphenosquamous Pivot (SS Pivot)
- Hiatus Fallopii (Petrosal Canal)
- Boney Eustachian Tube
- Tensor Tympani Canal
- Trigeminal Impression (Meckels Cave)
- Carotid Canal
- Sigmoid Sinus Impression
- Superior Petrosal Sinus Impression
- Internal Auditory Meatus
- Vestibular Aqueduct

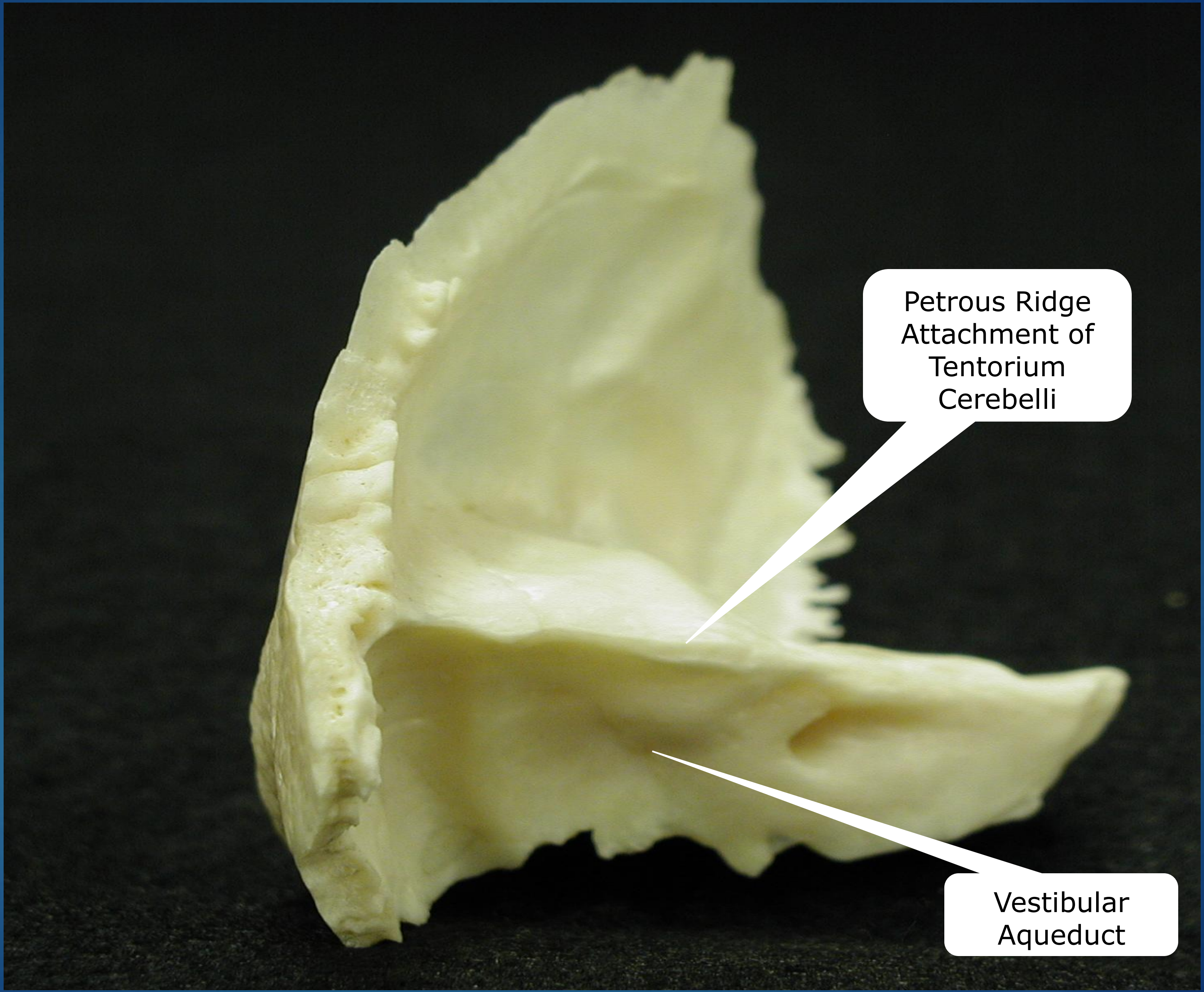

Parietal Notch
Squamous portion
Mastoid portion
Zygomatic Process
TMJ Fossa
External Auditory Meatus

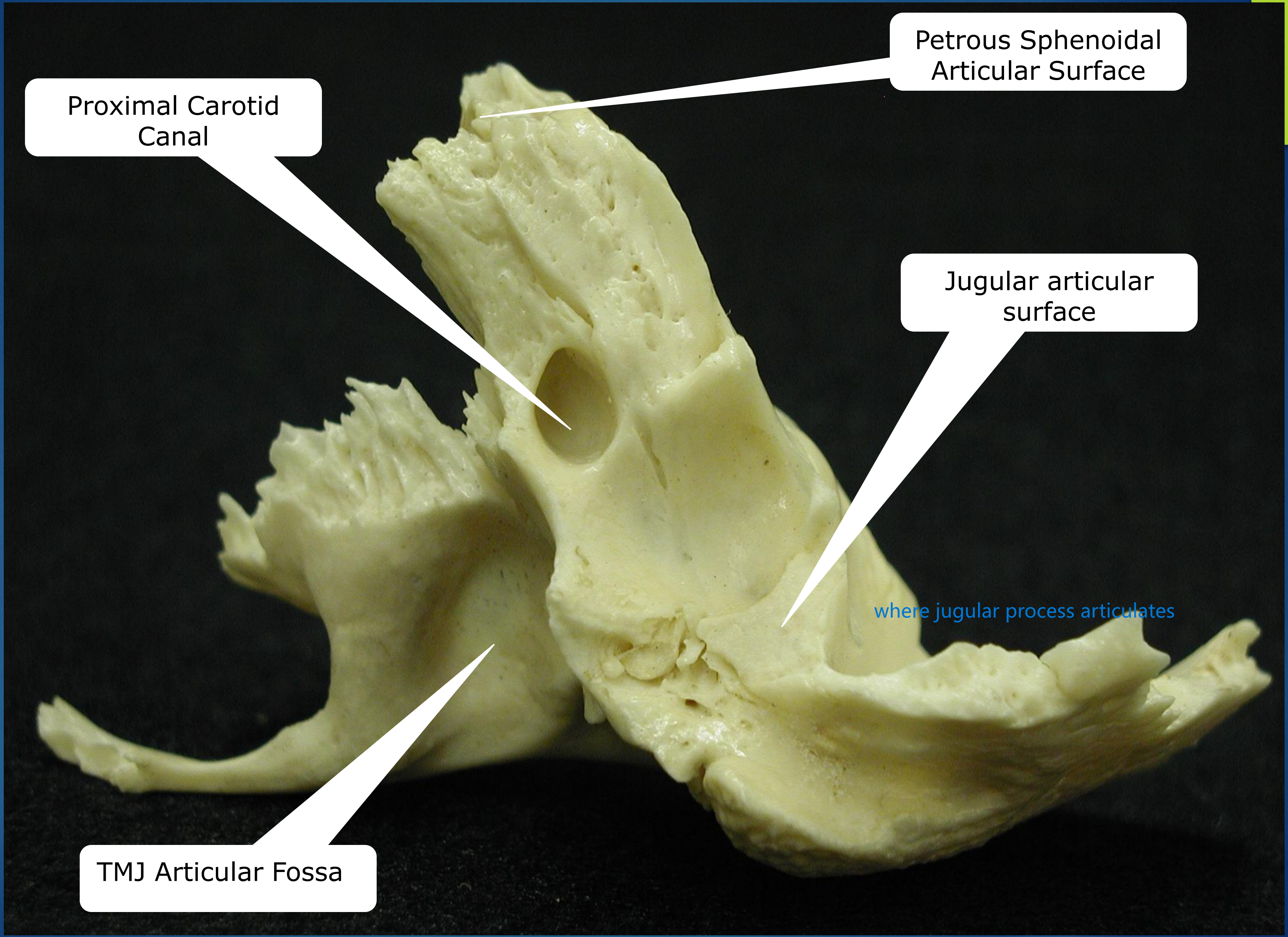

Parietosquamous Articulation
Parietomastoid Articulation
Sigmoid Sinus Impression
Petrous Portion
Vestibular Aqueduct

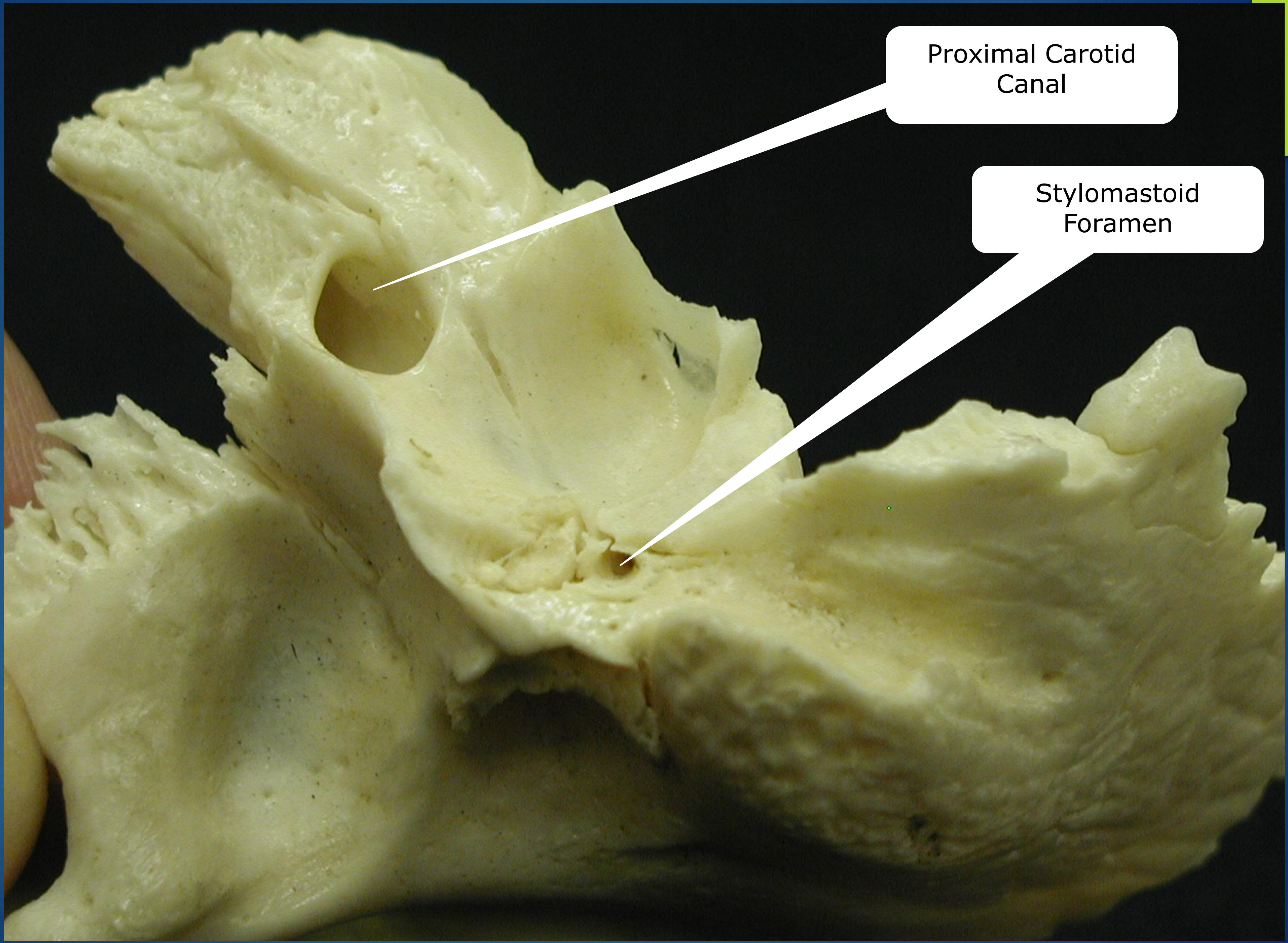

Sphenosquamous
(SS) Pivot

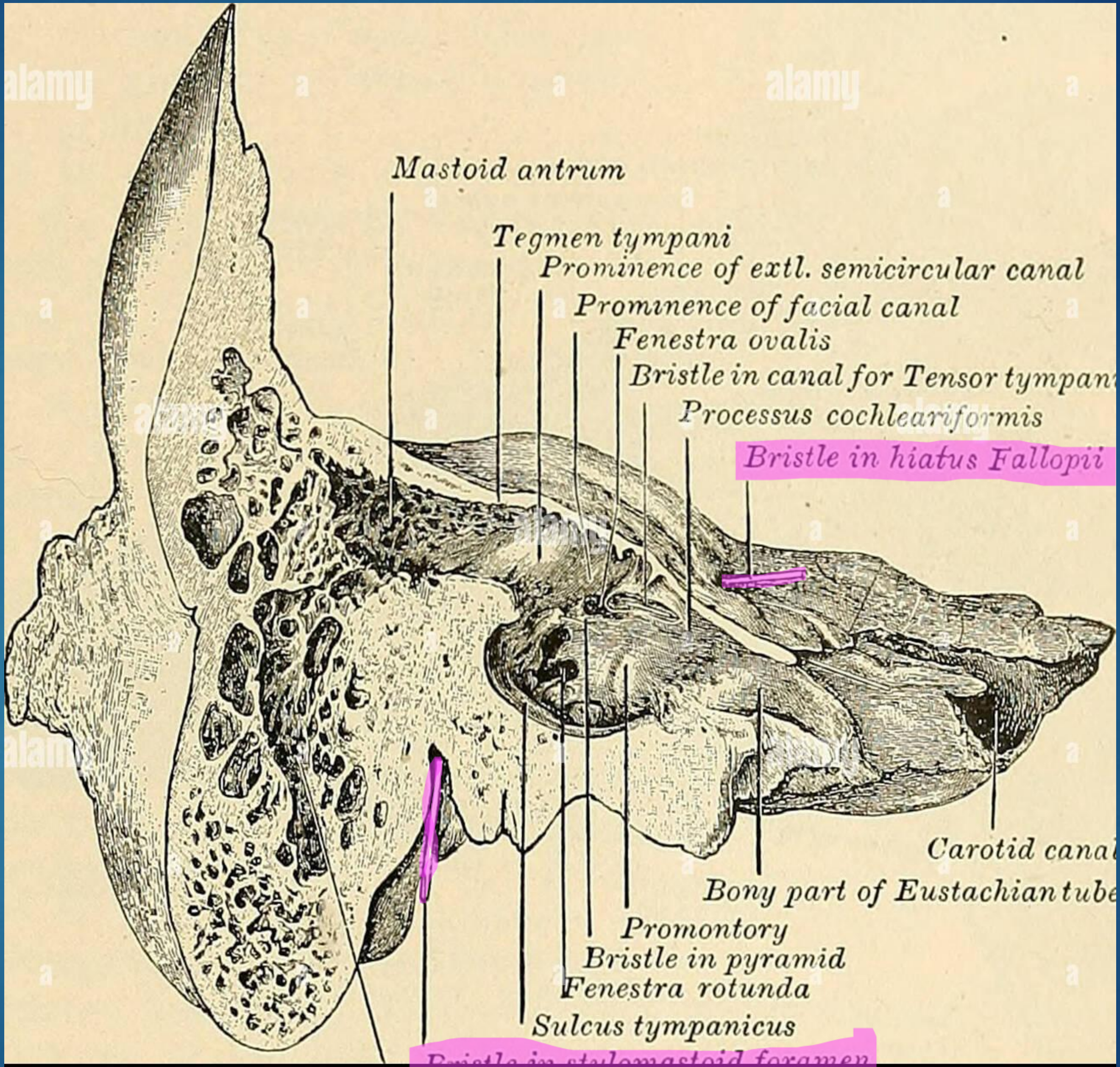

Distal Carotid Canal
Petrous Sphenoidal
Articular Surface


Distal Carotid Canal
Boney Eustachian Canal (audio canal)
Tensor Tympani Canal
Petrous Sphenoidal Articular Surface

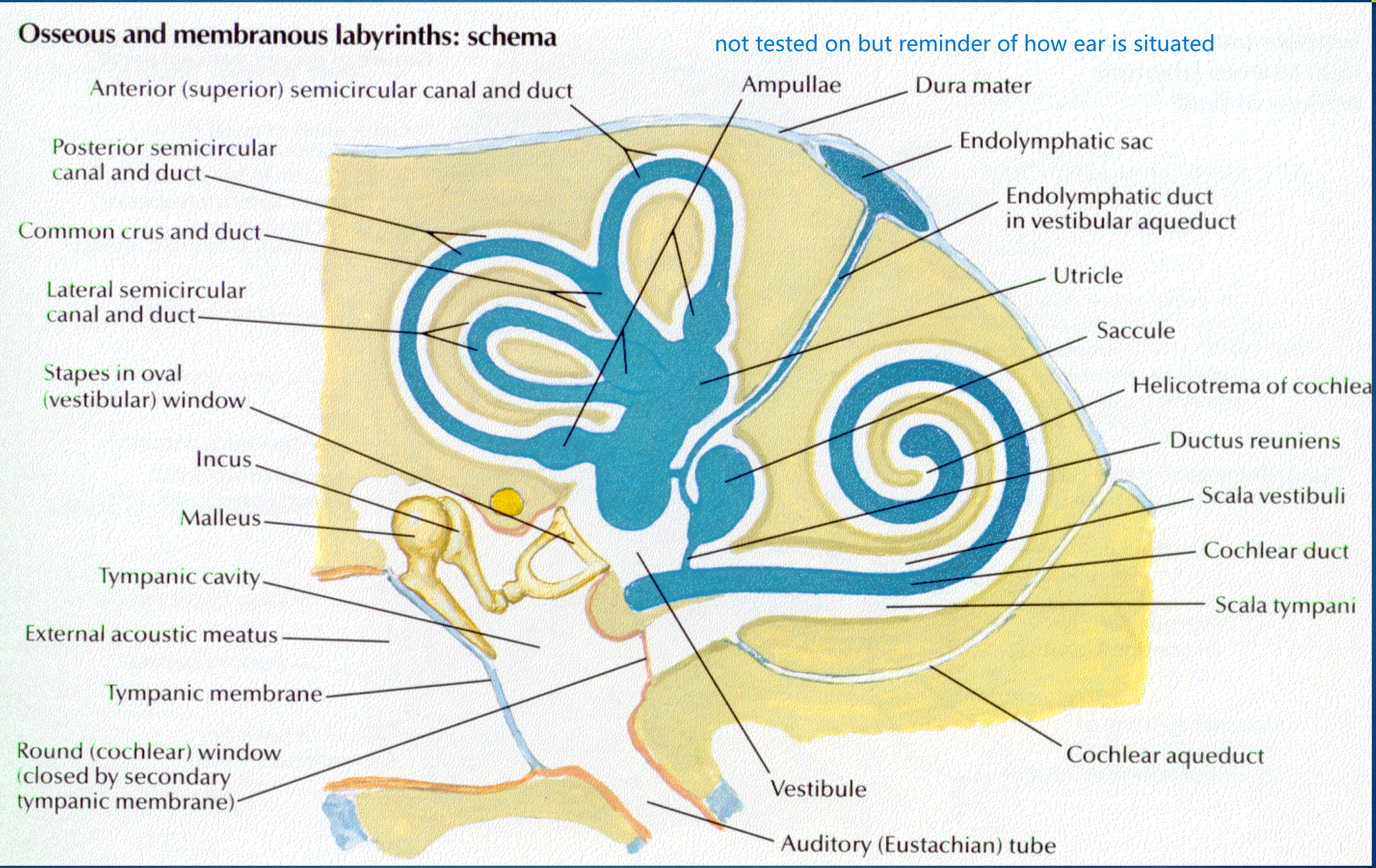

Petrous Ridge Attachment of Tentorium Cerebelli
Vestibular Aqueduct

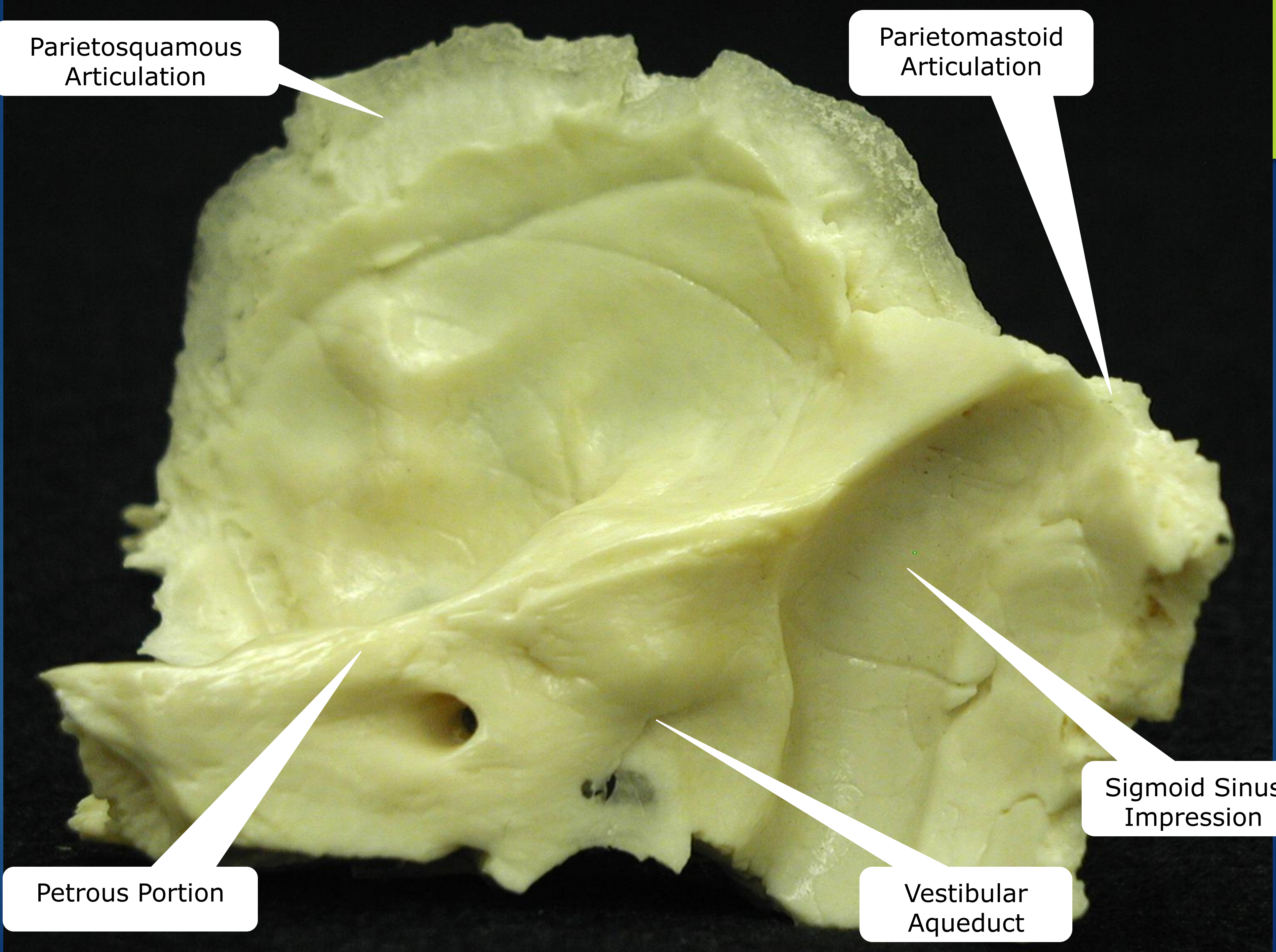

Petrous Sphenoidal Articular Surface
Proximal Carotid Canal
Jugular articular surface
where jugular process articulates
TMJ Articular Fossa


Proximal Carotid Canal
Stylomastoid Foramen

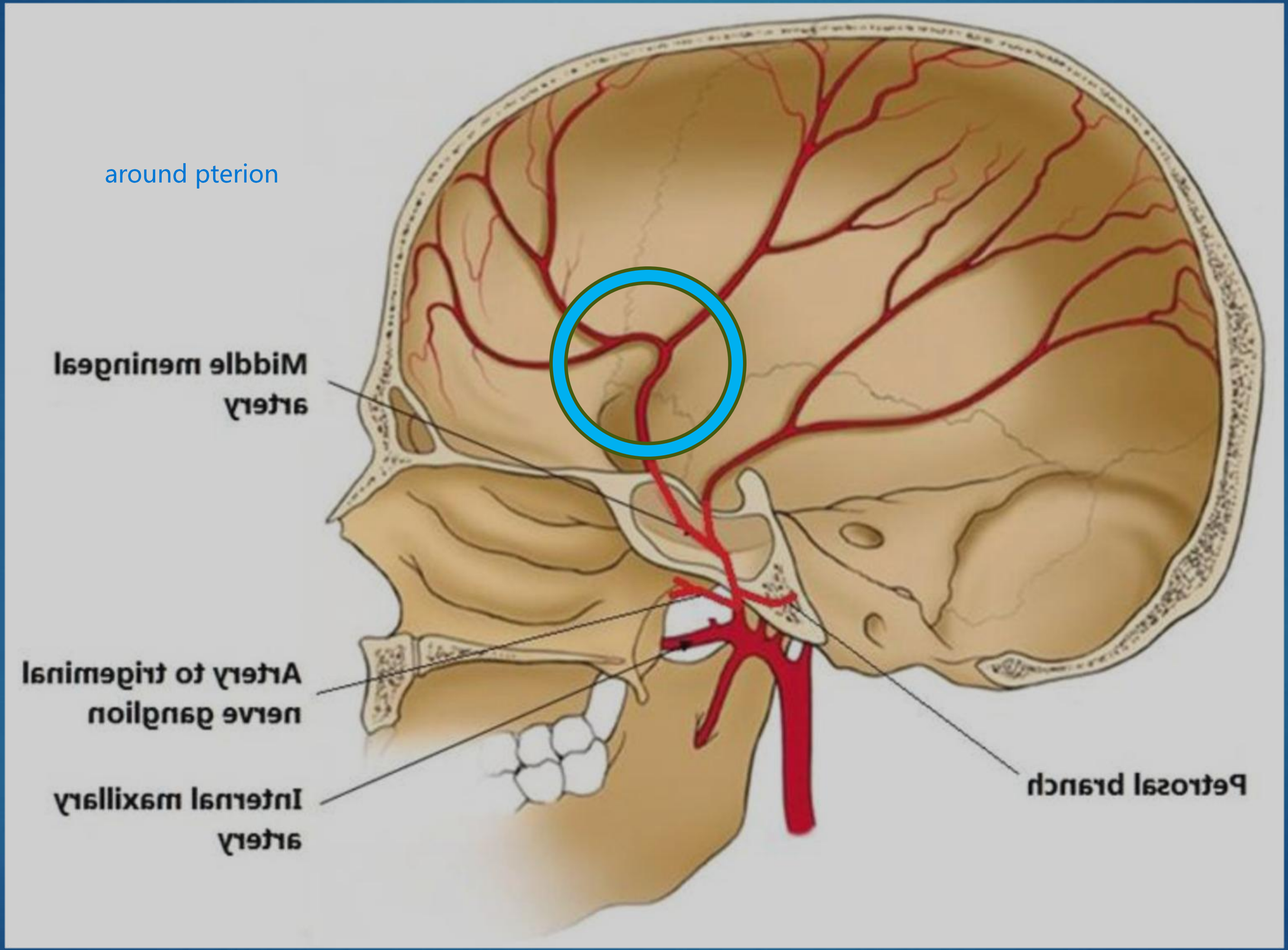

Mastoid antrum
Tegmen tympani
Prominence of extl. semicircular canal
Prominence of facial canal
Fenestra ovalis
Bristle in canal for Tensor tympani
Processus cochleariformis
Bristle in hiatus Fallopii
Carotid canal
Bony part of Eustachian tube
Promontory
Bristle in pyramid
Fenestra rotunda
Sulcus tympanicus
Bristle in stylomastoid foramen

Cross section of Petrous Portion of the Temporal Bone

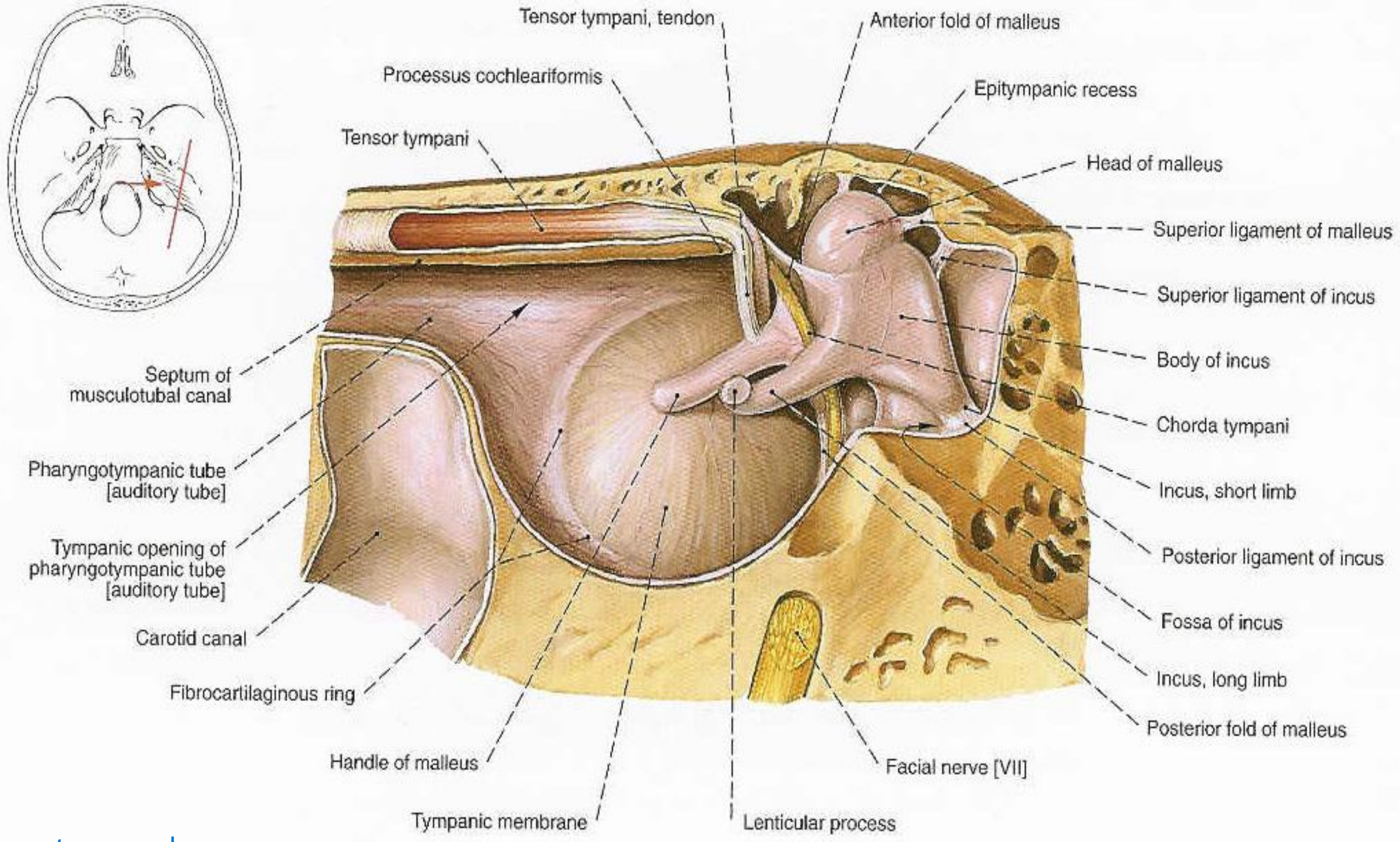



Parietosquamous Articulation
Parietomastoid Articulation
Sigmoid Sinus Impression
Petrous Portion
Vestibular Aqueduct

Middle Meningeal Artery



Tensor Tympani Articular Relationship



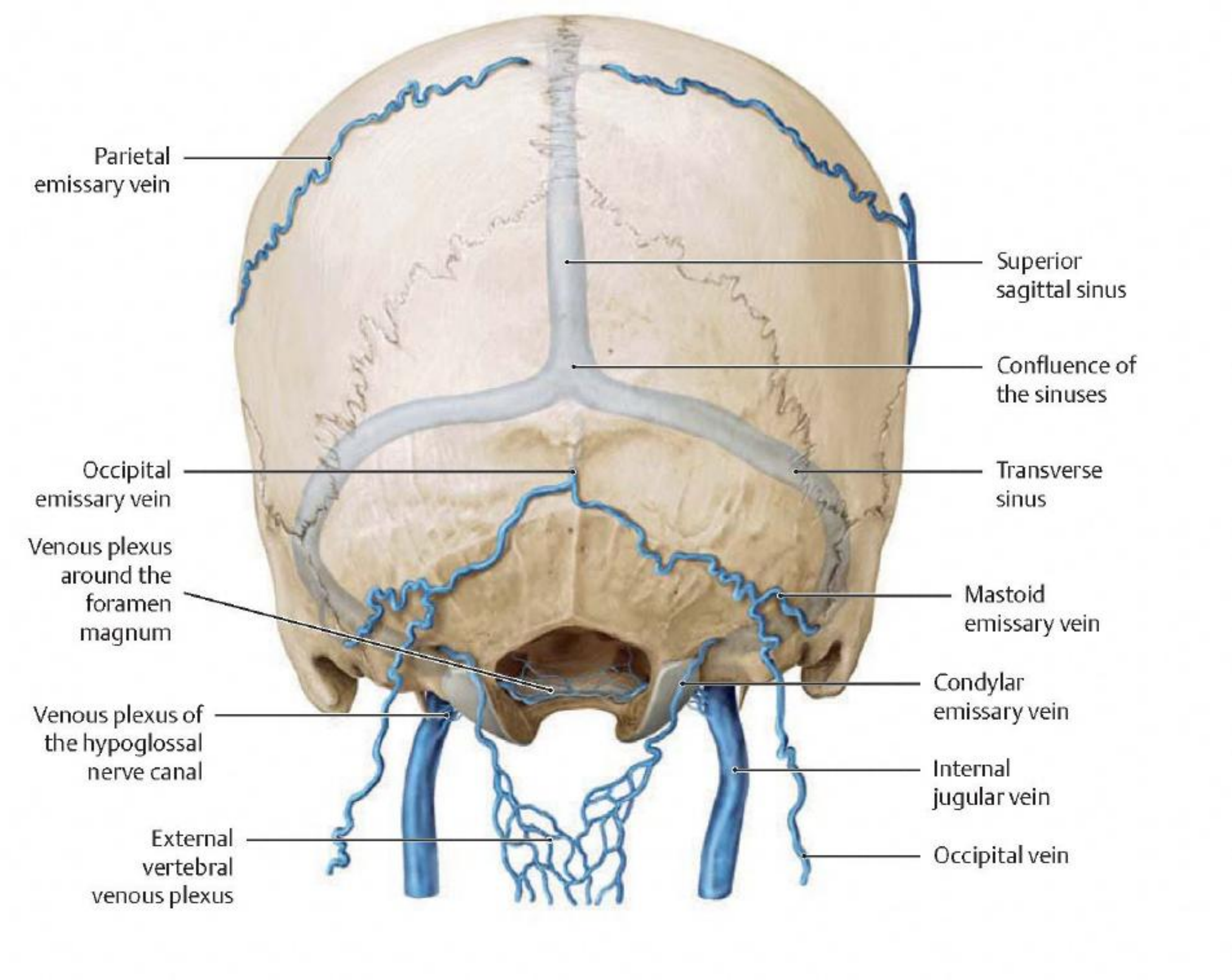
attenuates sound

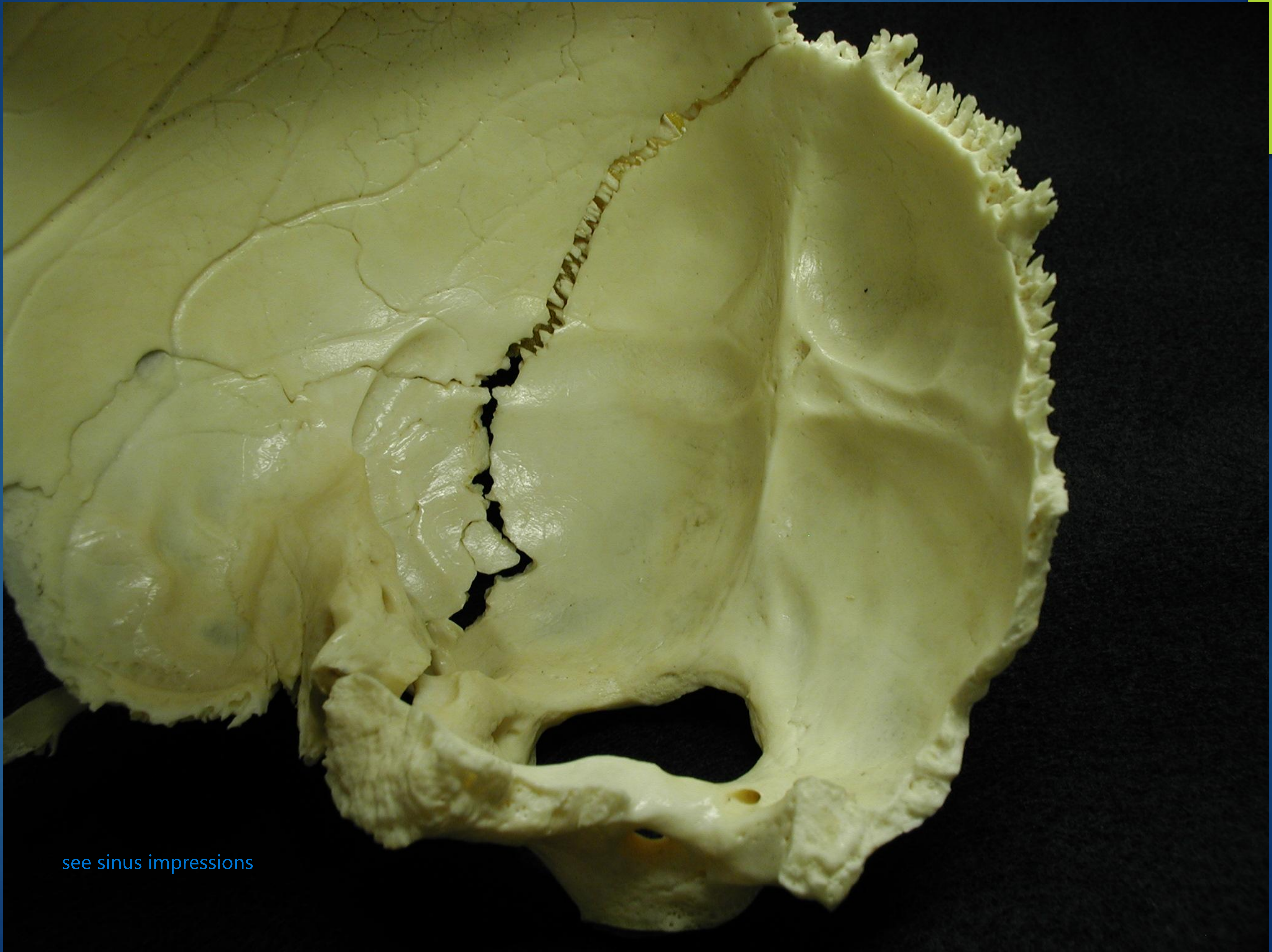
Fig. 677 Membranous wall [lateral wall] of right tympanic cavity; coronal section after removal of most of the pharyngotympanic tube [auditory tube] and removal of the fascia of tensor tympani; medial aspect (400%).

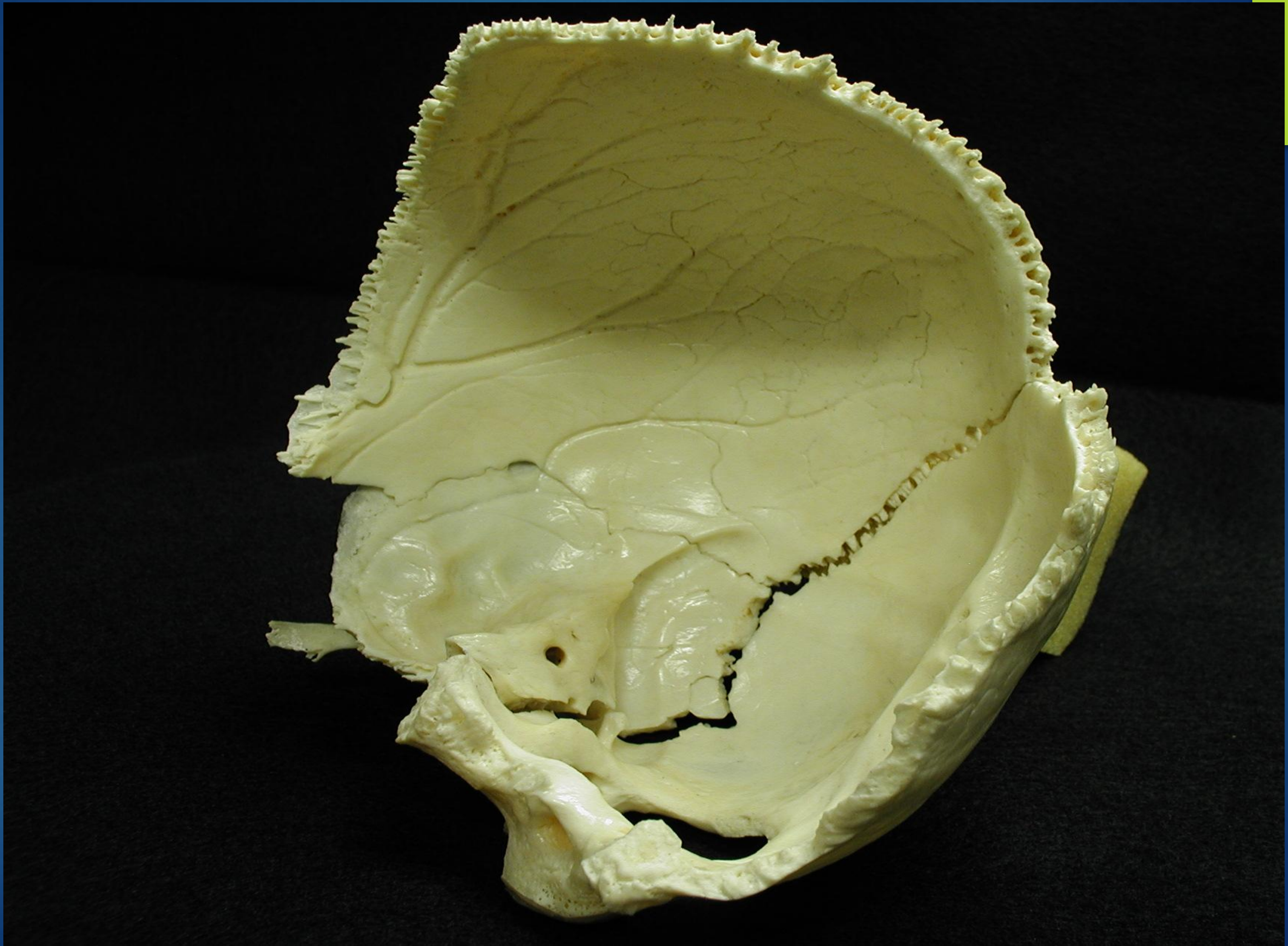
Temporal Bone Vascular Associations

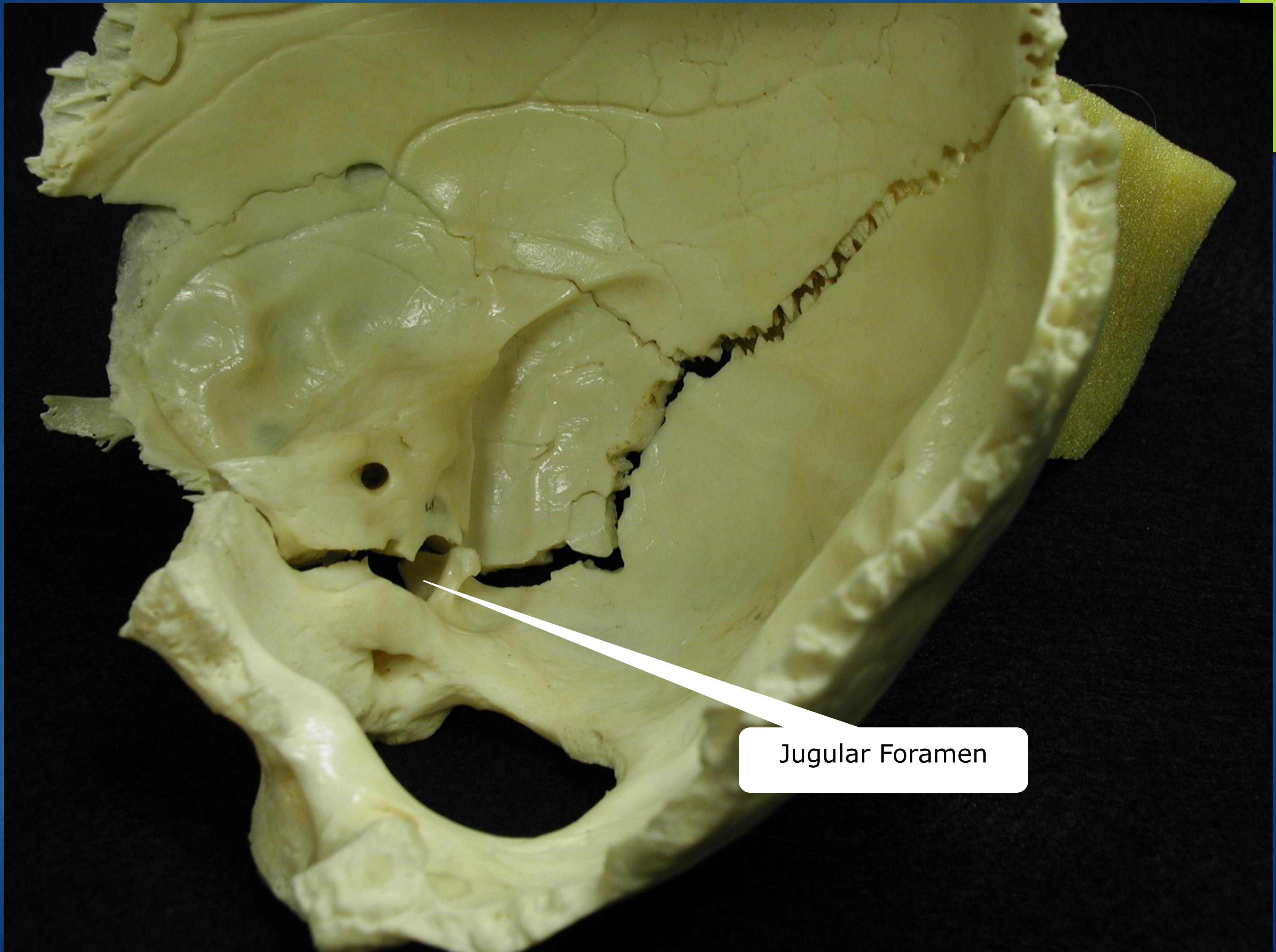
- Sigmoid Sinus
- Superior Petrosal Sinus
- Inferior Petrosal Sinus
- Jugular Vein (Occipital-Mastoid Foramen)
- Internal Carotid Artery

Temporal Bone Venous Drainage Associations

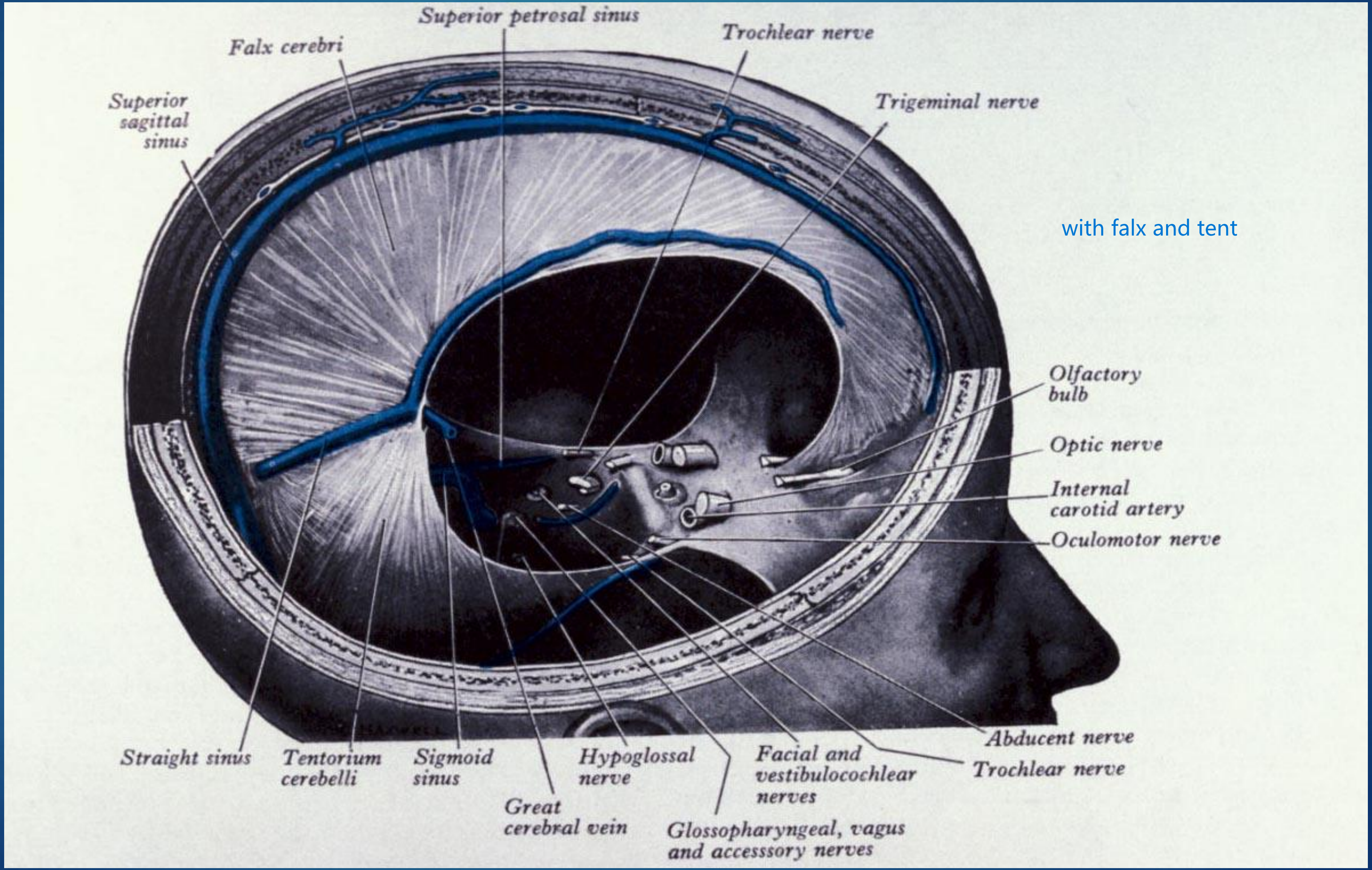



see sinus impressions

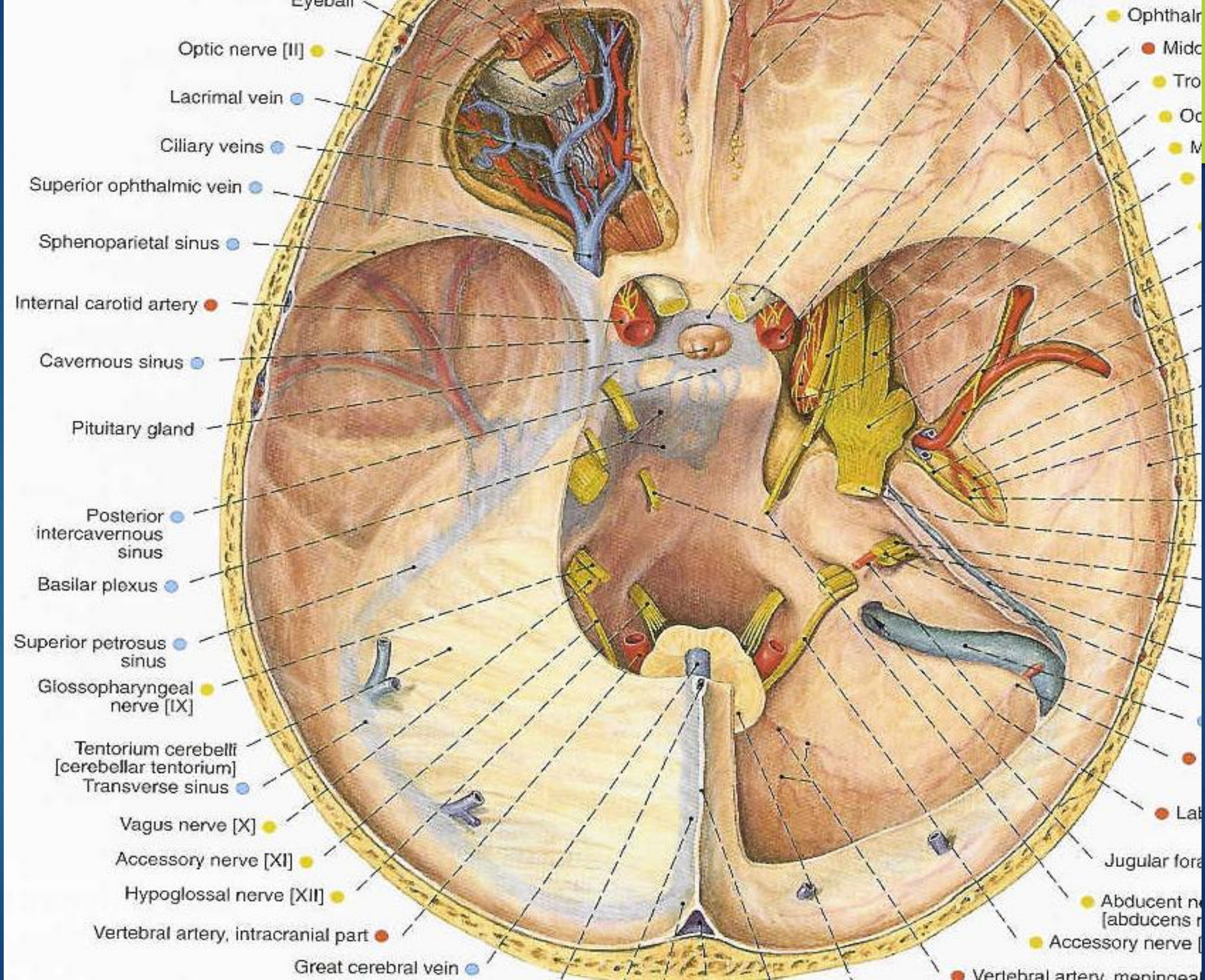



Jugular Foramen

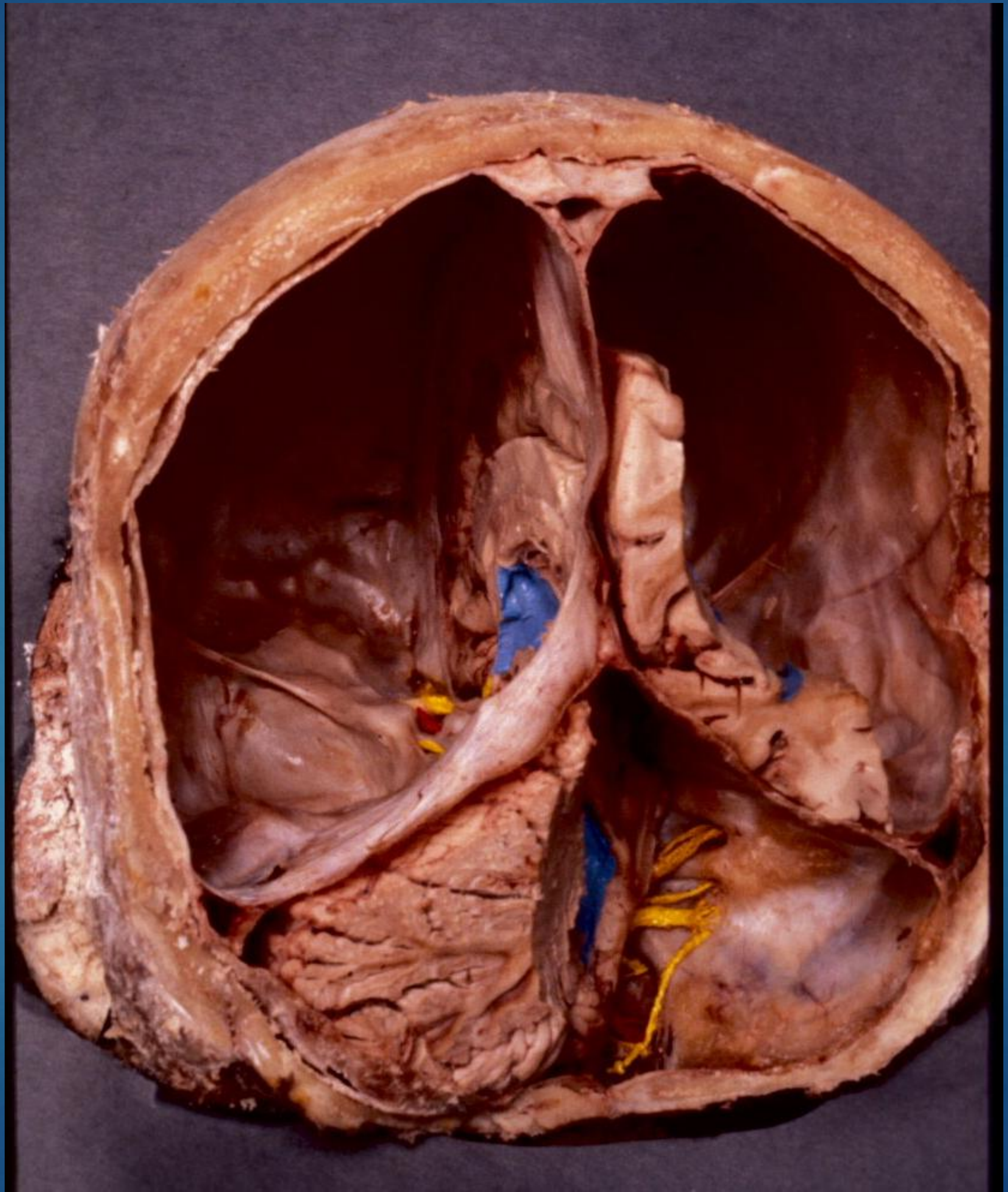
Temporal Bone Dural Associations



with falx and tent



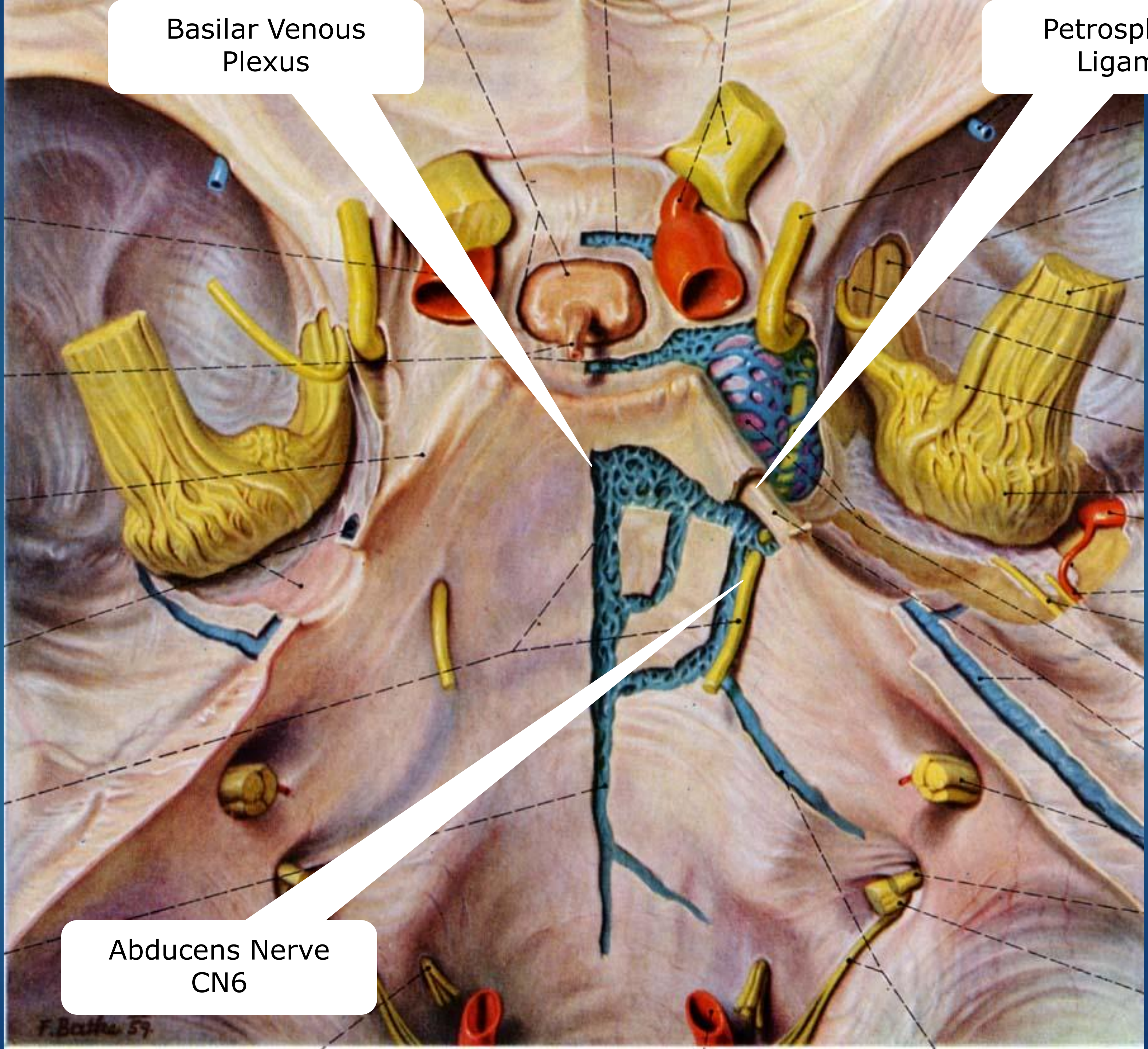
Optic nerve [II]
Lacrimal vein
Ciliary veins
Superior ophthalmic vein
Sphenoparietal sinus
Internal carotid artery
Cavernous sinus
Pituitary gland
Posterior intercavernous sinus
Basilar plexus
Superior petrosus sinus
Glossopharyngeal nerve [IX]
Tentorium cerebelli [cerebellar tentorium]
Transverse sinus
Vagus nerve [X]
Accessory nerve [XI]
Hypoglossal nerve [XII]
Vertebral artery, intracranial part
Great cerebral vein

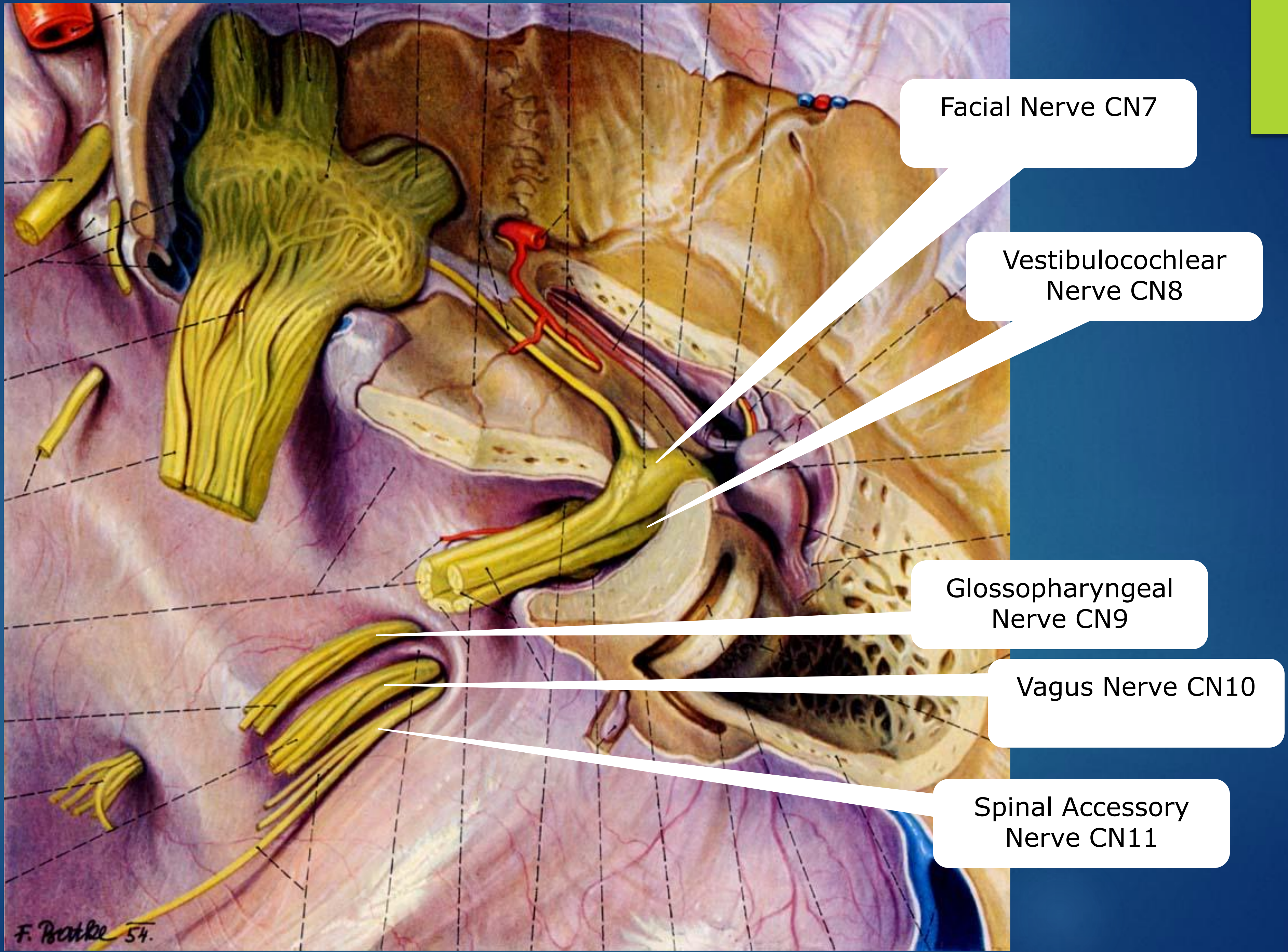


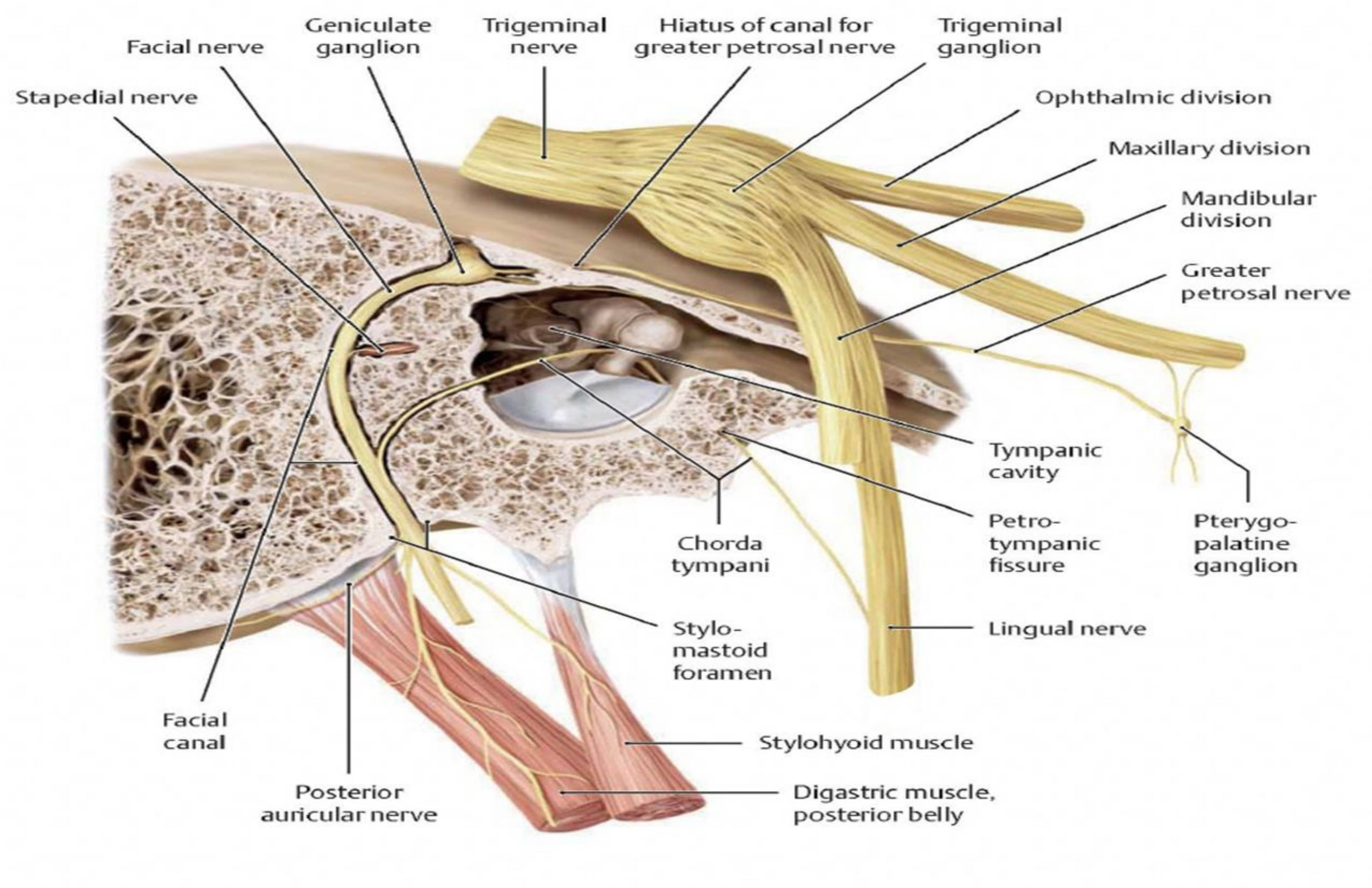
Temporal Bone Neural Associations

- V (Trigeminal-ganglion portion)
- VI (Abducent)
- VII (Facial)
- VIII (Vestibulo-cochlear)
- IX (Glossopharyngeal)
- X (Vagus)
- XI (Spinal Accessory)
- Greater Petrosal

Occipital-
Mastoid
Foramen

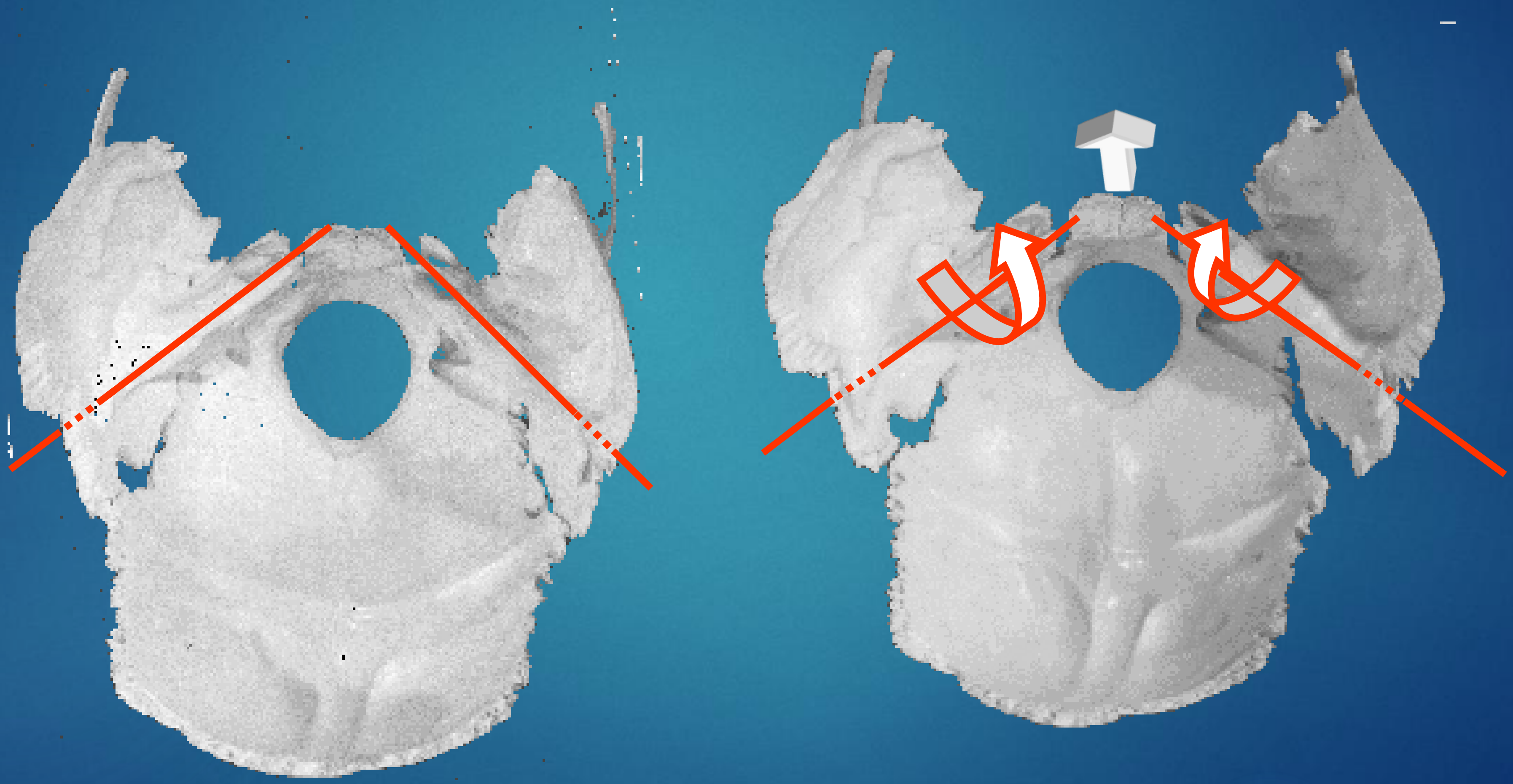

Basilar Venous Plexus
Petrosphenoid Ligament
Abducens Nerve CN6


F. Batke 54.
Facial Nerve CN7
Vestibulocochlear Nerve CN8
Glossopharyngeal Nerve CN9
Vagus Nerve CN10
Spinal Accessory Nerve CN11


Geniculate ganglion
Trigeminal nerve
Hiatus of canal for greater petrosal nerve
Trigeminal ganglion
Facial nerve
Stapedial nerve
Ophthalmic division
Maxillary division
Mandibular division
Greater petrosal nerve
Tympanic cavity
Petro-tympanic fissure
Pterygo-palatine ganglion
Chorda tympani
Stylo-mastoid foramen
Lingual nerve
Facial canal
Stylohyoid muscle
Posterior auricular nerve
Digastric muscle, posterior belly

Temporal Bone PRM Motion

Paired Bone Motion



axis around 3 digit

Clinical Indications for Temporal Bone Evaluation and/or Treatment

- Otitis (especially otitis media)
- Eustachian Tube Dysfunction
- Vertigo
- Tinnitus
- Hyper/Hypoacusis
- GI Autonomic Dysfunction
- CN7 palsy (Bell's Palsy)
- Migraine cephalgia
- Extraocular muscle dysfunction
- Secondary Respiratory Compromise
- TMJ Dysfunction

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